# Chapter 6



(© Tim Scrivener/Alamy)

In order to design the many parts of this boom assembly it is required that we know the forces that they must support. In this chapter we will show how to analyze such structures using the equations of equilibrium.

# Structural Analysis

## CHAPTER OBJECTIVES

- To show how to determine the forces in the members of a truss using the method of joints and the method of sections.
- To analyze the forces acting on the members of frames and machines composed of pin-connected members.

## 6.1 Simple Trusses

A ***truss*** is a structure composed of slender members joined together at their end points. The members commonly used in construction consist of wooden struts or metal bars. In particular, *planar* trusses lie in a single plane and are often used to support roofs and bridges. The truss shown in Fig. 6–1*a* is an example of a typical roof-supporting truss. In this figure, the roof load is transmitted to the truss *at the joints* by means of a series of *purlins*. Since this loading acts in the same plane as the truss, Fig. 6–1*b,* the analysis of the forces developed in the truss members will be two-dimensional.

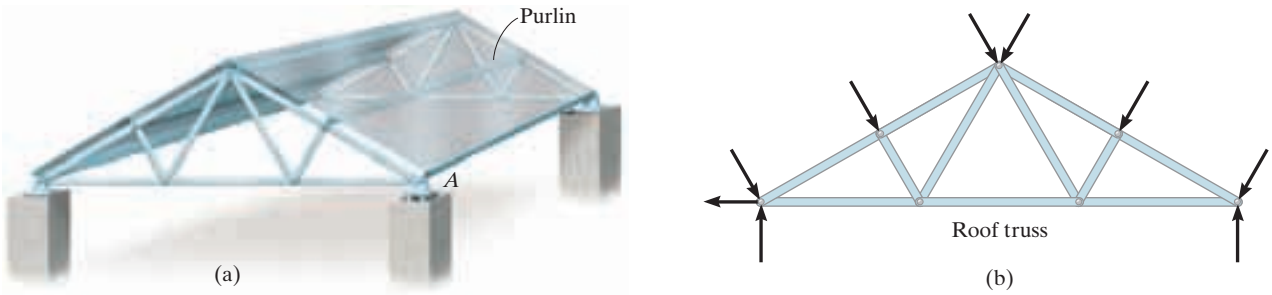



**Fig. 6–1**

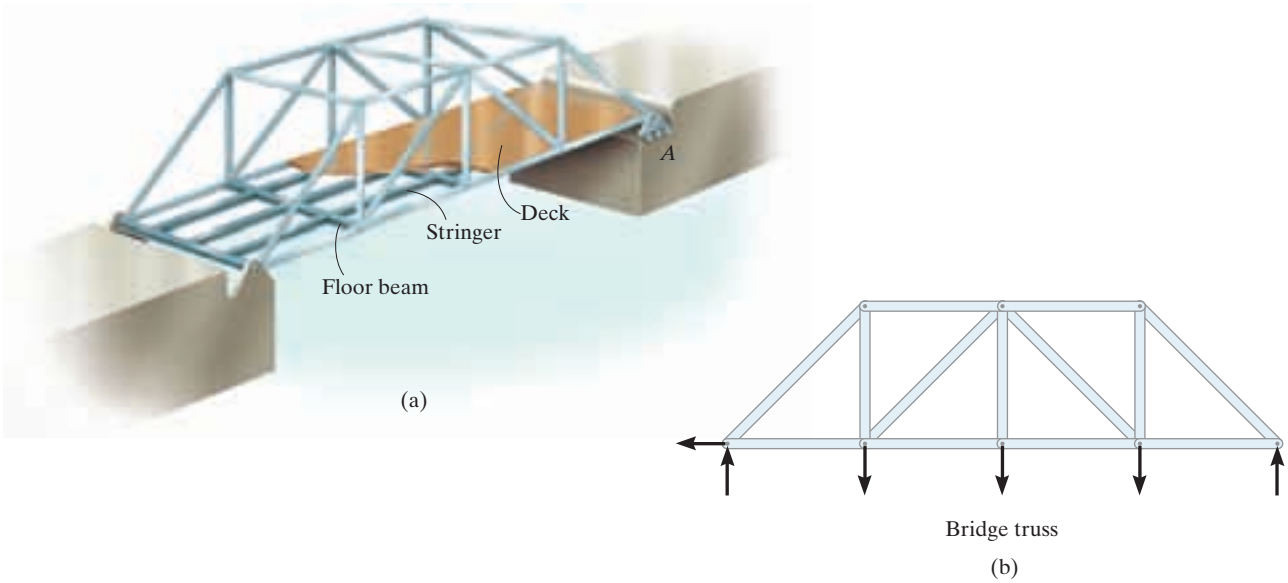



**Fig. 6–2**

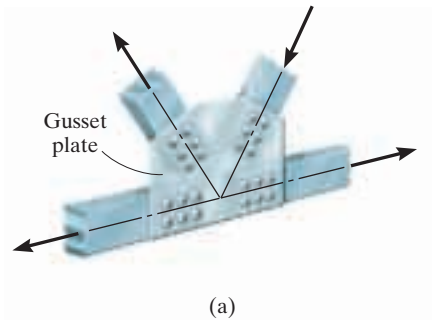



**Fig. 6–3**

In the case of a bridge, such as shown in Fig. 6–2*a*, the load on the deck is first transmitted to *stringers*, then to *floor beams*, and finally to the *joints* of the two supporting side trusses. Like the roof truss, the bridge truss loading is also coplanar, Fig. 6–2*b*.

When bridge or roof trusses extend over large distances, a rocker or roller is commonly used for supporting one end, for example, joint *A* in Figs. 6–1*a* and 6–2*a*. This type of support allows freedom for expansion or contraction of the members due to a change in temperature or application of loads.

**Assumptions for Design.** To design both the members and the connections of a truss, it is necessary first to determine the *force* developed in each member when the truss is subjected to a given loading. To do this we will make two important assumptions:

- ***All loadings are applied at the joints.*** In most situations, such as for bridge and roof trusses, this assumption is true. Frequently the weight of the members is neglected because the force supported by each member is usually much larger than its weight. However, if the weight is to be included in the analysis, it is generally satisfactory to apply it as a vertical force, with half of its magnitude applied at each end of the member.
- ***The members are joined together by smooth pins.*** The joint connections are usually formed by bolting or welding the ends of the members to a common plate, called a *gusset plate*, as shown in Fig. 6–3*a*, or by simply passing a large bolt or pin through each of the members, Fig. 6–3*b*. We can assume these connections act as pins provided the center lines of the joining members are *concurrent*, as in Fig. 6–3.

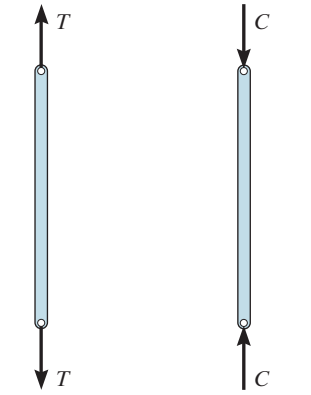



Fig. 6–4

Because of these two assumptions, *each truss member will act as a two-force member*, and therefore the force acting at each end of the member will be directed along the axis of the member. If the force tends to *elongate* the member, it is a ***tensile force*** (T), Fig. 6–4*a*; whereas if it tends to *shorten* the member, it is a ***compressive force*** (C), Fig. 6–4*b*. In the actual design of a truss it is important to state whether the nature of the force is tensile or compressive. Often, compression members must be made *thicker* than tension members because of the buckling or column effect that occurs when a member is in compression.



The use of metal gusset plates in the construction of these Warren trusses is clearly evident. (© Russell C. Hibbeler)

**Simple Truss.** If three members are pin connected at their ends, they form a *triangular truss* that will be *rigid*, Fig. 6–5. Attaching two more members and connecting these members to a new joint *D* forms a larger truss, Fig. 6–6. This procedure can be repeated as many times as desired to form an even larger truss. If a truss can be constructed by expanding the basic triangular truss in this way, it is called a ***simple truss***.

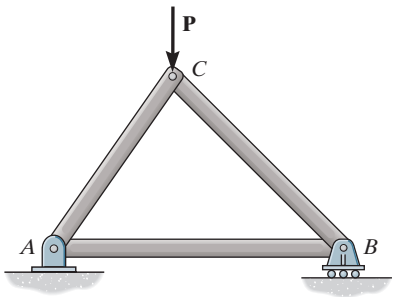



Fig. 6–5

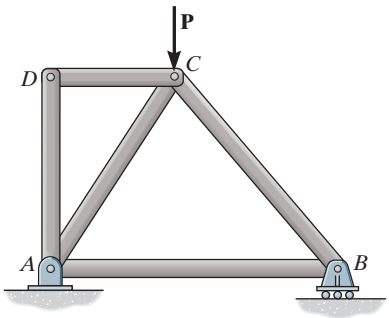
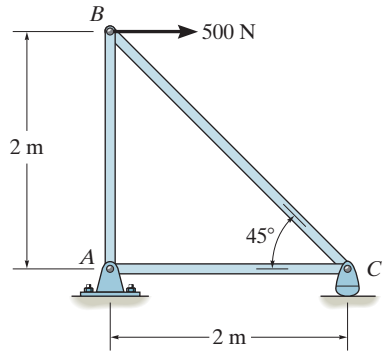
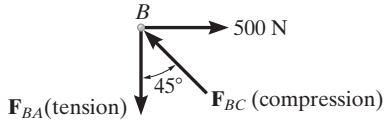


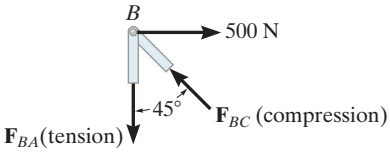

Fig. 6–6




(a)




(b)




(c)

**Fig. 6–7**



The forces in the members of this simple roof truss can be determined using the method of joints. (© Russell C. Hibbeler)

## 6.2 The Method of Joints

In order to analyze or design a truss, it is necessary to determine the force in each of its members. One way to do this is to use the ***method of joints***. This method is based on the fact that if the entire truss is in equilibrium, then each of its joints is also in equilibrium. Therefore, if the free-body diagram of each joint is drawn, the force equilibrium equations can then be used to obtain the member forces acting on each joint. Since the members of a *plane truss* are straight two-force members lying in a single plane, each joint is subjected to a force system that is *coplanar and concurrent*. As a result, only $\Sigma F_x = 0$ and $\Sigma F_y = 0$ need to be satisfied for equilibrium.

For example, consider the pin at joint $B$ of the truss in Fig. 6–7*a*. Three forces act on the pin, namely, the 500-N force and the forces exerted by members $BA$ and $BC$. The free-body diagram of the pin is shown in Fig. 6–7*b*. Here, $\mathbf{F}_{BA}$ is "pulling" on the pin, which means that member $BA$ is in *tension;* whereas $\mathbf{F}_{BC}$ is "pushing" on the pin, and consequently member $BC$ is in *compression.* These effects are clearly demonstrated by isolating the joint with small segments of the member connected to the pin, Fig. 6–7*c*. The pushing or pulling on these small segments indicates the effect of the member being either in compression or tension.

When using the method of joints, always start at a joint having at least one known force and at most two unknown forces, as in Fig. 6–7*b*. In this way, application of $\Sigma F_x = 0$ and $\Sigma F_y = 0$ yields two algebraic equations which can be solved for the two unknowns. When applying these equations, the correct sense of an unknown member force can be determined using one of two possible methods.

- The *correct* sense of direction of an unknown member force can, in many cases, be determined "by inspection." For example, $\mathbf{F}_{BC}$ in Fig. 6–7*b* must push on the pin (compression) since its horizontal component, $F_{BC} \sin 45°$, must balance the 500-N force ($\Sigma F_x = 0$). Likewise, $\mathbf{F}_{BA}$ is a tensile force since it balances the vertical component, $F_{BC} \cos 45°$ ($\Sigma F_y = 0$). In more complicated cases, the sense of an unknown member force can be *assumed*; then, after applying the equilibrium equations, the assumed sense can be verified from the numerical results. A *positive* answer indicates that the sense is *correct*, whereas a *negative* answer indicates that the sense shown on the free-body diagram must be *reversed.*
- *Always assume* the *unknown member forces* acting on the joint's free-body diagram to be in *tension*; i.e., the forces "pull" on the pin. If this is done, then numerical solution of the equilibrium equations will yield *positive scalars for members in tension and negative scalars for members in compression.* Once an unknown member force is found, use its *correct* magnitude and sense (T or C) on subsequent joint free-body diagrams.

## Important Points

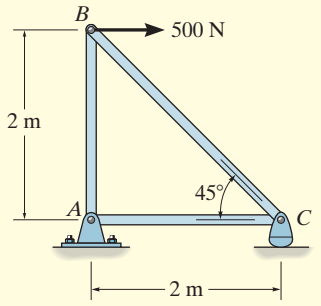
- Simple trusses are composed of triangular elements. The members are assumed to be pin connected at their ends and loads applied at the joints.
- If a truss is in equilibrium, then each of its joints is in equilibrium. The internal forces in the members become external forces when the free-body diagram of each joint of the truss is drawn. A force pulling on a joint is caused by tension in a member, and a force pushing on a joint is caused by compression.

## Procedure for Analysis

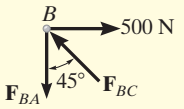
The following procedure provides a means for analyzing a truss using the method of joints.

- Draw the free-body diagram of a joint having at least one known force and at most two unknown forces. (If this joint is at one of the supports, then it may be necessary first to calculate the external reactions at the support.)
- Use one of the two methods described above for establishing the sense of an unknown force.
- Orient the $x$ and $y$ axes such that the forces on the free-body diagram can be easily resolved into their $x$ and $y$ components and then apply the two force equilibrium equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$. Solve for the two unknown member forces and verify their correct sense.
- Using the calculated results, continue to analyze each of the other joints. Remember that a member in *compression* "pushes" on the joint and a member in *tension* "pulls" on the joint. Also, be sure to choose a joint having at most two unknowns and at least one known force.

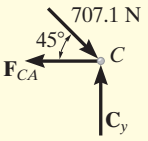
## EXAMPLE 6.1



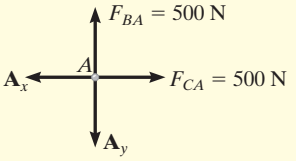

(a)



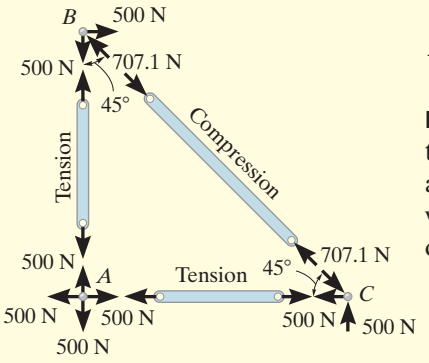

(b)




(c)




(d)




(e)

**Fig. 6–8**

Determine the force in each member of the truss shown in Fig. 6–8*a* and indicate whether the members are in tension or compression.

### SOLUTION

Since we should have no more than two unknown forces at the joint and at least one known force acting there, we will begin our analysis at joint *B*.

**Joint B.** The free-body diagram of the joint at *B* is shown in Fig. 6–8*b*. Applying the equations of equilibrium, we have

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \quad 500\ \text{N} - F_{BC}\sin 45° = 0 \quad F_{BC} = 707.1\ \text{N (C)} \quad \textit{Ans.}$$

$$+\uparrow\Sigma F_y = 0; \quad F_{BC}\cos 45° - F_{BA} = 0 \quad F_{BA} = 500\ \text{N (T)} \quad \textit{Ans.}$$

Since the force in member *BC* has been calculated, we can proceed to analyze joint *C* to determine the force in member *CA* and the support reaction at the rocker.

**Joint C.** From the free-body diagram of joint *C*, Fig. 6–8*c*, we have

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \quad -F_{CA} + 707.1\cos 45°\ \text{N} = 0 \quad F_{CA} = 500\ \text{N (T)} \quad \textit{Ans.}$$

$$+\uparrow\Sigma F_y = 0; \quad C_y - 707.1\sin 45°\ \text{N} = 0 \quad C_y = 500\ \text{N} \quad \textit{Ans.}$$

**Joint A.** Although it is not necessary, we can determine the components of the support reactions at joint *A* using the results of $F_{CA}$ and $F_{BA}$. From the free-body diagram, Fig. 6–8*d*, we have

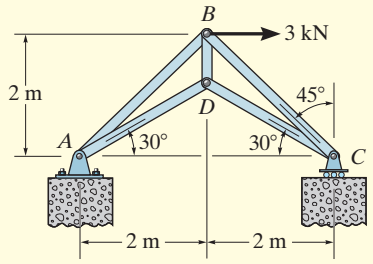
$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \quad 500\ \text{N} - A_x = 0 \quad A_x = 500\ \text{N}$$

$$+\uparrow\Sigma F_y = 0; \quad 500\ \text{N} - A_y = 0 \quad A_y = 500\ \text{N}$$

**NOTE:** The results of the analysis are summarized in Fig. 6–8*e*. Note that the free-body diagram of each joint (or pin) shows the effects of all the connected members and external forces applied to the joint, whereas the free-body diagram of each member shows only the effects of the end joints on the member.

## EXAMPLE 6.2

Determine the forces acting in all the members of the truss shown in Fig. 6–9*a*.



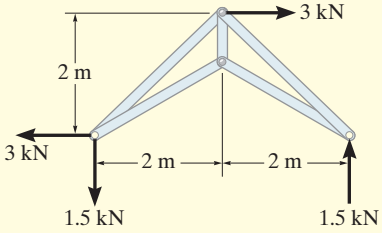
(a)

### SOLUTION

By inspection, there are more than two unknowns at each joint. Consequently, the support reactions on the truss must first be determined. Show that they have been correctly calculated on the free-body diagram in Fig. 6–9*b*. We can now begin the analysis at joint *C*. Why?

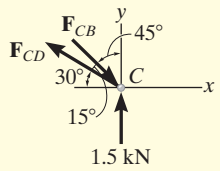
**Joint C.** From the free-body diagram, Fig. 6–9*c*,

$$\pm\rightarrow \Sigma F_x = 0; \qquad -F_{CD}\cos 30° + F_{CB}\sin 45° = 0$$
$$+\uparrow \Sigma F_y = 0; \qquad 1.5\text{ kN} + F_{CD}\sin 30° - F_{CB}\cos 45° = 0$$



(b)

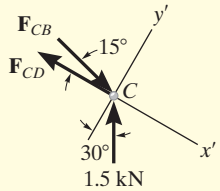
These two equations must be solved *simultaneously* for each of the two unknowns. Note, however, that a *direct solution* for one of the unknown forces may be obtained by applying a force summation along an axis that is *perpendicular* to the direction of the other unknown force. For example, summing forces along the $y'$ axis, which is perpendicular to the direction of $\mathbf{F}_{CD}$, Fig. 6–9*d*, yields a *direct solution* for $F_{CB}$.



(c)

$$+\nearrow \Sigma F_{y'} = 0; \qquad 1.5\cos 30°\text{ kN} - F_{CB}\sin 15° = 0$$
$$F_{CB} = 5.019\text{ kN} = 5.02\text{ kN (C)} \qquad \textit{Ans.}$$

Then,



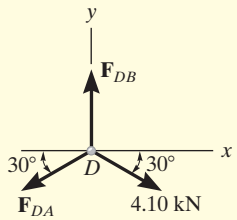
(d)

$$+\searrow \Sigma F_{x'} = 0;$$
$$-F_{CD} + 5.019\cos 15° - 1.5\sin 30° = 0; \qquad F_{CD} = 4.10\text{ kN (T)} \qquad \textit{Ans.}$$

**Joint D.** We can now proceed to analyze joint *D*. The free-body diagram is shown in Fig. 6–9*e*.

$$\pm\rightarrow \Sigma F_x = 0; \qquad -F_{DA}\cos 30° + 4.10\cos 30°\text{ kN} = 0$$
$$F_{DA} = 4.10\text{ kN} \quad \text{(T)} \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \qquad F_{DB} - 2(4.10\sin 30°\text{ kN}) = 0$$
$$F_{DB} = 4.10\text{ kN} \quad \text{(T)} \qquad \textit{Ans.}$$



(e)

**NOTE:** The force in the last member, *BA*, can be obtained from joint *B* or joint *A*. As an exercise, draw the free-body diagram of joint *B*, sum the forces in the horizontal direction, and show that $F_{BA} = 0.776$ kN (C).

**Fig. 6–9**

Determine the force in each member of the truss shown in Fig. 6–10*a*. Indicate whether the members are in tension or compression.

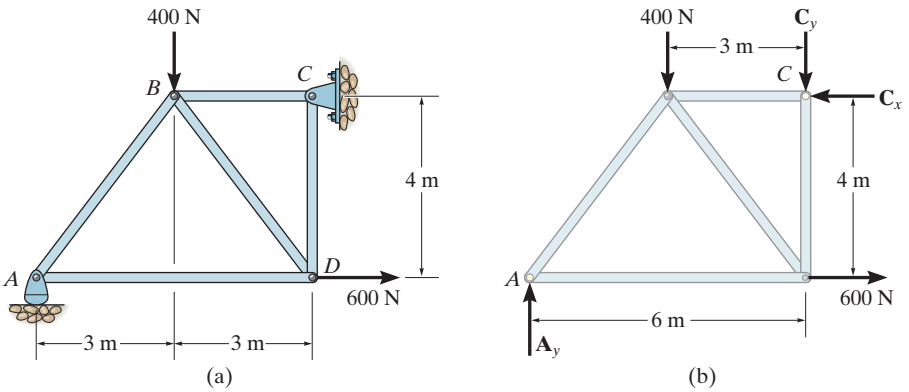


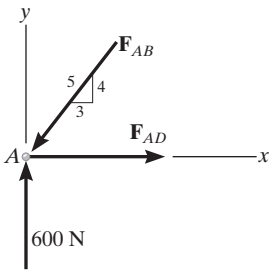

**Fig. 6–10**

## SOLUTION

**Support Reactions.** No joint can be analyzed until the support reactions are determined, because each joint has at least three unknown forces acting on it. A free-body diagram of the entire truss is given in Fig. 6–10*b*. Applying the equations of equilibrium, we have

$$\begin{aligned}
\overset{+}{\rightarrow}\Sigma F_x &= 0; & 600\ \text{N} - C_x &= 0 & C_x &= 600\ \text{N}\\
\circlearrowleft +\Sigma M_C &= 0; & -A_y(6\ \text{m}) + 400\ \text{N}(3\ \text{m}) &+ 600\ \text{N}(4\ \text{m}) = 0\\
& & A_y &= 600\ \text{N}\\
+\uparrow\Sigma F_y &= 0; & 600\ \text{N} - 400\ \text{N} - C_y &= 0 & C_y &= 200\ \text{N}
\end{aligned}$$

The analysis can now start at either joint $A$ or $C$. The choice is arbitrary since there are one known and two unknown member forces acting on the pin at each of these joints.




**Joint A.** (Fig. 6–10*c*). As shown on the free-body diagram, $\mathbf{F}_{AB}$ is assumed to be compressive and $\mathbf{F}_{AD}$ is tensile. Applying the equations of equilibrium, we have

$$\begin{aligned}
+\uparrow\Sigma F_y &= 0; & 600\ \text{N} - \tfrac{4}{5}F_{AB} &= 0 & F_{AB} &= 750\ \text{N}\ \ (\text{C}) && \textit{Ans.}\\
\overset{+}{\rightarrow}\Sigma F_x &= 0; & F_{AD} - \tfrac{3}{5}(750\ \text{N}) &= 0 & F_{AD} &= 450\ \text{N}\ \ (\text{T}) && \textit{Ans.}
\end{aligned}$$

**Joint *D*.** (Fig. 6–10*d*). Using the result for $F_{AD}$ and summing forces in the horizontal direction, Fig. 6–10*d*, we have

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \qquad -450\text{ N} + \tfrac{3}{5}F_{DB} + 600\text{ N} = 0 \qquad F_{DB} = -250\text{ N}$$

The negative sign indicates that $\mathbf{F}_{DB}$ acts in the *opposite sense* to that shown in Fig. 6–10*d*.* Hence,

$$F_{DB} = 250\text{ N (T)} \qquad \textit{Ans.}$$

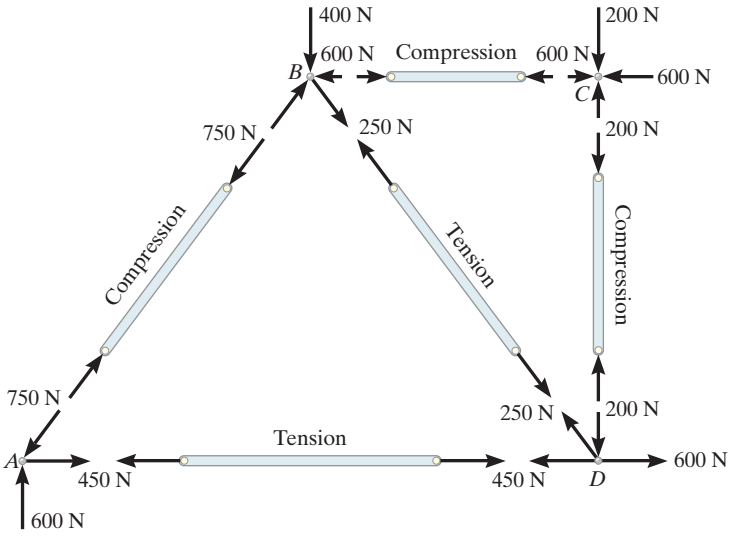
To determine $\mathbf{F}_{DC}$, we can either correct the sense of $\mathbf{F}_{DB}$ on the free-body diagram, and then apply $\Sigma F_y = 0$, or apply this equation and retain the negative sign for $F_{DB}$, i.e.,

$$+\uparrow\Sigma F_y = 0; \qquad -F_{DC} - \tfrac{4}{5}(-250\text{ N}) = 0 \qquad F_{DC} = 200\text{ N (C)} \qquad \textit{Ans.}$$

**Joint *C*.** (Fig. 6–10*e*).

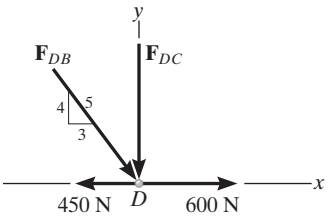
$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \qquad F_{CB} - 600\text{ N} = 0 \qquad F_{CB} = 600\text{ N (C)} \qquad \textit{Ans.}$$
$$+\uparrow\Sigma F_y = 0; \qquad 200\text{ N} - 200\text{ N} \equiv 0 \text{ (check)}$$

**NOTE:** The analysis is summarized in Fig. 6–10*f*, which shows the free-body diagram for each joint and member.

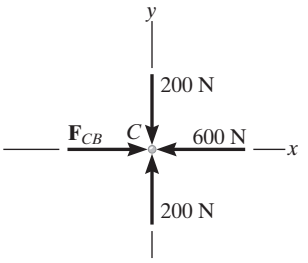



(f)

**Fig. 6–10 (cont.)**




(d)



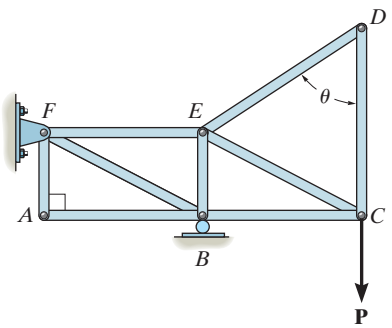

(e)

*The proper sense could have been determined by inspection, prior to applying $\Sigma F_x = 0$.

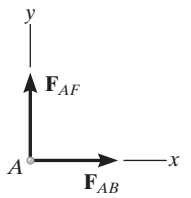
## 6.3 Zero-Force Members

Truss analysis using the method of joints is greatly simplified if we can first identify those members which support *no loading*. These *zero-force members* are used to increase the stability of the truss during construction and to provide added support if the loading is changed.

The zero-force members of a truss can generally be found *by inspection* of each of the joints. For example, consider the truss shown in Fig. 6–11*a*. If a free-body diagram of the pin at joint $A$ is drawn, Fig. 6–11*b*, it is seen that members $AB$ and $AF$ are zero-force members. (We could not have come to this conclusion if we had considered the free-body diagrams of joints $F$ or $B$ simply because there are five unknowns at each of these joints.) In a similar manner, consider the free-body diagram of joint $D$, Fig. 6–11*c*. Here again it is seen that $DC$ and $DE$ are zero-force members. From these observations, we can conclude that *if only two non-collinear members form a truss joint and no external load or support reaction is applied to the joint, the two members must be zero-force members.* The load on the truss in Fig. 6–11*a* is therefore supported by only five members as shown in Fig. 6–11*d*.



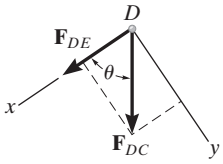

(a)




$+\rightarrow \Sigma F_x = 0;\ F_{AB} = 0$

$+\uparrow \Sigma F_y = 0;\ F_{AF} = 0$

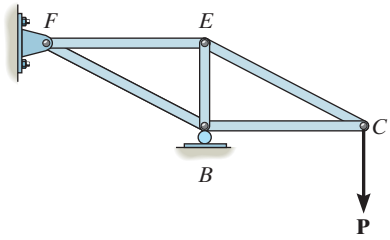
(b)




$+\searrow \Sigma F_y = 0;\ F_{DC} \sin\theta = 0;\quad F_{DC} = 0 \text{ since } \sin\theta \neq 0$

$+\swarrow \Sigma F_x = 0;\ F_{DE} + 0 = 0;\quad F_{DE} = 0$

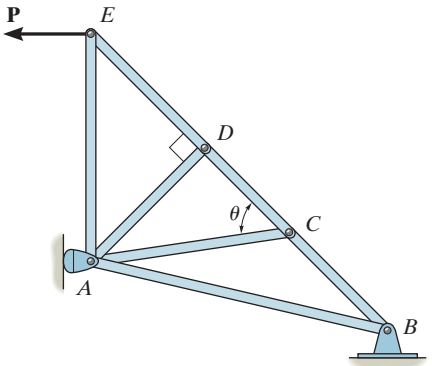
(c)



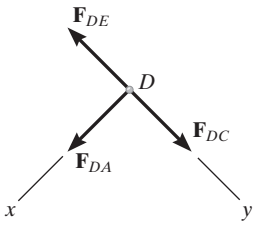

(d)

**Fig. 6–11**

Now consider the truss shown in Fig. 6–12*a*. The free-body diagram of the pin at joint *D* is shown in Fig. 6–12*b*. By orienting the *y* axis along members *DC* and *DE* and the *x* axis along member *DA*, it is seen that *DA* is a zero-force member. Note that this is also the case for member *CA*, Fig. 6–12*c*. In general then, *if three members form a truss joint for which two of the members are collinear, the third member is a zero-force member provided no external force or support reaction has a component that acts along this member.* The truss shown in Fig. 6–12*d* is therefore suitable for supporting the load **P**.



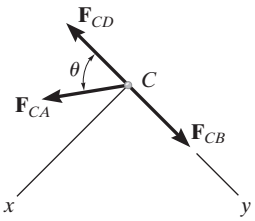

(a)




$+\swarrow \Sigma F_x = 0; \quad F_{DA} = 0$

$+\searrow \Sigma F_y = 0; \quad F_{DC} = F_{DE}$

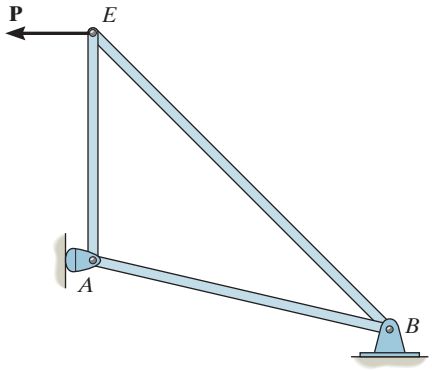
(b)




$+\swarrow \Sigma F_x = 0; \quad F_{CA} \sin\theta = 0; \quad F_{CA} = 0 \text{ since } \sin\theta \neq 0;$

$+\searrow \Sigma F_y = 0; \quad F_{CB} = F_{CD}$

(c)

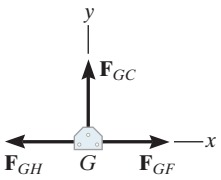



(d)

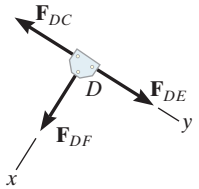
**Fig. 6–12**

## Important Point

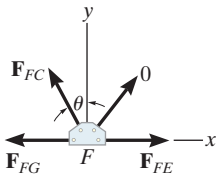
- Zero-force members support no load; however, they are necessary for stability, and are available when additional loadings are applied to the joints of the truss. These members can usually be identified by inspection. They occur at joints where only two members are connected and no external load acts along either member. Also, at joints having two collinear members, a third member will be a zero-force member if no external force components act along this member.



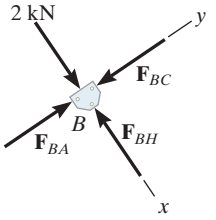

(b)



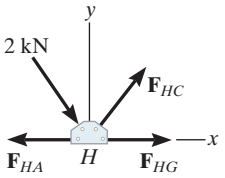

(c)




(d)

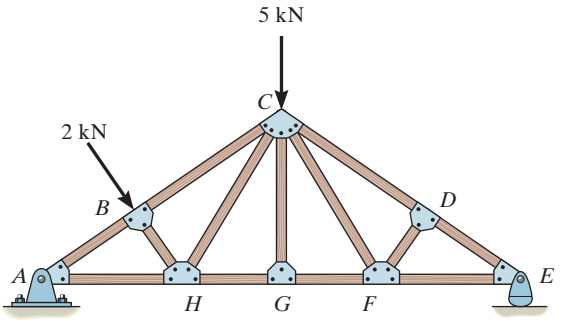



(e)




(f)

Using the method of joints, determine all the zero-force members of the *Fink roof truss* shown in Fig. 6–13*a*. Assume all joints are pin connected.




(a)

**Fig. 6–13**

## SOLUTION

Look for joint geometries that have three members for which two are collinear. We have

**Joint *G*.** (Fig. 6–13*b*).

$$+\uparrow \Sigma F_y = 0; \qquad F_{GC} = 0 \qquad \textit{Ans.}$$

Realize that we could not conclude that *GC* is a zero-force member by considering joint *C*, where there are five unknowns. The fact that *GC* is a zero-force member means that the 5-kN load at *C* must be supported by members *CB*, *CH*, *CF*, and *CD*.

**Joint *D*.** (Fig. 6–13*c*).

$$+\swarrow \Sigma F_x = 0; \qquad F_{DF} = 0 \qquad \textit{Ans.}$$

**Joint *F*.** (Fig. 6–13*d*).

$$+\uparrow \Sigma F_y = 0; \qquad F_{FC}\cos\theta = 0 \quad \text{Since } \theta \neq 90°, \quad F_{FC} = 0 \qquad \textit{Ans.}$$

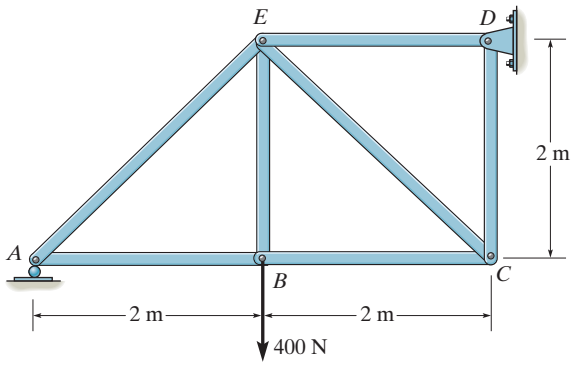
**NOTE:** If joint *B* is analyzed, Fig. 6–13*e*,

$$+\searrow \Sigma F_x = 0; \qquad 2\text{ kN} - F_{BH} = 0 \quad F_{BH} = 2\text{ kN} \quad \text{(C)}$$

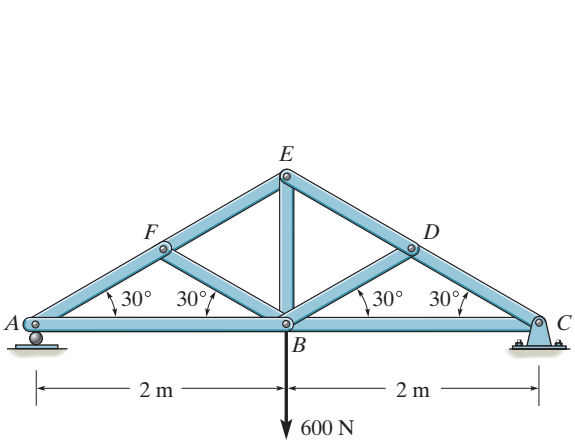
Also, $F_{HC}$ must satisfy $\Sigma F_y = 0$, Fig. 6–13*f*, and therefore *HC* is *not* a zero-force member.

## PRELIMINARY PROBLEMS

**P6–1.** In each case, calculate the support reactions and then draw the free-body diagrams of joints $A$, $B$, and $C$ of the truss.



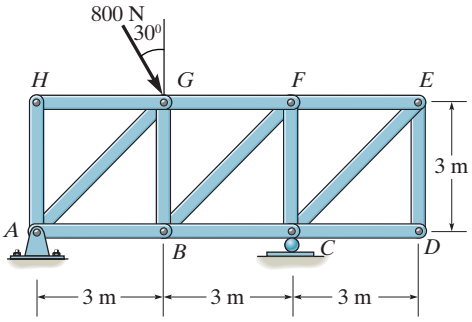

(a)



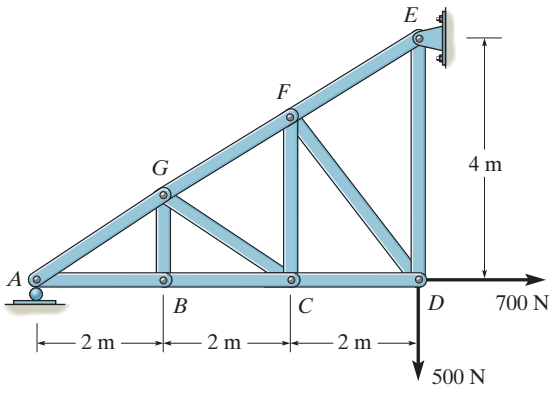

(b)

**Prob. P6–1**

**P6–2.** Identify the zero-force members in each truss.




(a)

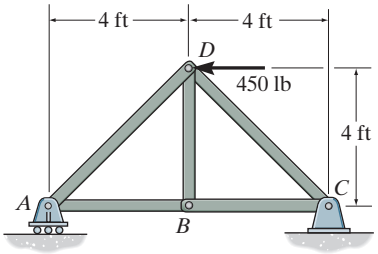



(b)

**Prob. P6–2**

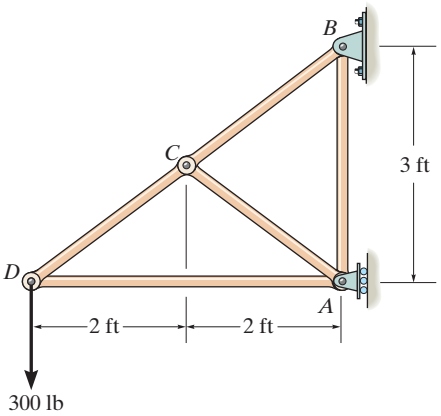
***All problem solutions must include FBDs.***

**F6–1.** Determine the force in each member of the truss. State if the members are in tension or compression.



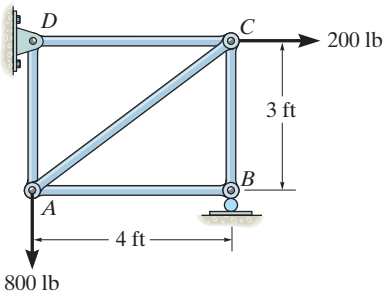

**Prob. F6–1**

**F6–2.** Determine the force in each member of the truss. State if the members are in tension or compression.



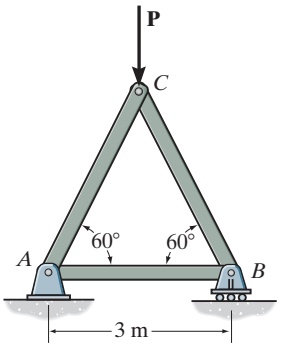

**Prob. F6–2**

**F6–3.** Determine the force in each member of the truss. State if the members are in tension or compression.



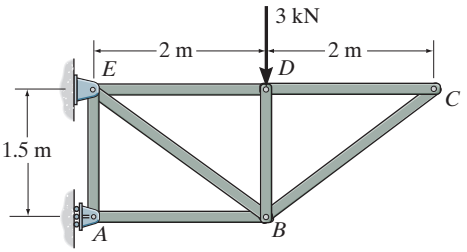

**Prob. F6–3**

**F6–4.** Determine the greatest load $P$ that can be applied to the truss so that none of the members are subjected to a force exceeding either 2 kN in tension or 1.5 kN in compression.



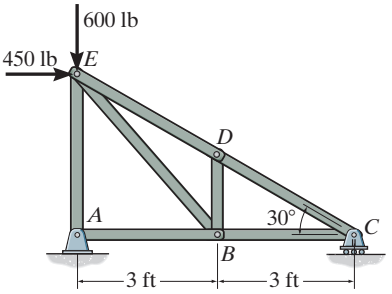

**Prob. F6–4**

**F6–5.** Identify the zero-force members in the truss.




**Prob. F6–5**

**F6–6.** Determine the force in each member of the truss. State if the members are in tension or compression.

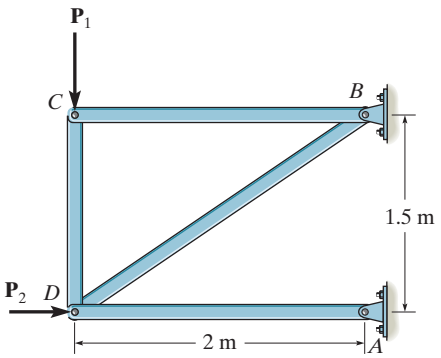



**Prob. F6–6**

***All problem solutions must include FBDs.***

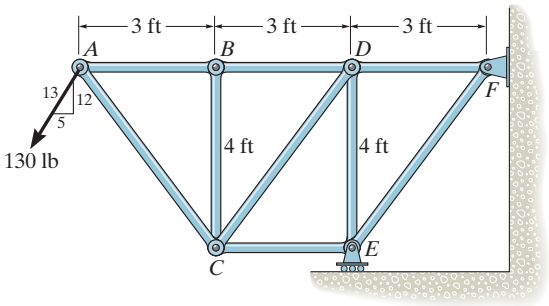
**6–1.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 20$ kN, $P_2 = 10$ kN.

**6–2.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 45$ kN, $P_2 = 30$ kN.



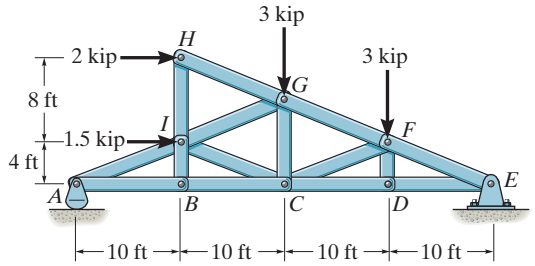

**Probs. 6–1/2**

**6–3.** Determine the force in each member of the truss. State if the members are in tension or compression.




**Prob. 6–3**

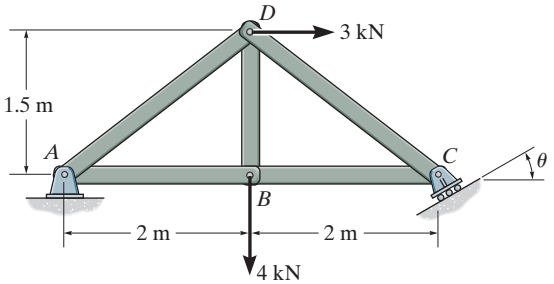
***6–4.** Determine the force in each member of the truss and state if the members are in tension or compression.




**Prob. 6–4**

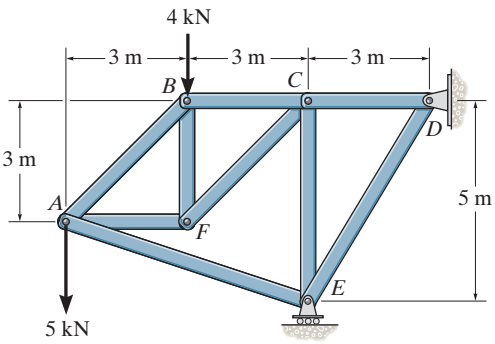
**6–5.** Determine the force in each member of the truss, and state if the members are in tension or compression. Set $\theta = 0°$.

**6–6.** Determine the force in each member of the truss, and state if the members are in tension or compression. Set $\theta = 30°$.



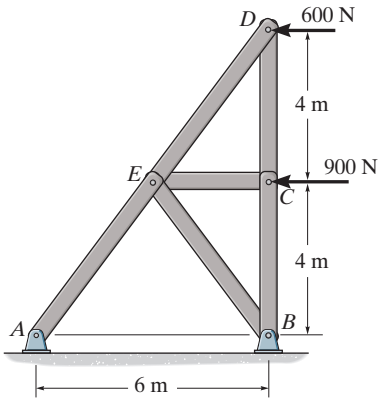

**Probs. 6–5/6**

**6–7.** Determine the force in each member of the truss and state if the members are in tension or compression.




**Prob. 6–7**

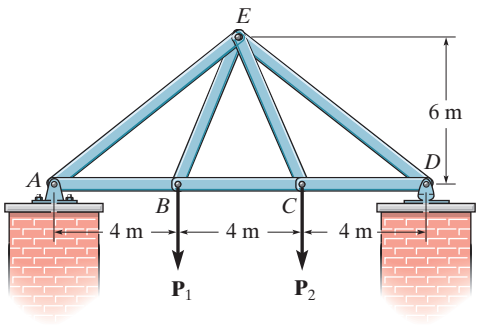
***6–8.** Determine the force in each member of the truss and state if the members are in tension or compression.




**Prob. 6–8**

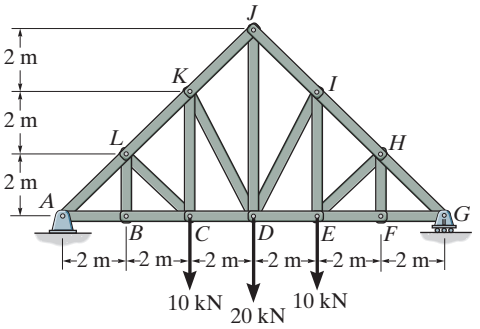
**6–9.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 3$ kN, $P_2 = 6$ kN.

**6–10.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 6$ kN, $P_2 = 9$ kN.



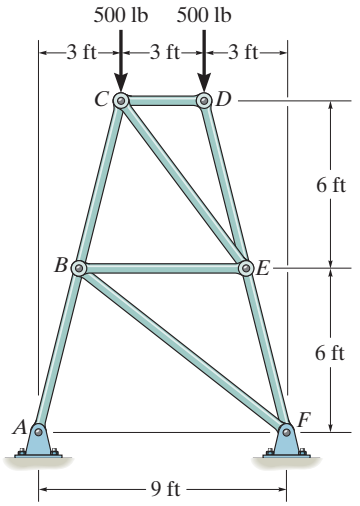

**Probs. 6–9/10**

**6–11.** Determine the force in each member of the *Pratt truss*, and state if the members are in tension or compression.




**Prob. 6–11**

***6–12.** Determine the force in each member of the truss and state if the members are in tension or compression.

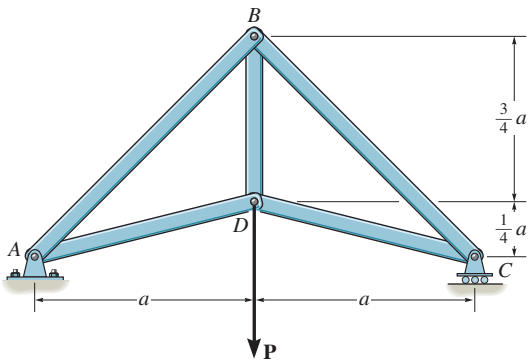



**Prob. 6–12**

**6–13.** Determine the force in each member of the truss in terms of the load $P$ and state if the members are in tension or compression.

**6–14.** Members $AB$ and $BC$ can each support a maximum compressive force of 800 lb, and members $AD$, $DC$, and $BD$ can support a maximum tensile force of 1500 lb. If $a = 10$ ft, determine the greatest load $P$ the truss can support.

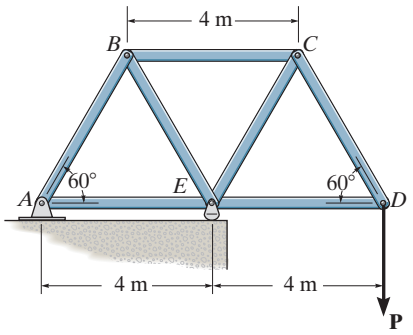
**6–15.** Members $AB$ and $BC$ can each support a maximum compressive force of 800 lb, and members $AD$, $DC$, and $BD$ can support a maximum tensile force of 2000 lb. If $a = 6$ ft, determine the greatest load $P$ the truss can support.




**Probs. 6–13/14/15**

***6–16.** Determine the force in each member of the truss. State whether the members are in tension or compression. Set $P = 8$ kN.

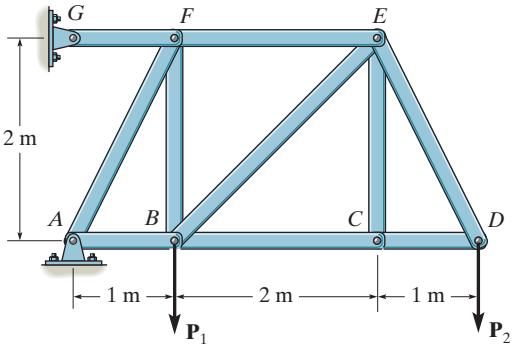
**6–17.** If the maximum force that any member can support is 8 kN in tension and 6 kN in compression, determine the maximum force $P$ that can be supported at joint $D$.




**Probs. 6–16/17**

**6–18.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 10$ kN, $P_2 = 8$ kN.

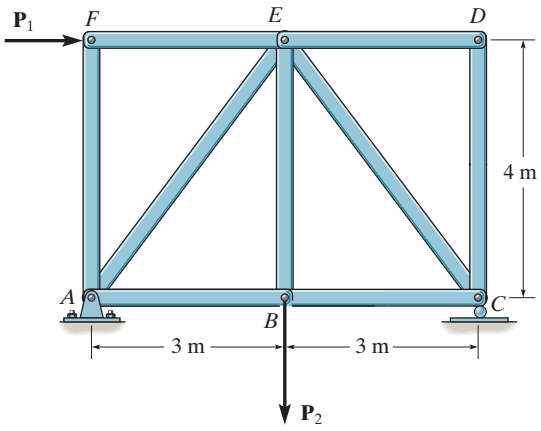
**6–19.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 8$ kN, $P_2 = 12$ kN.




**Probs. 6–18/19**

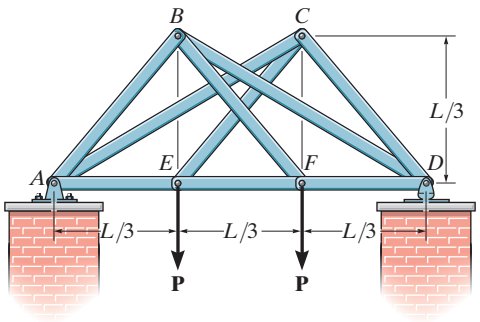
***6–20.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 9$ kN, $P_2 = 15$ kN.

**6–21.** Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 30$ kN, $P_2 = 15$ kN.



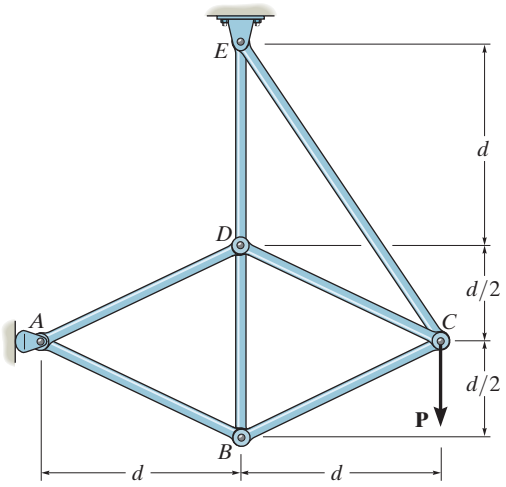

**Probs. 6–20/21**

**6–22.** Determine the force in each member of the double scissors truss in terms of the load $P$ and state if the members are in tension or compression.



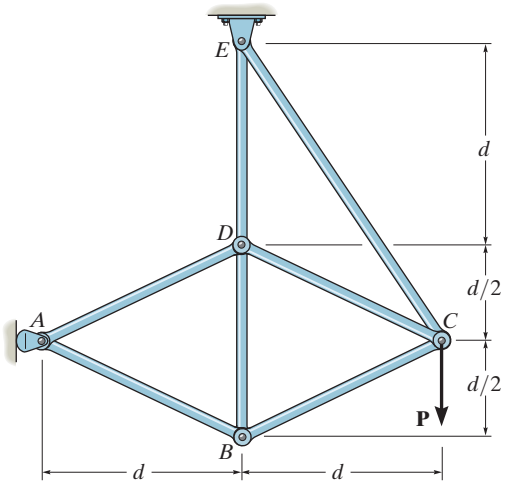

**Prob. 6–22**

**6–23.** Determine the force in each member of the truss in terms of the load $P$ and state if the members are in tension or compression.




**Prob. 6–23**

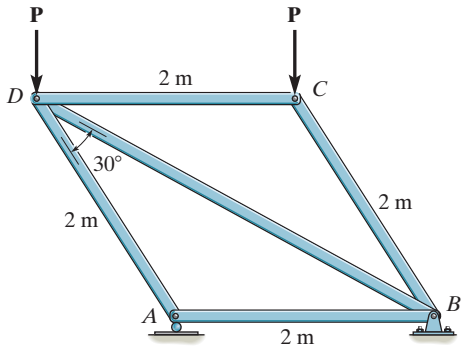
***6–24.** The maximum allowable tensile force in the members of the truss is $(F_t)_{max} = 5$ kN, and the maximum allowable compressive force is $(F_c)_{max} = 3$ kN. Determine the maximum magnitude of load **P** that can be applied to the truss. Take $d = 2$ m.




**Prob. 6–24**

**6–25.** Determine the force in each member of the truss in terms of the external loading and state if the members are in tension or compression. Take $P = 2$ kN.

**6–26.** The maximum allowable tensile force in the members of the truss is $(F_t)_{max} = 5$ kN, and the maximum allowable compressive force is $(F_c)_{max} = 3$ kN. Determine the maximum magnitude $P$ of the two loads that can be applied to the truss.




**Probs. 6–25/26**

## 6.4 The Method of Sections

When we need to find the force in only a few members of a truss, we can analyze the truss using the ***method of sections***. It is based on the principle that if the truss is in equilibrium then any segment of the truss is also in equilibrium. For example, consider the two truss members shown on the left in Fig. 6–14. If the forces within the members are to be determined, then an imaginary section, indicated by the blue line, can be used to cut each member into two parts and thereby "expose" each internal force as "external" to the free-body diagrams shown on the right. Clearly, it can be seen that equilibrium requires that the member in tension (T) be subjected to a "pull," whereas the member in compression (C) is subjected to a "push."

The method of sections can also be used to "cut" or section the members of an entire truss. If the section passes through the truss and the free-body diagram of either of its two parts is drawn, we can then apply the equations of equilibrium to that part to determine the member forces at the "cut section." Since only *three* independent equilibrium equations ($\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M_O = 0$) can be applied to the free-body diagram of any segment, then we should try to select a section that, in general, passes through not more than *three* members in which the forces are unknown. For example, consider the truss in Fig. 6–15*a*. If the forces in members *BC*, *GC*, and *GF* are to be determined, then section *aa* would be appropriate. The free-body diagrams of the two segments are shown in Figs. 6–15*b* and 6–15*c*. Note that the line of action of each member force is specified from the *geometry* of the truss, since the force in a member is along its axis. Also, the member forces acting on one part of the truss are equal but opposite to those acting on the other part—Newton's third law. Members *BC* and *GC* are assumed to be in *tension* since they are subjected to a "pull," whereas *GF* in *compression* since it is subjected to a "push."

The three unknown member forces $\mathbf{F}_{BC}$, $\mathbf{F}_{GC}$, and $\mathbf{F}_{GF}$ can be obtained by applying the three equilibrium equations to the free-body diagram in Fig. 6–15*b*. If, however, the free-body diagram in Fig. 6–15*c* is considered, the three support reactions $\mathbf{D}_x$, $\mathbf{D}_y$ and $\mathbf{E}_x$ will have to be known, because only three equations of equilibrium are available. (This, of course, is done in the usual manner by considering a free-body diagram of the *entire truss*.)

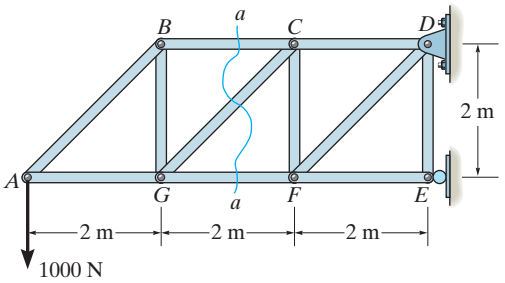



Fig. 6–15

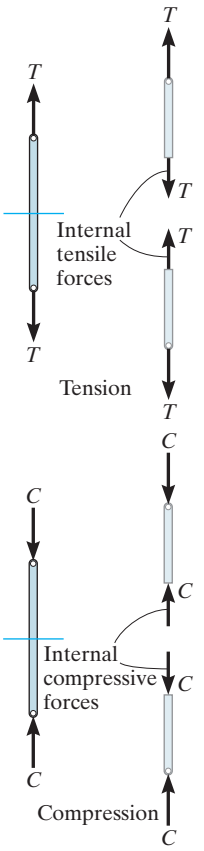



Fig. 6–14



The forces in selected members of this Pratt truss can readily be determined using the method of sections. (© Russell C. Hibbeler)

When applying the equilibrium equations, we should carefully consider ways of writing the equations so as to yield a *direct solution* for each of the unknowns, rather than having to solve simultaneous equations. For example, using the truss segment in Fig. 6–15*b* and summing moments about $C$ would yield a direct solution for $\mathbf{F}_{GF}$ since $\mathbf{F}_{BC}$ and $\mathbf{F}_{GC}$ create zero moment about $C$. Likewise, $\mathbf{F}_{BC}$ can be directly obtained by summing moments about $G$. Finally, $\mathbf{F}_{GC}$ can be found directly from a force summation in the vertical direction since $\mathbf{F}_{GF}$ and $\mathbf{F}_{BC}$ have no vertical components. This ability to *determine directly* the force in a particular truss member is one of the main advantages of using the method of sections.*

As in the method of joints, there are two ways in which we can determine the correct sense of an unknown member force:

- The correct sense of an unknown member force can in many cases be determined "by inspection." For example, $\mathbf{F}_{BC}$ is a tensile force as represented in Fig. 6–15*b* since moment equilibrium about $G$ requires that $\mathbf{F}_{BC}$ create a moment opposite to that of the 1000-N force. Also, $\mathbf{F}_{GC}$ is tensile since its vertical component must balance the 1000-N force which acts downward. In more complicated cases, the sense of an unknown member force may be *assumed.* If the solution yields a *negative* scalar, it indicates that the force's sense is *opposite* to that shown on the free-body diagram.
- *Always assume* that the unknown member forces at the cut section are *tensile* forces, i.e., "pulling" on the member. By doing this, the numerical solution of the equilibrium equations will yield *positive scalars for members in tension and negative scalars for members in compression.*

*Notice that if the method of joints were used to determine, say, the force in member $GC$, it would be necessary to analyze joints $A$, $B$, and $G$ in sequence.

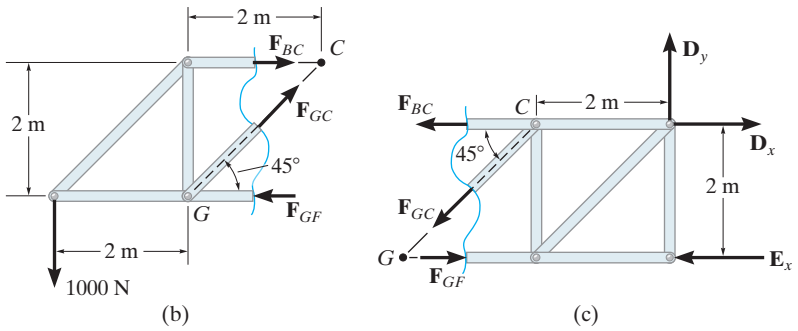



**Fig. 6–15 (cont.)**

## Important Point

- If a truss is in equilibrium, then each of its segments is in equilibrium. The internal forces in the members become external forces when the free-body diagram of a segment of the truss is drawn. A force pulling on a member causes tension in the member, and a force pushing on a member causes compression.



Simple trusses are often used in the construction of large cranes in order to reduce the weight of the boom and tower. (© Russell C. Hibbeler)

## Procedure for Analysis

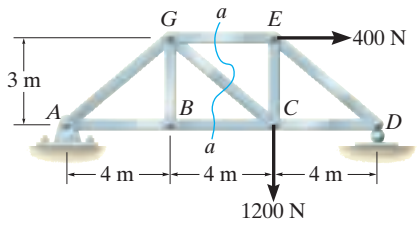
The forces in the members of a truss may be determined by the method of sections using the following procedure.

### Free-Body Diagram.

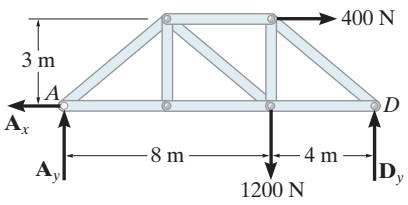
- Make a decision on how to "cut" or section the truss through the members where forces are to be determined.
- Before isolating the appropriate section, it may first be necessary to determine the truss's support reactions. If this is done then the three equilibrium equations will be available to solve for member forces at the section.
- Draw the free-body diagram of that segment of the sectioned truss which has the least number of forces acting on it.
- Use one of the two methods described above for establishing the sense of the unknown member forces.

### Equations of Equilibrium.

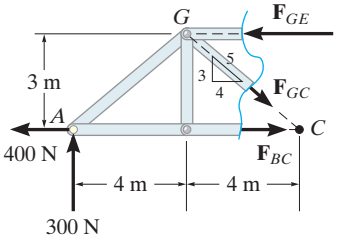
- Moments should be summed about a point that lies at the intersection of the lines of action of two unknown forces, so that the third unknown force can be determined directly from the moment equation.
- If two of the unknown forces are *parallel*, forces may be summed *perpendicular* to the direction of these unknowns to determine *directly* the third unknown force.




(a)




(b)




(c)

**Fig. 6–16**

Determine the force in members $GE$, $GC$, and $BC$ of the truss shown in Fig. 6–16*a*. Indicate whether the members are in tension or compression.

## SOLUTION

Section *aa* in Fig. 6–16*a* has been chosen since it cuts through the *three* members whose forces are to be determined. In order to use the method of sections, however, it is *first* necessary to determine the external reactions at $A$ or $D$. Why? A free-body diagram of the entire truss is shown in Fig. 6–16*b*. Applying the equations of equilibrium, we have

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \qquad 400\ \text{N} - A_x = 0 \qquad A_x = 400\ \text{N}$$

$$\circlearrowleft +\Sigma M_A = 0; \qquad -1200\ \text{N}(8\ \text{m}) - 400\ \text{N}(3\ \text{m}) + D_y(12\ \text{m}) = 0$$

$$D_y = 900\ \text{N}$$

$$+\uparrow\Sigma F_y = 0; \qquad A_y - 1200\ \text{N} + 900\ \text{N} = 0 \qquad A_y = 300\ \text{N}$$

**Free-Body Diagram.** For the analysis the free-body diagram of the left portion of the sectioned truss will be used, since it involves the least number of forces, Fig. 6–16*c*.

**Equations of Equilibrium.** Summing moments about point $G$ eliminates $\mathbf{F}_{GE}$ and $\mathbf{F}_{GC}$ and yields a direct solution for $F_{BC}$.

$$\circlearrowleft +\Sigma M_G = 0; \quad -300\ \text{N}(4\ \text{m}) - 400\ \text{N}(3\ \text{m}) + F_{BC}(3\ \text{m}) = 0$$

$$F_{BC} = 800\ \text{N} \quad (\text{T}) \qquad \textit{Ans.}$$

In the same manner, by summing moments about point $C$ we obtain a direct solution for $F_{GE}$.

$$\circlearrowleft +\Sigma M_C = 0; \quad -300\ \text{N}(8\ \text{m}) + F_{GE}(3\ \text{m}) = 0$$

$$F_{GE} = 800\ \text{N} \quad (\text{C}) \qquad \textit{Ans.}$$

Since $\mathbf{F}_{BC}$ and $\mathbf{F}_{GE}$ have no vertical components, summing forces in the $y$ direction directly yields $F_{GC}$, i.e.,

$$+\uparrow\Sigma F_y = 0; \qquad 300\ \text{N} - \tfrac{3}{5}F_{GC} = 0$$

$$F_{GC} = 500\ \text{N} \quad (\text{T}) \qquad \textit{Ans.}$$

**NOTE:** Here it is possible to tell, by inspection, the proper direction for each unknown member force. For example, $\Sigma M_C = 0$ requires $\mathbf{F}_{GE}$ to be *compressive* because it must balance the moment of the 300-N force about $C$.

## EXAMPLE 6.6

Determine the force in member *CF* of the truss shown in Fig. 6–17*a*. Indicate whether the member is in tension or compression. Assume each member is pin connected.

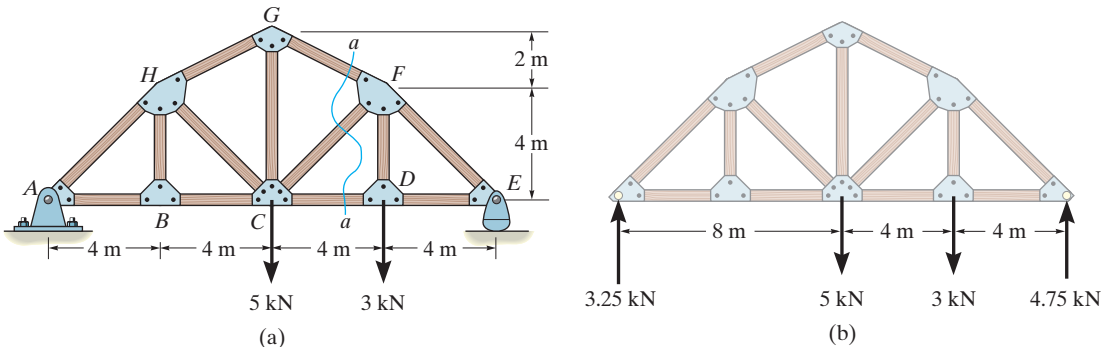


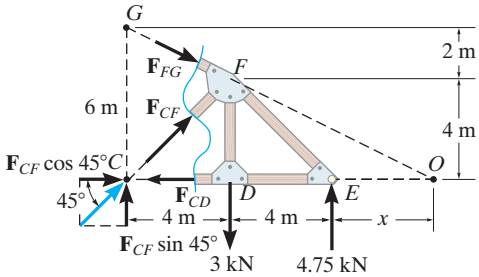

Fig. 6–17

### SOLUTION

**Free-Body Diagram.** Section *aa* in Fig. 6–17*a* will be used since this section will "expose" the internal force in member *CF* as "external" on the free-body diagram of either the right or left portion of the truss. It is first necessary, however, to determine the support reactions on either the left or right side. Verify the results shown on the free-body diagram in Fig. 6–17*b*.

The free-body diagram of the right portion of the truss, which is the easiest to analyze, is shown in Fig. 6–17*c*. There are three unknowns, $F_{FG}$, $F_{CF}$, and $F_{CD}$.

**Equations of Equilibrium.** We will apply the moment equation about point *O* in order to eliminate the two unknowns $F_{FG}$ and $F_{CD}$. The location of point *O* measured from *E* can be determined from proportional triangles, i.e., $4/(4 + x) = 6/(8 + x)$, $x = 4$ m. Or, stated in another manner, the slope of member *GF* has a drop of 2 m to a horizontal distance of 4 m. Since *FD* is 4 m, Fig. 6–17*c*, then from *D* to *O* the distance must be 8 m.



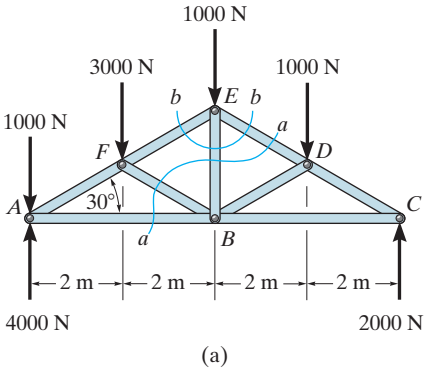

(c)

An easy way to determine the moment of $\mathbf{F}_{CF}$ about point *O* is to use the principle of transmissibility and slide $\mathbf{F}_{CF}$ to point *C*, and then resolve $\mathbf{F}_{CF}$ into its two rectangular components. We have

$$\zeta + \Sigma M_O = 0;$$

$$-F_{CF} \sin 45°(12 \text{ m}) + (3 \text{ kN})(8 \text{ m}) - (4.75 \text{ kN})(4 \text{ m}) = 0$$

$$F_{CF} = 0.589 \text{ kN} \quad \text{(C)} \qquad \textit{Ans.}$$

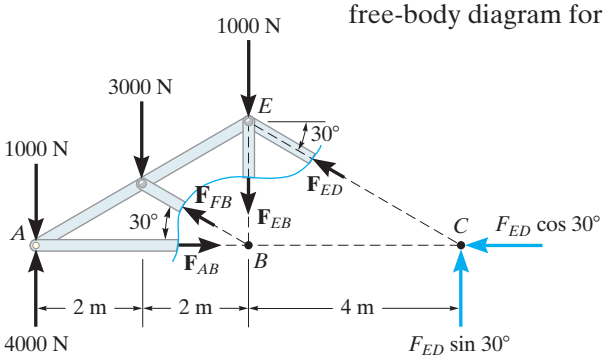



(a)

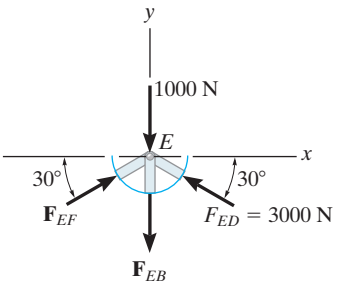
Determine the force in member *EB* of the roof truss shown in Fig. 6–18*a*. Indicate whether the member is in tension or compression.

## SOLUTION

**Free-Body Diagrams.** By the method of sections, any imaginary section that cuts through *EB*, Fig. 6–18*a*, will also have to cut through three other members for which the forces are unknown. For example, section *aa* cuts through *ED, EB, FB*, and *AB*. If a free-body diagram of the left side of this section is considered, Fig. 6–18*b*, it is possible to obtain $\mathbf{F}_{ED}$ by summing moments about *B* to eliminate the other three unknowns; however, $\mathbf{F}_{EB}$ cannot be determined from the remaining two equilibrium equations. One possible way of obtaining $\mathbf{F}_{EB}$ is first to determine $\mathbf{F}_{ED}$ from section *aa*, then use this result on section *bb*, Fig. 6–18*a*, which is shown in Fig. 6–18*c*. Here the force system is concurrent and our sectioned free-body diagram is the same as the free-body diagram for the joint at *E*.




(b)




(c)

**Fig. 6–18**

**Equations of Equilibrium.** In order to determine the moment of $\mathbf{F}_{ED}$ about point *B*, Fig. 6–18*b*, we will use the principle of transmissibility and slide the force to point *C* and then resolve it into its rectangular components as shown. Therefore,

$$\curvearrowright +\Sigma M_B = 0; \quad 1000\ \text{N}(4\ \text{m}) + 3000\ \text{N}(2\ \text{m}) - 4000\ \text{N}(4\ \text{m}) + F_{ED} \sin 30°(4\ \text{m}) = 0$$

$$F_{ED} = 3000\ \text{N} \quad (\text{C})$$

Considering now the free-body diagram of section *bb*, Fig. 6–18*c*, we have

$$\pm\rightarrow \Sigma F_x = 0; \quad F_{EF} \cos 30° - 3000 \cos 30°\ \text{N} = 0$$

$$F_{EF} = 3000\ \text{N} \quad (\text{C})$$

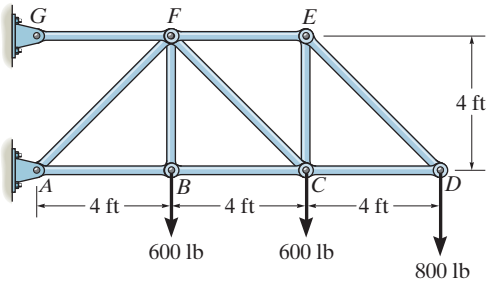
$$+\uparrow \Sigma F_y = 0; \quad 2(3000 \sin 30°\ \text{N}) - 1000\ \text{N} - F_{EB} = 0$$

$$F_{EB} = 2000\ \text{N} \quad (\text{T}) \qquad \textit{Ans.}$$

## FUNDAMENTAL PROBLEMS

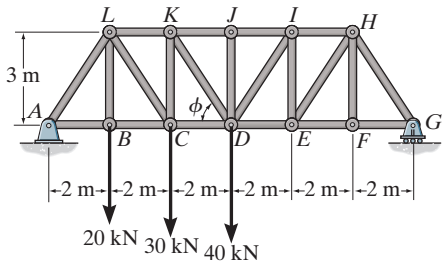
***All problem solutions must include FBDs.***

**F6–7.** Determine the force in members *BC*, *CF*, and *FE*. State if the members are in tension or compression.



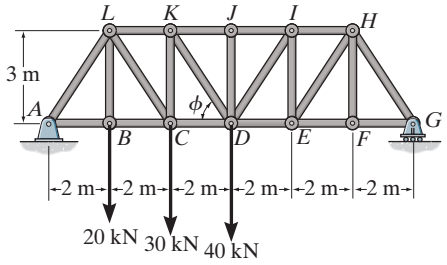

**Prob. F6–7**

**F6–8.** Determine the force in members *LK*, *KC*, and *CD* of the Pratt truss. State if the members are in tension or compression.



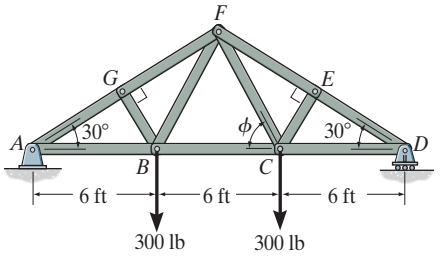

**Prob. F6–8**

**F6–9.** Determine the force in members *KJ*, *KD*, and *CD* of the Pratt truss. State if the members are in tension or compression.



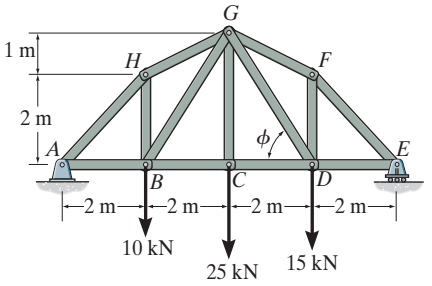

**Prob. F6–9**

**F6–10.** Determine the force in members *EF*, *CF*, and *BC* of the truss. State if the members are in tension or compression.



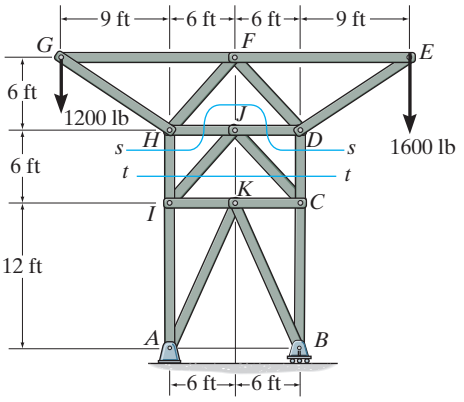

**Prob. F6–10**

**F6–11.** Determine the force in members *GF*, *GD*, and *CD* of the truss. State if the members are in tension or compression.




**Prob. F6–11**

**F6–12.** Determine the force in members *DC*, *HI*, and *JI* of the truss. State if the members are in tension or compression. *Suggestion:* Use the sections shown.



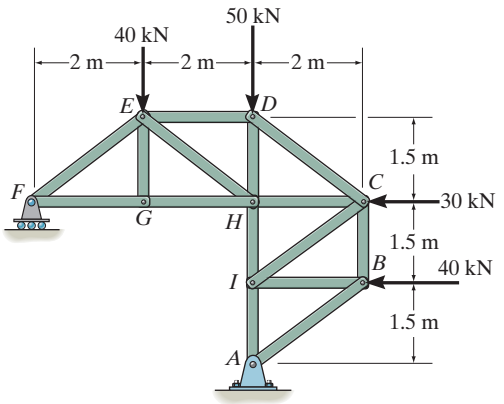

**Prob. F6–12**

## PROBLEMS

***All problem solutions must include FBDs.***

**6–27.** Determine the force in members *DC*, *HC*, and *HI* of the truss, and state if the members are in tension or compression.

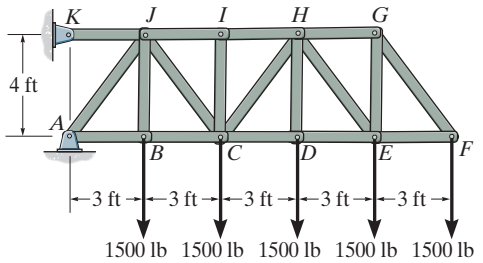
***6–28.** Determine the force in members *ED*, *EH*, and *GH* of the truss, and state if the members are in tension or compression.




**Probs. 6–27/28**

**6–29.** Determine the force in members *HG*, *HE* and *DE* of the truss, and state if the members are in tension or compression.

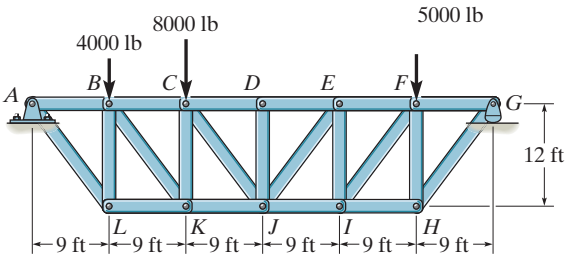
**6–30.** Determine the force in members *CD*, *HI*, and *CH* of the truss, and state if the members are in tension or compression.




**Probs. 6–29/30**

**6–31.** Determine the force in members *CD*, *CJ*, *KJ*, and *DJ* of the truss which serves to support the deck of a bridge. State if these members are in tension or compression.

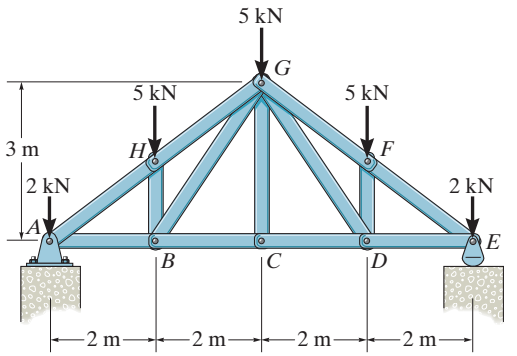
***6–32.** Determine the force in members *EI* and *JI* of the truss which serves to support the deck of a bridge. State if these members are in tension or compression.




**Probs. 6–31/32**

**6–33.** The *Howe truss* is subjected to the loading shown. Determine the force in members *GF*, *CD*, and *GC*, and state if the members are in tension or compression.

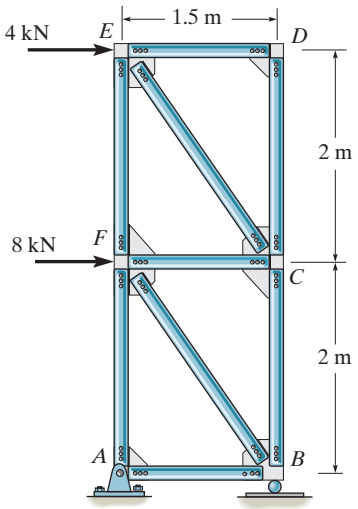
**6–34.** The *Howe truss* is subjected to the loading shown. Determine the force in members *GH*, *BC*, and *BG* of the truss and state if the members are in tension or compression.




**Probs. 6–33/34**

**6–35.** Determine the force in members *EF*, *CF*, and *BC*, and state if the members are in tension or compression.

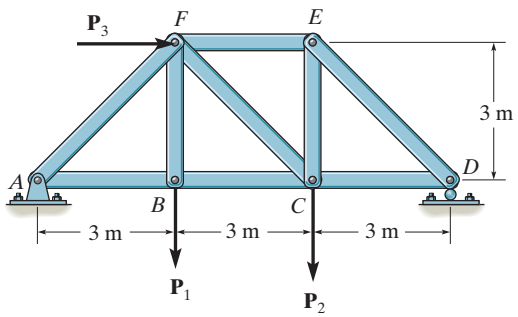
***6–36.** Determine the force in members *AF*, *BF*, and *BC*, and state if the members are in tension or compression.




**Probs. 6–35/36**

**6–37.** Determine the force in members *EF*, *BE*, *BC* and *BF* of the truss and state if these members are in tension or compression. Set $P_1 = 9$ kN, $P_2 = 12$ kN, and $P_3 = 6$ kN.

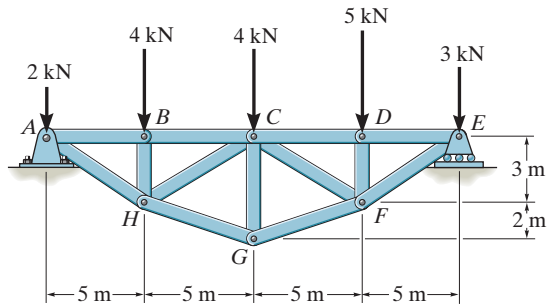
**6–38.** Determine the force in members *BC*, *BE*, and *EF* of the truss and state if these members are in tension or compression. Set $P_1 = 6$ kN, $P_2 = 9$ kN, and $P_3 = 12$ kN.




**Probs. 6–37/38**

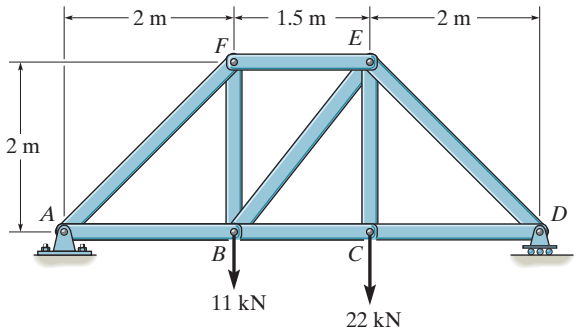
**6–39.** Determine the force in members *BC*, *HC*, and *HG*. After the truss is sectioned use a single equation of equilibrium for the calculation of each force. State if these members are in tension or compression.

***6–40.** Determine the force in members *CD*, *CF*, and *CG* and state if these members are in tension or compression.




**Probs. 6–39/40**

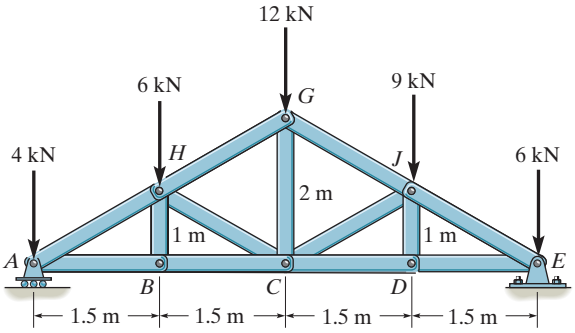
**6–41.** Determine the force developed in members *FE*, *EB*, and *BC* of the truss and state if these members are in tension or compression.




**Prob. 6–41**

**6–42.** Determine the force in members *BC*, *HC*, and *HG*. State if these members are in tension or compression.

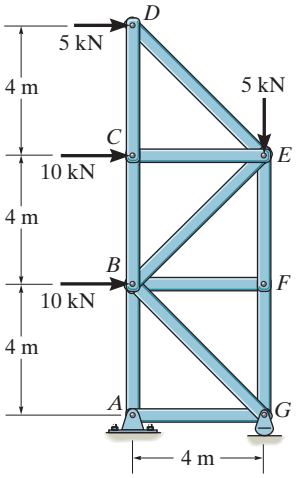
**6–43.** Determine the force in members *CD*, *CJ*, *GJ*, and *CG* and state if these members are in tension or compression.




**Probs. 6–42/43**

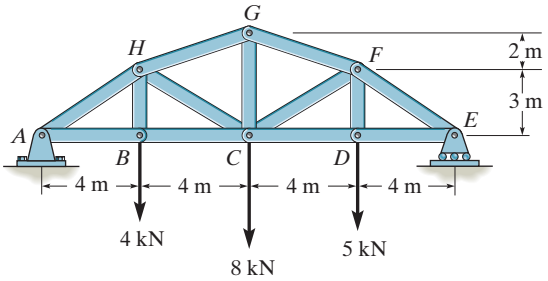
*__6–44.__ Determine the force in members *BE*, *EF*, and *CB*, and state if the members are in tension or compression.

**6–45.** Determine the force in members *BF*, *BG*, and *AB*, and state if the members are in tension or compression.



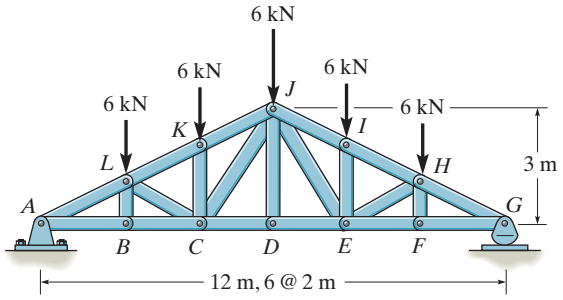

**Probs. 6–44/45**

**6–46.** Determine the force in members *BC*, *CH*, *GH*, and *CG* of the truss and state if the members are in tension or compression.




**Prob. 6–46**

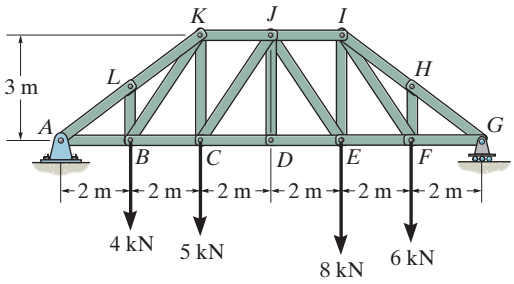
**6–47.** Determine the force in members *CD*, *CJ*, and *KJ* and state if these members are in tension or compression.




**Prob. 6–47**

*__6–48.__ Determine the force in members *JK*, *CJ*, and *CD* of the truss, and state if the members are in tension or compression.

**6–49.** Determine the force in members *HI*, *FI*, and *EF* of the truss, and state if the members are in tension or compression.




**Probs. 6–48/49**

## *6.5 Space Trusses

A ***space truss*** consists of members joined together at their ends to form a stable three-dimensional structure. The simplest form of a space truss is a ***tetrahedron***, formed by connecting six members together, as shown in Fig. 6–19. Any additional members added to this basic element would be redundant in supporting the force **P**. A *simple space truss* can be built from this basic tetrahedral element by adding three additional members and a joint, and continuing in this manner to form a system of multiconnected tetrahedrons.

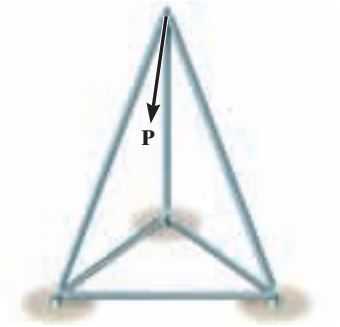



Fig. 6–19

**Assumptions for Design.** The members of a space truss may be treated as two-force members provided the external loading is applied at the joints and the joints consist of ball-and-socket connections. These assumptions are justified if the welded or bolted connections of the joined members intersect at a common point and the weight of the members can be neglected. In cases where the weight of a member is to be included in the analysis, it is generally satisfactory to apply it as a vertical force, half of its magnitude applied at each end of the member.



Typical roof-supporting space truss. Notice the use of ball-and-socket joints for the connections. (© Russell C. Hibbeler)

### Procedure for Analysis

Either the method of joints or the method of sections can be used to determine the forces developed in the members of a simple space truss.

#### Method of Joints.

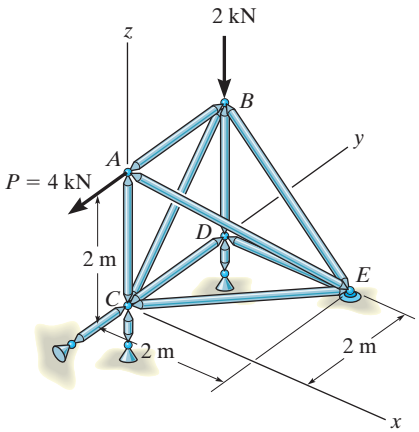
If the forces in *all* the members of the truss are to be determined, then the method of joints is most suitable for the analysis. Here it is necessary to apply the three equilibrium equations $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma F_z = 0$ to the forces acting at each joint. Remember that the solution of many simultaneous equations can be avoided if the force analysis begins at a joint having at least one known force and at most three unknown forces. Also, if the three-dimensional geometry of the force system at the joint is hard to visualize, it is recommended that a Cartesian vector analysis be used for the solution.

#### Method of Sections.

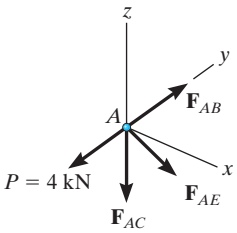
If only a *few* member forces are to be determined, the method of sections can be used. When an imaginary section is passed through a truss and the truss is separated into two parts, the force system acting on one of the segments must satisfy the *six* equilibrium equations: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma F_z = 0$, $\Sigma M_x = 0$, $\Sigma M_y = 0$, $\Sigma M_z = 0$ (Eqs. 5–6). By proper choice of the section and axes for summing forces and moments, many of the unknown member forces in a space truss can be computed *directly*, using a single equilibrium equation.



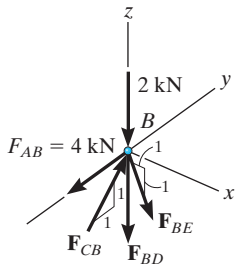
For economic reasons, large electrical transmission towers are often constructed using space trusses. (© Russell C. Hibbeler)




(a)




(b)




(c)

**Fig. 6–20**

Determine the forces acting in the members of the space truss shown in Fig. 6–20*a*. Indicate whether the members are in tension or compression.

## SOLUTION

Since there are one known force and three unknown forces acting at joint $A$, the force analysis of the truss will begin at this joint.

**Joint A.** (Fig. 6–20*b*). Expressing each force acting on the free-body diagram of joint $A$ as a Cartesian vector, we have

$$\mathbf{P} = \{-4\mathbf{j}\}\ \text{kN}, \qquad \mathbf{F}_{AB} = F_{AB}\mathbf{j}, \quad \mathbf{F}_{AC} = -F_{AC}\mathbf{k},$$

$$\mathbf{F}_{AE} = F_{AE}\left(\frac{\mathbf{r}_{AE}}{r_{AE}}\right) = F_{AE}(0.577\mathbf{i} + 0.577\mathbf{j} - 0.577\mathbf{k})$$

For equilibrium,

$$\Sigma\mathbf{F} = \mathbf{0}; \qquad \mathbf{P} + \mathbf{F}_{AB} + \mathbf{F}_{AC} + \mathbf{F}_{AE} = \mathbf{0}$$

$$-4\mathbf{j} + F_{AB}\mathbf{j} - F_{AC}\mathbf{k} + 0.577F_{AE}\mathbf{i} + 0.577F_{AE}\mathbf{j} - 0.577F_{AE}\mathbf{k} = \mathbf{0}$$

$$\Sigma F_x = 0; \qquad 0.577F_{AE} = 0$$

$$\Sigma F_y = 0; \qquad -4 + F_{AB} + 0.577F_{AE} = 0$$

$$\Sigma F_z = 0; \qquad -F_{AC} - 0.577F_{AE} = 0$$

$$F_{AC} = F_{AE} = 0 \qquad \textit{Ans.}$$

$$F_{AB} = 4\ \text{kN} \quad (\text{T}) \qquad \textit{Ans.}$$

Since $F_{AB}$ is known, joint $B$ can be analyzed next.

**Joint B.** (Fig. 6–20*c*).

$$\Sigma F_x = 0; \qquad F_{BE}\frac{1}{\sqrt{2}} = 0$$

$$\Sigma F_y = 0; \qquad -4 + F_{CB}\frac{1}{\sqrt{2}} = 0$$

$$\Sigma F_z = 0; \qquad -2 + F_{BD} - F_{BE}\frac{1}{\sqrt{2}} + F_{CB}\frac{1}{\sqrt{2}} = 0$$

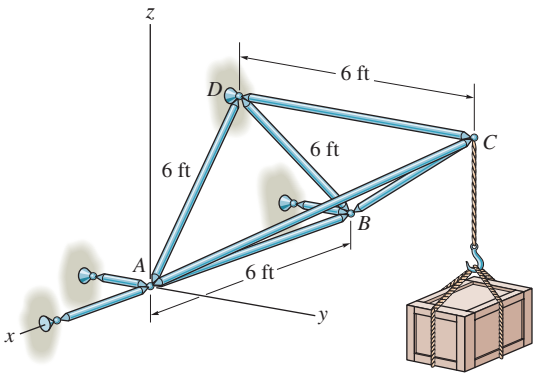
$$F_{BE} = 0, \qquad F_{CB} = 5.65\ \text{kN (C)} \qquad F_{BD} = 2\ \text{kN (T)} \qquad \textit{Ans.}$$

The *scalar* equations of equilibrium can now be applied to the forces acting on the free-body diagrams of joints $D$ and $C$. Show that

$$F_{DE} = F_{DC} = F_{CE} = 0 \qquad \textit{Ans.}$$

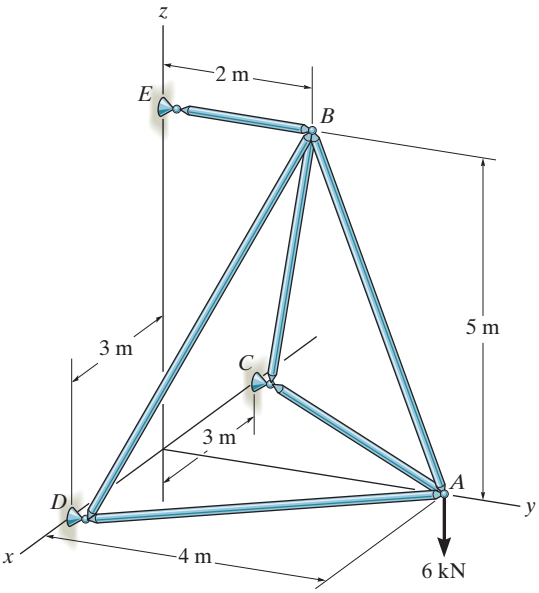
***All problem solutions must include FBDs.***

**6–50.** Determine the force developed in each member of the space truss and state if the members are in tension or compression. The crate has a weight of 150 lb.



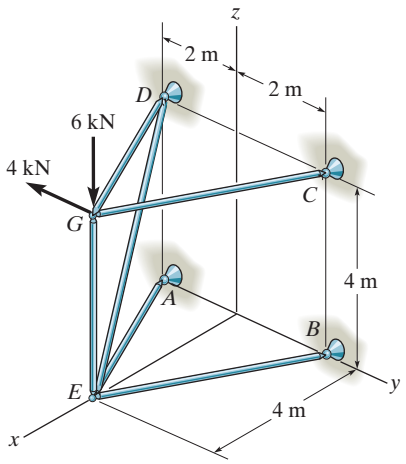

**Prob. 6–50**

**6–51.** Determine the force in each member of the space truss and state if the members are in tension or compression. *Hint*: The support reaction at $E$ acts along member $EB$. Why?




**Prob. 6–51**

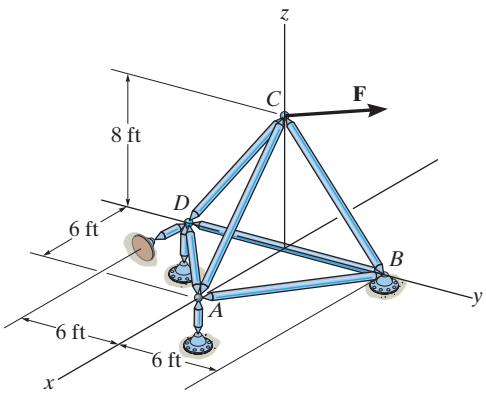
***6–52.** Determine the force in each member of the space truss and state if the members are in tension or compression. The truss is supported by ball-and-socket joints at $A$, $B$, $C$, and $D$.




**Prob. 6–52**

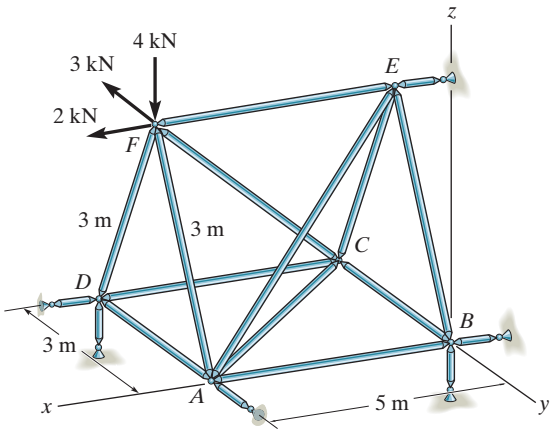
**6–53.** The space truss supports a force $\mathbf{F} = \{-500\mathbf{i} + 600\mathbf{j} + 400\mathbf{k}\}$ lb. Determine the force in each member, and state if the members are in tension or compression.

**6–54.** The space truss supports a force $\mathbf{F} = \{600\mathbf{i} + 450\mathbf{j} - 750\mathbf{k}\}$ lb. Determine the force in each member, and state if the members are in tension or compression.



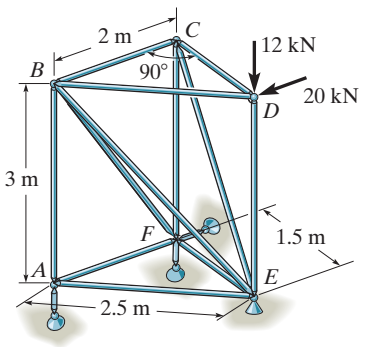

**Probs. 6–53/54**

**6–55.** Determine the force in members $EF$, $AF$, and $DF$ of the space truss and state if the members are in tension or compression. The truss is supported by short links at $A$, $B$, $D$, and $E$.




Prob. 6–55

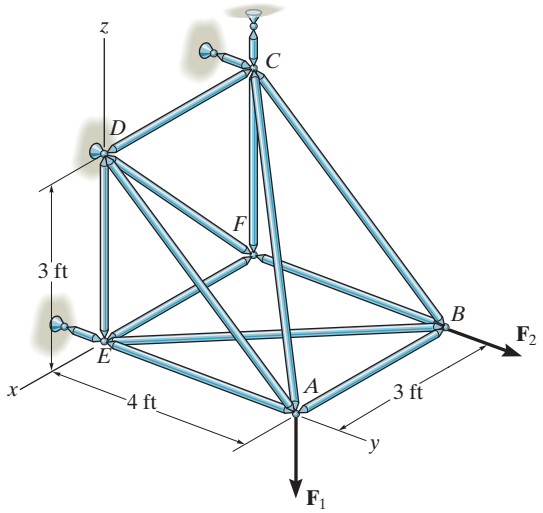
*6–56. The space truss is used to support the forces at joints $B$ and $D$. Determine the force in each member and state if the members are in tension or compression.




Prob. 6–56

**6–57.** The space truss is supported by a ball-and-socket joint at $D$ and short links at $C$ and $E$. Determine the force in each member and state if the members are in tension or compression. Take $\mathbf{F}_1 = \{-500\mathbf{k}\}$ lb and $\mathbf{F}_2 = \{400\mathbf{j}\}$ lb.

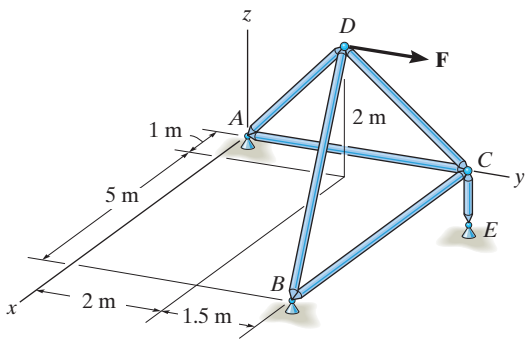
**6–58.** The space truss is supported by a ball-and-socket joint at $D$ and short links at $C$ and $E$. Determine the force in each member and state if the members are in tension or compression. Take $\mathbf{F}_1 = \{200\mathbf{i} + 300\mathbf{j} - 500\mathbf{k}\}$ lb and $\mathbf{F}_2 = \{400\mathbf{j}\}$ lb.




Probs. 6–57/58

**6–59.** Determine the force in each member of the space truss and state if the members are in tension or compression. The truss is supported by ball-and-socket joints at $A$, $B$, and $E$. Set $\mathbf{F} = \{800\mathbf{j}\}$ N. *Hint*: The support reaction at $E$ acts along member $EC$. Why?

*6–60. Determine the force in each member of the space truss and state if the members are in tension or compression. The truss is supported by ball-and-socket joints at $A$, $B$, and $E$. Set $\mathbf{F} = \{-200\mathbf{i} + 400\mathbf{j}\}$ N. *Hint*: The support reaction at $E$ acts along member $EC$. Why?




Probs. 6–59/60

## 6.6 Frames and Machines

Frames and machines are two types of structures which are often composed of pin-connected ***multiforce members***, i.e., members that are subjected to more than two forces. ***Frames*** are used to support loads, whereas ***machines*** contain moving parts and are designed to transmit and alter the effect of forces. Provided a frame or machine contains no more supports or members than are necessary to prevent its collapse, the forces acting at the joints and supports can be determined by applying the equations of equilibrium to each of its members. Once these forces are obtained, it is then possible to *design* the size of the members, connections, and supports using the theory of mechanics of materials and an appropriate engineering design code.



This crane is a typical example of a framework. (© Russell C. Hibbeler)

**Free-Body Diagrams.** In order to determine the forces acting at the joints and supports of a frame or machine, the structure must be disassembled and the free-body diagrams of its parts must be drawn. The following important points *must* be observed:

- Isolate each part by drawing its *outlined shape*. Then show all the forces and/or couple moments that act on the part. Make sure to *label* or *identify* each known and unknown force and couple moment with reference to an established *x, y* coordinate system. Also, indicate any dimensions used for taking moments. Most often the equations of equilibrium are easier to apply if the forces are represented by their rectangular components. As usual, the sense of an unknown force or couple moment can be assumed.
- Identify all the two-force members in the structure and represent their free-body diagrams as having two equal but opposite collinear forces acting at their points of application. (See Sec. 5.4.) By recognizing the two-force members, we can avoid solving an unnecessary number of equilibrium equations.
- Forces common to *any* two *contacting* members act with equal magnitudes but opposite sense on the respective members. If the two members are treated as a *"system" of connected members*, then these forces are *"internal"* and are *not shown* on the *free-body diagram of the system*; however, if the free-body diagram of *each member* is drawn, the forces are *"external"* and *must* be shown as equal in magnitude and opposite in direction on each of the two free-body diagrams.

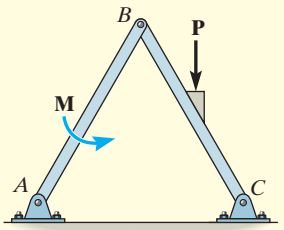


Common tools such as these pliers act as simple machines. Here the applied force on the handles creates a much larger force at the jaws. (© Russell C. Hibbeler)

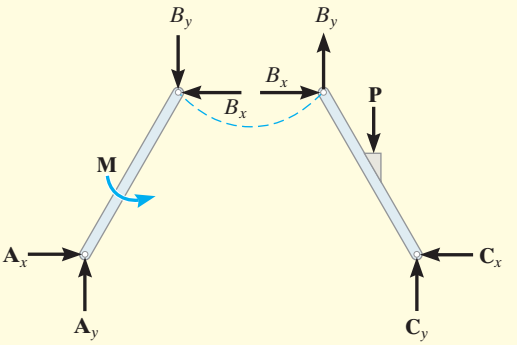
The following examples graphically illustrate how to draw the free-body diagrams of a dismembered frame or machine. In all cases, the weight of the members is neglected.

## EXAMPLE 6.9

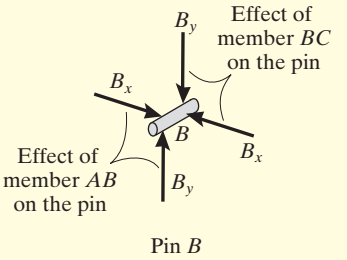
For the frame shown in Fig. 6–21*a*, draw the free-body diagram of (a) each member, (b) the pins at *B* and *A*, and (c) the two members connected together.




(a)

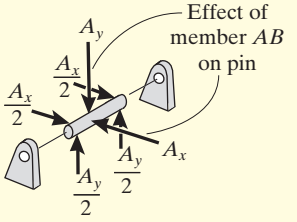



(b)



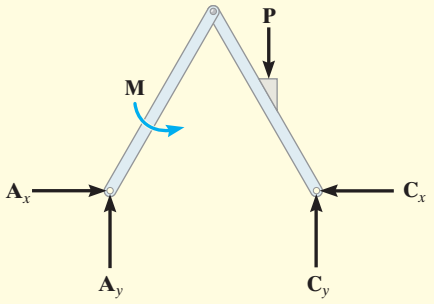

Pin *B*

(c)




Pin *A*

(d)




(e)

**Fig. 6–21**

### SOLUTION

**Part (a).** By inspection, members *BA* and *BC* are *not* two-force members. Instead, as shown on the free-body diagrams, Fig. 6–21*b*, *BC* is subjected to a force from each of the pins at *B* and *C* and the external force **P**. Likewise, *AB* is subjected to a force from each of the pins at *A* and *B* and the external couple moment **M**. The pin forces are represented by their *x* and *y* components.

**Part (b).** The pin at *B* is subjected to only *two forces*, i.e., the force of member *BC* and the force of member *AB*. For *equilibrium* these forces (or their respective components) must be equal but opposite, Fig. 6–21*c*. Realize that Newton's third law is applied between the pin and its connected members, i.e., the effect of the pin on the two members, Fig. 6–21*b*, and the equal but opposite effect of the two members on the pin, Fig. 6–21*c*. In the same manner, there are three forces on pin *A*, Fig. 6–21*d*, caused by the force components of member *AB* and each of the two pin leafs.

**Part (c).** The free-body diagram of both members connected together, yet removed from the supporting pins at *A* and *C*, is shown in Fig. 6–21*e*. The force components $\mathbf{B}_x$ and $\mathbf{B}_y$ are *not shown* on this diagram since they are *internal* forces (Fig. 6–21*b*) and therefore cancel out. Also, to be consistent when later applying the equilibrium equations, the unknown force components at *A* and *C* must act in the *same sense* as those shown in Fig. 6–21*b*.

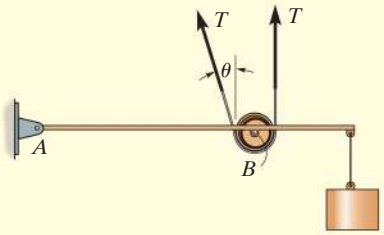
## EXAMPLE 6.10

A constant tension in the conveyor belt is maintained by using the device shown in Fig. 6–22*a*. Draw the free-body diagrams of the frame and the cylinder (or pulley) that the belt surrounds. The suspended block has a weight of $W$.

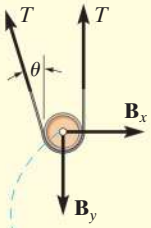


(a)

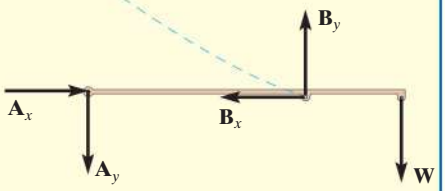
**Fig. 6–22** (© Russell C. Hibbeler)




(b)




(c)



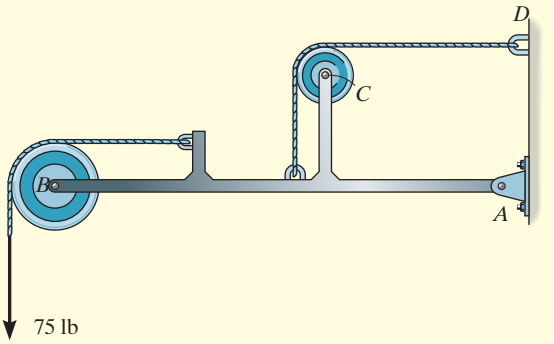

(d)

### SOLUTION

The idealized model of the device is shown in Fig. 6–22*b*. Here the angle $\theta$ is assumed to be known. From this model, the free-body diagrams of the pulley and frame are shown in Figs. 6–22*c* and 6–22*d*, respectively. Note that the force components $\mathbf{B}_x$ and $\mathbf{B}_y$ that the pin at $B$ exerts on the pulley must be equal but opposite to the ones acting on the frame. See Fig. 6–21*c* of Example 6.9.

## EXAMPLE 6.11

For the frame shown in Fig. 6–23*a*, draw the free-body diagrams of (a) the entire frame including the pulleys and cords, (b) the frame without the pulleys and cords, and (c) each of the pulleys.



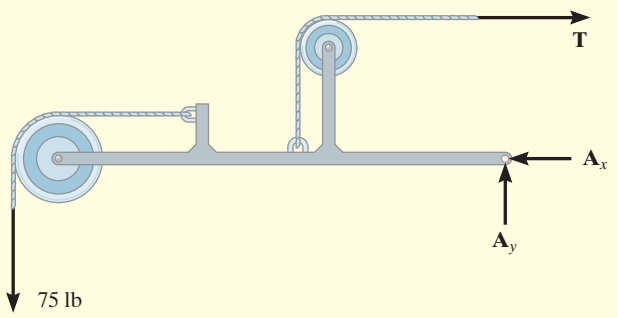

(a)

### SOLUTION

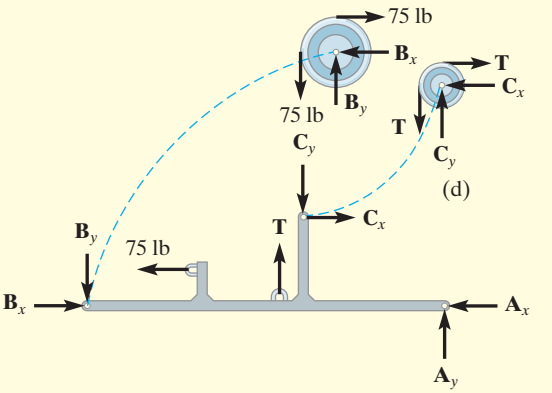
**Part (a).** When the entire frame including the pulleys and cords is considered, the interactions at the points where the pulleys and cords are connected to the frame become pairs of *internal* forces which cancel each other and therefore are not shown on the free-body diagram, Fig. 6–23*b*.

**Part (b).** When the cords and pulleys are removed, their effect *on the frame* must be shown, Fig. 6–23*c*.

**Part (c).** The force components $\mathbf{B}_x$, $\mathbf{B}_y$, $\mathbf{C}_x$, $\mathbf{C}_y$ of the pins on the pulleys, Fig. 6–23*d*, are equal but opposite to the force components exerted by the pins on the frame, Fig. 6–23*c*. See Example 6.9.




(b)




(c)

**Fig. 6–23**

## EXAMPLE 6.12

Draw the free-body diagrams of the members of the backhoe, shown in the photo, Fig. 6–24*a*. The bucket and its contents have a weight $W$.

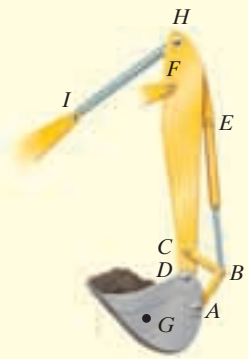


(a)

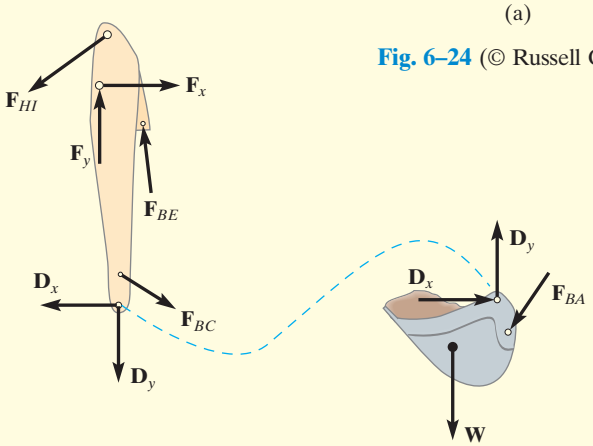
**Fig. 6–24** (© Russell C. Hibbeler)

### SOLUTION

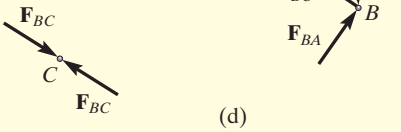
The idealized model of the assembly is shown in Fig. 6–24*b*. By inspection, members $AB$, $BC$, $BE$, and $HI$ are all two-force members since they are pin connected at their end points and no other forces act on them. The free-body diagrams of the bucket and the stick are shown in Fig. 6–24*c*. Note that pin $C$ is subjected to only two forces, whereas the pin at $B$ is subjected to three forces, Fig. 6–24*d*. The free-body diagram of the entire assembly is shown in Fig. 6–24*e*.



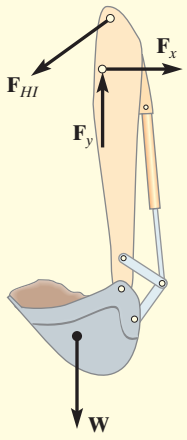

(b)




(c)




(d)




(e)

## EXAMPLE 6.13

Draw the free-body diagram of each part of the smooth piston and link mechanism used to crush recycled cans, Fig. 6–25*a*.

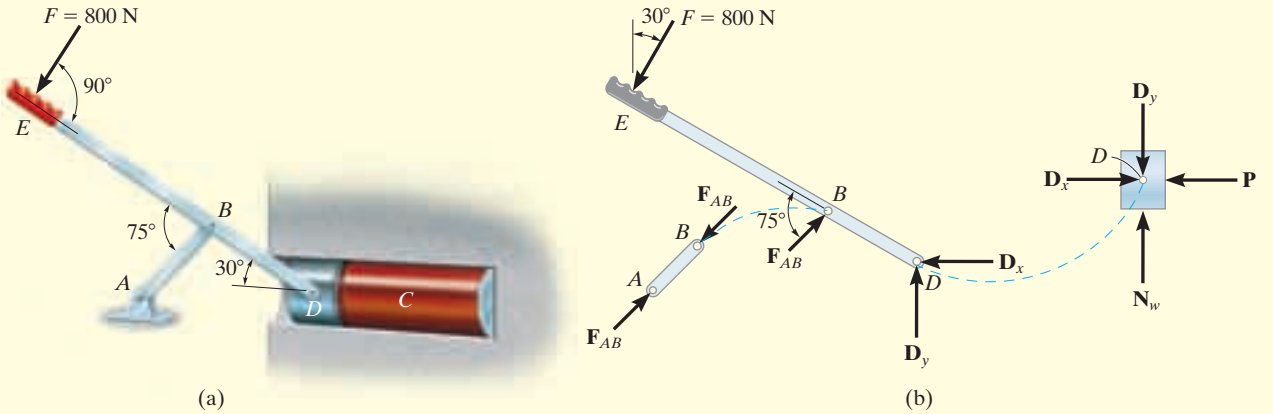
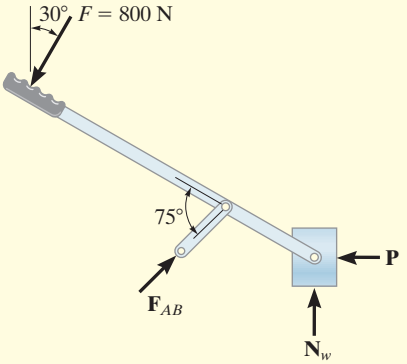



Fig. 6–25

### SOLUTION

By inspection, member *AB* is a two-force member. The free-body diagrams of the three parts are shown in Fig. 6–25*b*. Since the pins at *B* and *D* *connect only two parts together*, the forces there are shown as equal but opposite on the separate free-body diagrams of their connected members. In particular, four components of force act on the piston: $\mathbf{D}_x$ and $\mathbf{D}_y$ represent the effect of the pin (or lever *EBD*), $\mathbf{N}_w$ is the *resultant force* of the wall support, and **P** is the resultant compressive force caused by the can *C*. The directional sense of each of the unknown forces is assumed, and the correct sense will be established after the equations of equilibrium are applied.




**NOTE:** A free-body diagram of the entire assembly is shown in Fig. 6–25*c*. Here the forces between the components are internal and are not shown on the free-body diagram.

*Before proceeding, it is highly recommended that you cover the solutions of these examples and attempt to draw the requested free-body diagrams. When doing so, make sure the work is neat and that all the forces and couple moments are properly labeled.*

## Procedure for Analysis

The joint reactions on frames or machines (structures) composed of multiforce members can be determined using the following procedure.

### Free-Body Diagram.

- Draw the free-body diagram of the entire frame or machine, a portion of it, or each of its members. The choice should be made so that it leads to the most direct solution of the problem.
- Identify the two-force members. Remember that regardless of their shape, they have equal but opposite collinear forces acting at their ends.
- When the free-body diagram of a group of members of a frame or machine is drawn, the forces between the connected parts of this group are internal forces and are not shown on the free-body diagram of the group.
- Forces common to two members which are in contact act with equal magnitude but opposite sense on the respective free-body diagrams of the members.
- In many cases it is possible to tell by inspection the proper sense of the unknown forces acting on a member; however, if this seems difficult, the sense can be assumed.
- Remember that once the free-body diagram is drawn, a couple moment is a free vector and can act at any point on the diagram. Also, a force is a sliding vector and can act at any point along its line of action.

### Equations of Equilibrium.

- Count the number of unknowns and compare it to the total number of equilibrium equations that are available. In two dimensions, there are three equilibrium equations that can be written for each member.
- Sum moments about a point that lies at the intersection of the lines of action of as many of the unknown forces as possible.
- If the solution of a force or couple moment magnitude is found to be negative, it means the sense of the force is the reverse of that shown on the free-body diagram.

## EXAMPLE 6.14

Determine the tension in the cables and also the force **P** required to support the 600-N force using the frictionless pulley system shown in Fig. 6–26*a*.

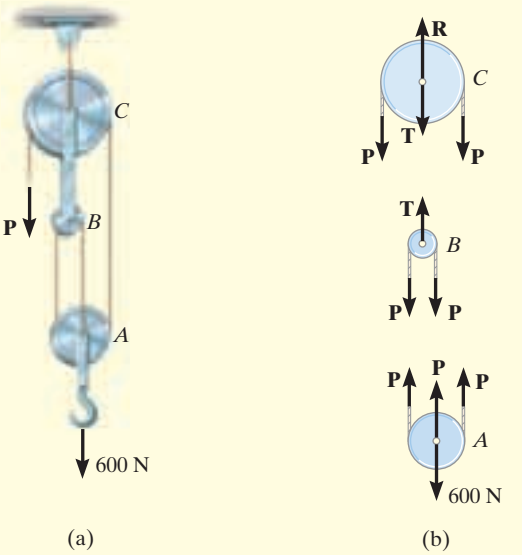



Fig. 6–26

### SOLUTION

**Free-Body Diagram.** A free-body diagram of each pulley *including* its pin and a portion of the contacting cable is shown in Fig. 6–26*b*. Since the cable is *continuous*, it has a *constant tension* $P$ acting throughout its length. The link connection between pulleys $B$ and $C$ is a two-force member, and therefore it has an unknown tension $T$ acting on it. Notice that the *principle of action, equal but opposite reaction* must be carefully observed for forces **P** and **T** when the *separate* free-body diagrams are drawn.

**Equations of Equilibrium.** The three unknowns are obtained as follows:

*Pulley A*

$$+\uparrow \Sigma F_y = 0; \qquad 3P - 600\ \text{N} = 0 \qquad P = 200\ \text{N} \qquad \textit{Ans.}$$

*Pulley B*

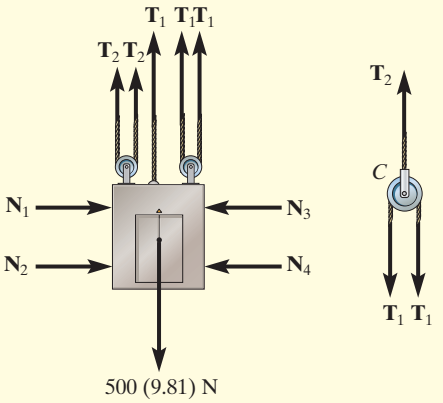
$$+\uparrow \Sigma F_y = 0; \qquad T - 2P = 0 \qquad T = 400\ \text{N} \qquad \textit{Ans.}$$

*Pulley C*

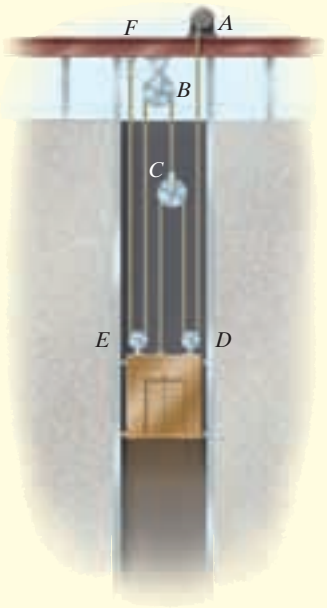
$$+\uparrow \Sigma F_y = 0; \qquad R - 2P - T = 0 \qquad R = 800\ \text{N} \qquad \textit{Ans.}$$

## EXAMPLE 6.15

A 500-kg elevator car in Fig. 6–27*a* is being hoisted by motor *A* using the pulley system shown. If the car is traveling with a constant speed, determine the force developed in the two cables. Neglect the mass of the cable and pulleys.




(b)




(a)

**Fig. 6–27**

### SOLUTION

**Free-Body Diagram.** We can solve this problem using the free-body diagrams of the elevator car and pulley *C*, Fig. 6–27*b*. The tensile forces developed in the cables are denoted as $T_1$ and $T_2$.

**Equations of Equilibrium.** For pulley *C*,

$$+\uparrow \Sigma F_y = 0; \qquad T_2 - 2T_1 = 0 \qquad \text{or} \qquad T_2 = 2T_1 \tag{1}$$

For the elevator car,

$$+\uparrow \Sigma F_y = 0; \qquad 3T_1 + 2T_2 - 500(9.81)\,\text{N} = 0 \tag{2}$$

Substituting Eq. (1) into Eq. (2) yields

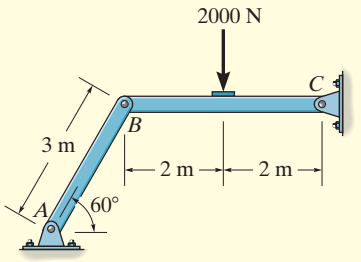
$$3T_1 + 2(2T_1) - 500(9.81)\,\text{N} = 0$$

$$T_1 = 700.71\,\text{N} = 701\,\text{N} \qquad \textit{Ans.}$$

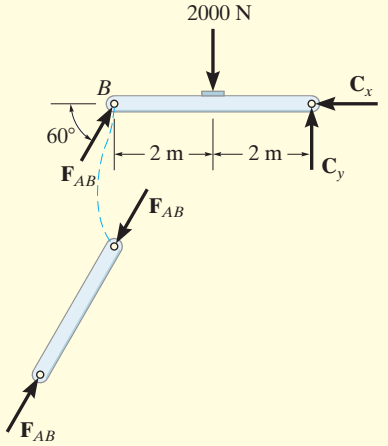
Substituting this result into Eq. (1),

$$T_2 = 2(700.71)\,\text{N} = 1401\,\text{N} = 1.40\,\text{kN} \qquad \textit{Ans.}$$

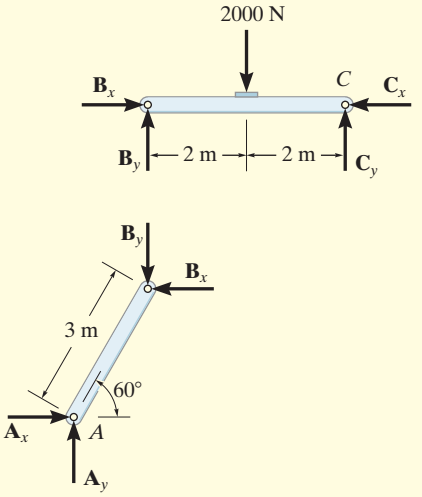
## EXAMPLE 6.16




(a)




(b)




(c)

**Fig. 6–28**

Determine the horizontal and vertical components of force which the pin at $C$ exerts on member $BC$ of the frame in Fig. 6–28*a*.

### SOLUTION I

**Free-Body Diagrams.** By inspection it can be seen that $AB$ is a two-force member. The free-body diagrams are shown in Fig. 6–28*b*.

**Equations of Equilibrium.** The *three unknowns* can be determined by applying the three equations of equilibrium to member $BC$.

$$\circlearrowleft + \Sigma M_C = 0;\ 2000\ \text{N}(2\ \text{m}) - (F_{AB} \sin 60°)(4\ \text{m}) = 0 \quad F_{AB} = 1154.7\ \text{N}$$

$$\pm \Sigma F_x = 0;\ 1154.7 \cos 60°\ \text{N} - C_x = 0 \quad C_x = 577\ \text{N} \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0;\ 1154.7 \sin 60°\ \text{N} - 2000\ \text{N} + C_y = 0$$

$$C_y = 1000\ \text{N} \qquad \textit{Ans.}$$

### SOLUTION II

**Free-Body Diagrams.** If one does not recognize that $AB$ is a two-force member, then more work is involved in solving this problem. The free-body diagrams are shown in Fig. 6–28*c*.

**Equations of Equilibrium.** The *six unknowns* are determined by applying the three equations of equilibrium to each member.

*Member AB*

$$\circlearrowleft + \Sigma M_A = 0; \quad B_x(3 \sin 60°\ \text{m}) - B_y(3 \cos 60°\ \text{m}) = 0 \qquad (1)$$

$$\pm \Sigma F_x = 0; \quad A_x - B_x = 0 \qquad (2)$$

$$+\uparrow \Sigma F_y = 0; \quad A_y - B_y = 0 \qquad (3)$$

*Member BC*

$$\circlearrowleft + \Sigma M_C = 0; \quad 2000\ \text{N}(2\ \text{m}) - B_y(4\ \text{m}) = 0 \qquad (4)$$

$$\pm \Sigma F_x = 0; \quad B_x - C_x = 0 \qquad (5)$$

$$+\uparrow \Sigma F_y = 0; \quad B_y - 2000\ \text{N} + C_y = 0 \qquad (6)$$

The results for $C_x$ and $C_y$ can be determined by solving these equations in the following sequence: 4, 1, 5, then 6. The results are

$$B_y = 1000\ \text{N}$$

$$B_x = 577\ \text{N}$$

$$C_x = 577\ \text{N} \qquad \textit{Ans.}$$

$$C_y = 1000\ \text{N} \qquad \textit{Ans.}$$

By comparison, Solution I is simpler since the requirement that $F_{AB}$ in Fig. 6–28*b* be equal, opposite, and collinear at the ends of member $AB$ automatically satisfies Eqs. 1, 2, and 3 above and therefore eliminates the need to write these equations. ***As a result, save yourself some time and effort by always identifying the two-force members before starting the analysis!***

## EXAMPLE 6.17

The compound beam shown in Fig. 6–29*a* is pin connected at *B*. Determine the components of reaction at its supports. Neglect its weight and thickness.

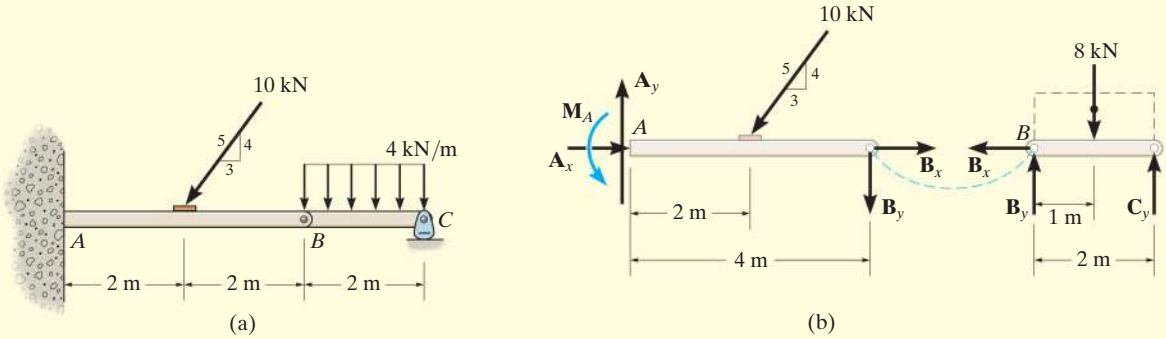



**Fig. 6–29**

### SOLUTION

**Free-Body Diagrams.** By inspection, if we consider a free-body diagram of the *entire beam ABC*, there will be three unknown reactions at *A* and one at *C*. These four unknowns cannot all be obtained from the three available equations of equilibrium, and so for the solution it will become necessary to dismember the beam into its two segments, as shown in Fig. 6–29*b*.

**Equations of Equilibrium.** The six unknowns are determined as follows:

*Segment BC*

$$\pm\leftarrow \Sigma F_x = 0; \qquad B_x = 0$$

$$\circlearrowleft +\Sigma M_B = 0; \qquad -8\ \text{kN}(1\ \text{m}) + C_y(2\ \text{m}) = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad B_y - 8\ \text{kN} + C_y = 0$$

*Segment AB*

$$\pm\rightarrow \Sigma F_x = 0; \qquad A_x - (10\ \text{kN})\left(\tfrac{3}{5}\right) + B_x = 0$$

$$\circlearrowleft +\Sigma M_A = 0; \qquad M_A - (10\ \text{kN})\left(\tfrac{4}{5}\right)(2\ \text{m}) - B_y(4\ \text{m}) = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad A_y - (10\ \text{kN})\left(\tfrac{4}{5}\right) - B_y = 0$$

Solving each of these equations successively, using previously calculated results, we obtain

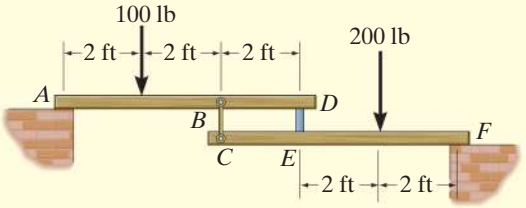
$$A_x = 6\ \text{kN} \qquad A_y = 12\ \text{kN} \qquad M_A = 32\ \text{kN}\cdot\text{m} \qquad \textit{Ans.}$$

$$B_x = 0 \qquad B_y = 4\ \text{kN}$$

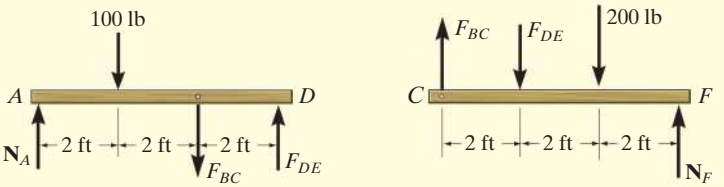
$$C_y = 4\ \text{kN} \qquad \textit{Ans.}$$

## EXAMPLE 6.18

The two planks in Fig. 6–30*a* are connected together by cable *BC* and a smooth spacer *DE*. Determine the reactions at the smooth supports *A* and *F*, and also find the force developed in the cable and spacer.




(a)




(b)

**Fig. 6–30**

### SOLUTION

**Free-Body Diagrams.** The free-body diagram of each plank is shown in Fig. 6–30*b*. It is important to apply Newton's third law to the interaction forces $F_{BC}$ and $F_{DE}$ as shown.

**Equations of Equilibrium.** For plank *AD*,

$$\circlearrowleft + \Sigma M_A = 0; \qquad F_{DE}(6\text{ ft}) - F_{BC}(4\text{ ft}) - 100\text{ lb}\,(2\text{ ft}) = 0$$

For plank *CF*,

$$\circlearrowleft + \Sigma M_F = 0; \qquad F_{DE}(4\text{ ft}) - F_{BC}(6\text{ ft}) + 200\text{ lb}\,(2\text{ ft}) = 0$$

Solving simultaneously,

$$F_{DE} = 140\text{ lb} \quad F_{BC} = 160\text{ lb} \qquad \textit{Ans.}$$

Using these results, for plank *AD*,

$$+\uparrow \Sigma F_y = 0; \qquad N_A + 140\text{ lb} - 160\text{ lb} - 100\text{ lb} = 0$$

$$N_A = 120\text{ lb} \qquad \textit{Ans.}$$

And for plank *CF*,

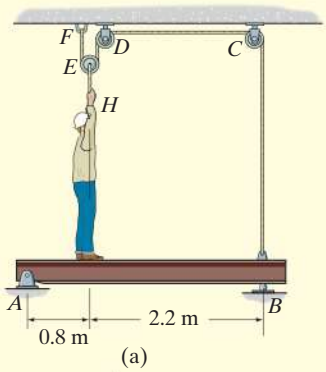
$$+\uparrow \Sigma F_y = 0; \qquad N_F + 160\text{ lb} - 140\text{ lb} - 200\text{ lb} = 0$$

$$N_F = 180\text{ lb} \qquad \textit{Ans.}$$

**NOTE:** Draw the free-body diagram of the system of *both* planks and apply $\Sigma M_A = 0$ to determine $N_F$. Then use the free-body diagram of *CEF* to determine $F_{DE}$ and $F_{BC}$.

## EXAMPLE 6.19

The 75-kg man in Fig. 6–31*a* attempts to lift the 40-kg uniform beam off the roller support at $B$. Determine the tension developed in the cable attached to $B$ and the normal reaction of the man on the beam when this is about to occur.



(a)

### SOLUTION

**Free-Body Diagrams.** The tensile force in the cable will be denoted as $T_1$. The free-body diagrams of the pulley $E$, the man, and the beam are shown in Fig. 6–31*b*. Since the man must lift the beam off the roller $B$ then $N_B = 0$. When drawing each of these diagrams, it is very important to apply Newton's third law.

**Equations of Equilibrium.** Using the free-body diagram of pulley $E$,

$$+\uparrow \Sigma F_y = 0; \qquad 2T_1 - T_2 = 0 \quad \text{or} \quad T_2 = 2T_1 \tag{1}$$

Referring to the free-body diagram of the man using this result,

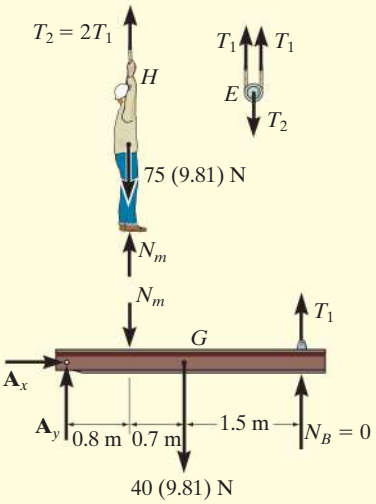
$$+\uparrow \Sigma F_y = 0 \qquad N_m + 2T_1 - 75(9.81)\,\text{N} = 0 \tag{2}$$

Summing moments about point $A$ on the beam,

$$\circlearrowleft +\Sigma M_A = 0;\ T_1(3\,\text{m}) - N_m(0.8\,\text{m}) - [40(9.81)\,\text{N}]\,(1.5\,\text{m}) = 0 \tag{3}$$

Solving Eqs. 2 and 3 simultaneously for $T_1$ and $N_m$, then using Eq. (1) for $T_2$, we obtain

$$T_1 = 256\,\text{N} \qquad N_m = 224\,\text{N} \qquad T_2 = 512\,\text{N} \qquad \textit{Ans.}$$



(b)

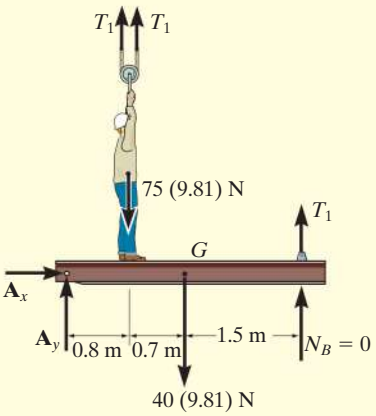
### SOLUTION II

A direct solution for $T_1$ can be obtained by considering the beam, the man, and pulley $E$ as a *single system*. The free-body diagram is shown in Fig. 6–31*c*. Thus,

$$\circlearrowleft +\Sigma M_A = 0; \qquad 2T_1(0.8\,\text{m}) - [75(9.81)\,\text{N}](0.8\,\text{m}) - [40(9.81)\,\text{N}](1.5\,\text{m}) + T_1(3\,\text{m}) = 0$$

$$T_1 = 256\,\text{N} \qquad \textit{Ans.}$$

With this result Eqs. 1 and 2 can then be used to find $N_m$ and $T_2$.

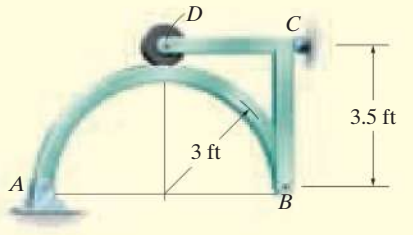


(c)

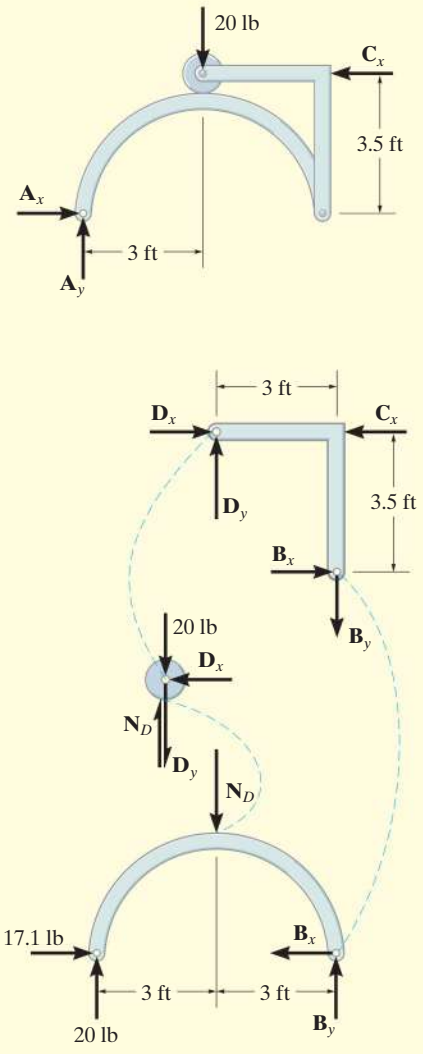
**Fig. 6–31**

## EXAMPLE 6.20

The smooth disk shown in Fig. 6–32*a* is pinned at $D$ and has a weight of 20 lb. Neglecting the weights of the other members, determine the horizontal and vertical components of reaction at pins $B$ and $D$.




(a)




(b)

**Fig. 6–32**

### SOLUTION

**Free-Body Diagrams.** The free-body diagrams of the entire frame and each of its members are shown in Fig. 6–32*b*.

**Equations of Equilibrium.** The eight unknowns can of course be obtained by applying the eight equilibrium equations to each member—three to member $AB$, three to member $BCD$, and two to the disk. (Moment equilibrium is automatically satisfied for the disk.) If this is done, however, all the results can be obtained only from a simultaneous solution of some of the equations. (Try it and find out.) To avoid this situation, it is best first to determine the three support reactions on the *entire* frame; then, using these results, the remaining five equilibrium equations can be applied to two other parts in order to solve successively for the other unknowns.

*Entire Frame*

$$\circlearrowleft +\Sigma M_A = 0; \quad -20\text{ lb}\,(3\text{ ft}) + C_x(3.5\text{ ft}) = 0 \qquad C_x = 17.1\text{ lb}$$

$$\pm\rightarrow \Sigma F_x = 0; \quad A_x - 17.1\text{ lb} = 0 \qquad A_x = 17.1\text{ lb}$$

$$+\uparrow \Sigma F_y = 0; \quad A_y - 20\text{ lb} = 0 \qquad A_y = 20\text{ lb}$$

*Member AB*

$$\pm\rightarrow \Sigma F_x = 0; \quad 17.1\text{ lb} - B_x = 0 \qquad B_x = 17.1\text{ lb} \quad \textit{Ans.}$$

$$\circlearrowleft +\Sigma M_B = 0; \quad -20\text{ lb}\,(6\text{ ft}) + N_D(3\text{ ft}) = 0 \qquad N_D = 40\text{ lb}$$

$$+\uparrow \Sigma F_y = 0; \quad 20\text{ lb} - 40\text{ lb} + B_y = 0 \qquad B_y = 20\text{ lb} \quad \textit{Ans.}$$

*Disk*

$$\pm\rightarrow \Sigma F_x = 0; \quad D_x = 0 \quad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad 40\text{ lb} - 20\text{ lb} - D_y = 0 \qquad D_y = 20\text{ lb} \quad \textit{Ans.}$$

## EXAMPLE 6.21

The frame in Fig. 6–33*a* supports the 50-kg cylinder. Determine the horizontal and vertical components of reaction at *A* and the force at *C*.

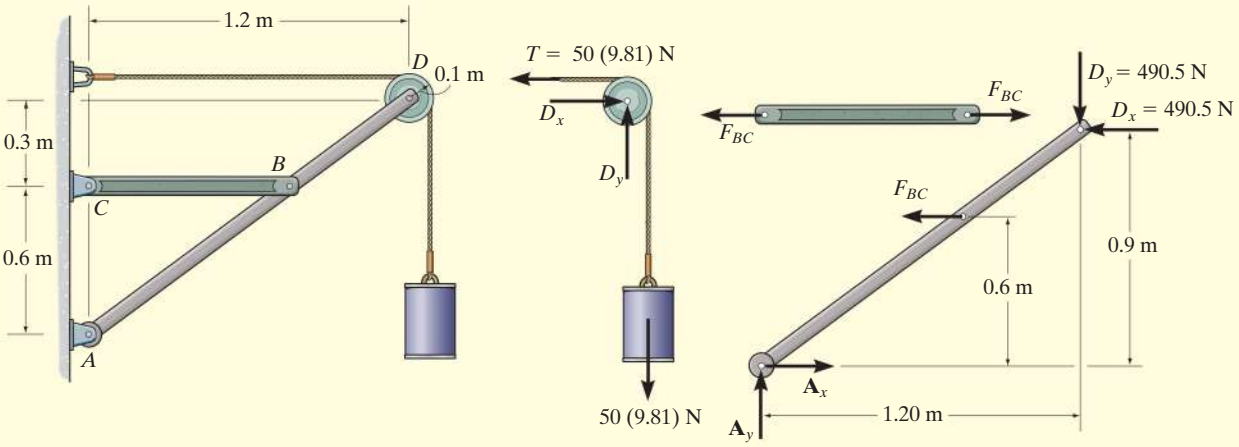



**Fig. 6–33**

### SOLUTION

**Free-Body Diagrams.** The free-body diagram of pulley *D*, along with the cylinder and a portion of the cord (a system), is shown in Fig. 6–33*b*. Member *BC* is a two-force member as indicated by its free-body diagram. The free-body diagram of member *ABD* is also shown.

**Equations of Equilibrium.** We will begin by analyzing the equilibrium of the pulley. The moment equation of equilibrium is automatically satisfied with $T = 50(9.81)$ N, and so

$$\stackrel{+}{\rightarrow} \Sigma F_x = 0; \quad D_x - 50(9.81)\,\text{N} = 0 \quad D_x = 490.5\,\text{N}$$

$$+\uparrow \Sigma F_y = 0; \quad D_y - 50(9.81)\,\text{N} = 0 \quad D_y = 490.5\,\text{N} \qquad \textit{Ans.}$$

Using these results, $F_{BC}$ can be determined by summing moments about point *A* on member *ABD*.

$$\circlearrowleft + \Sigma M_A = 0;\ F_{BC}(0.6\,\text{m}) + 490.5\,\text{N}(0.9\,\text{m}) - 490.5\,\text{N}(1.20\,\text{m}) = 0$$

$$F_{BC} = 245.25\ \text{N} \qquad \textit{Ans.}$$

Now $A_x$ and $A_y$ can be determined by summing forces.

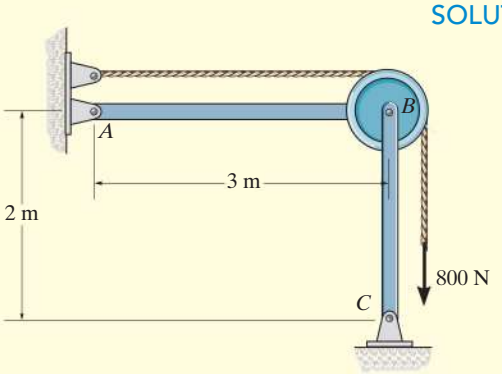
$$\stackrel{+}{\rightarrow} \Sigma F_x = 0; \quad A_x - 245.25\,\text{N} - 490.5\,\text{N} = 0 \quad A_x = 736\,\text{N} \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad A_y - 490.5\,\text{N} = 0 \quad A_y = 490.5\,\text{N} \qquad \textit{Ans.}$$

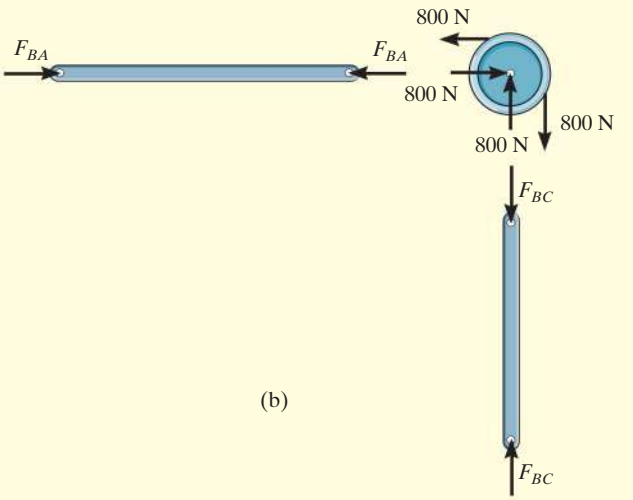
## EXAMPLE 6.22

Determine the force the pins at $A$ and $B$ exert on the two-member frame shown in Fig. 6–34*a*.

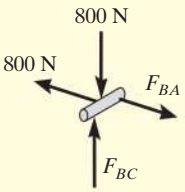
### SOLUTION I




(a)

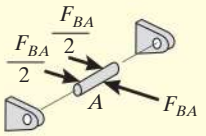



(b)



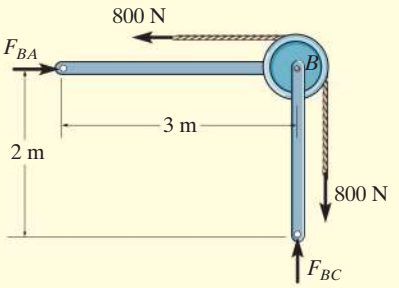

Pin $B$

(c)




Pin $A$

(d)




(e)

**Fig. 6–34**

**Free-Body Diagrams.** By inspection $AB$ and $BC$ are two-force members. Their free-body diagrams, along with that of the pulley, are shown in Fig. 6–34*b*. In order to solve this problem we must also include the free-body diagram of the pin at $B$ because this pin connects all *three members* together, Fig. 6–34*c*.

**Equations of Equilibrium:** Apply the equations of force equilibrium to pin $B$.

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \quad F_{BA} - 800\ \text{N} = 0; \quad F_{BA} = 800\ \text{N} \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad F_{BC} - 800\ \text{N} = 0; \quad F_{BC} = 800\ \text{N} \qquad \textit{Ans.}$$

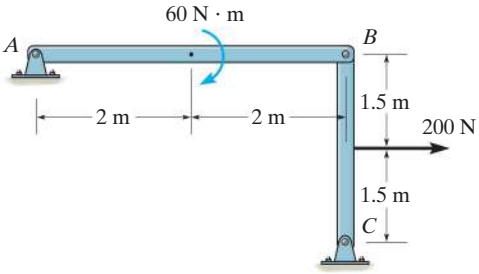
**NOTE:** The free-body diagram of the pin at $A$, Fig. 6–34*d*, indicates how the force $F_{AB}$ is balanced by the force $(F_{AB}/2)$ exerted on the pin by each of the two pin leaves.

### SOLUTION II

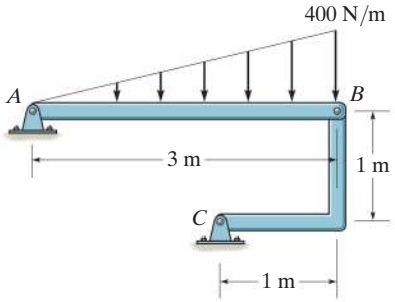
**Free-Body Diagram.** If we realize that $AB$ and $BC$ are two-force members, then the free-body diagram of the *entire frame* produces an easier solution, Fig. 6–34*e*. The force equations of equilibrium are the same as those above. Note that moment equilibrium will be satisfied, regardless of the radius of the pulley.

## PRELIMINARY PROBLEMS

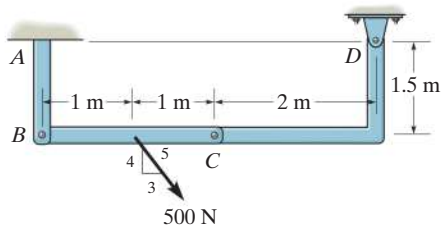
**P6–3.** In each case, identify any two-force members, and then draw the free-body diagrams of each member of the frame.



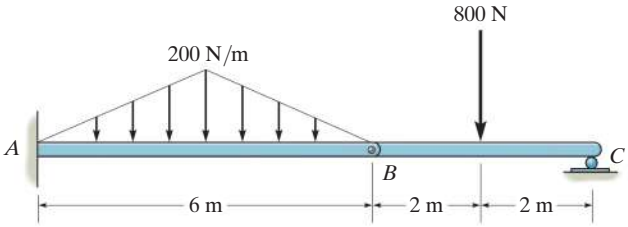

(a)



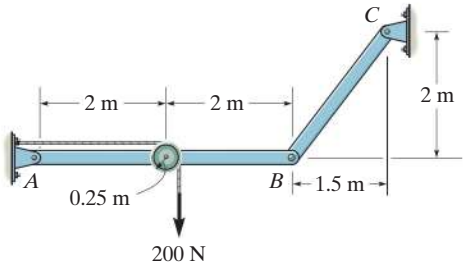

(b)



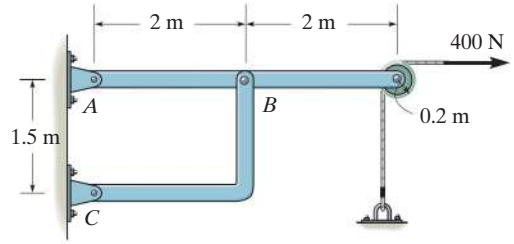

(c)




(d)




(e)



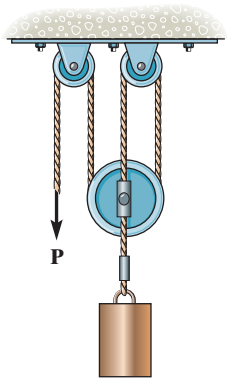

(f)

Prob. P6–3

## FUNDAMENTAL PROBLEMS

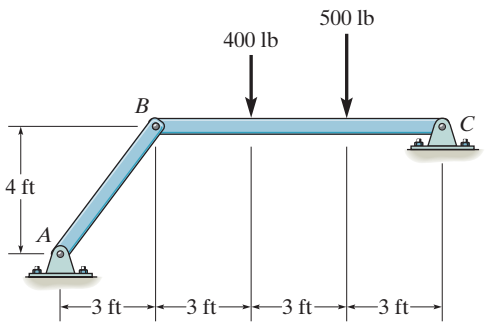
***All problem solutions must include FBDs.***

**F6–13.** Determine the force $P$ needed to hold the 60-lb weight in equilibrium.



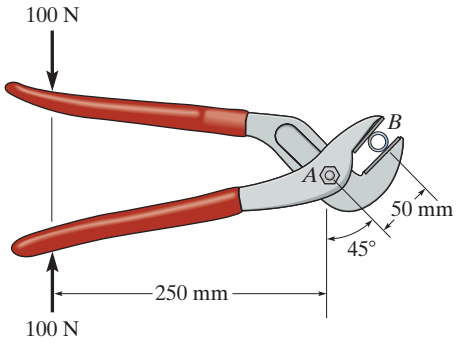

**Prob. F6–13**

**F6–14.** Determine the horizontal and vertical components of reaction at pin $C$.



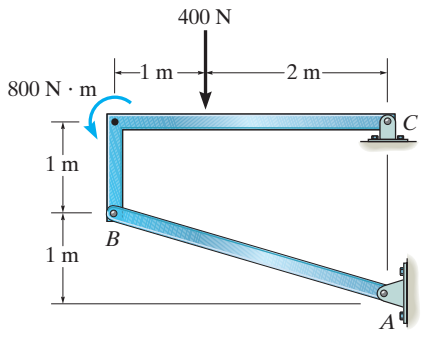

**Prob. F6–14**

**F6–15.** If a 100-N force is applied to the handles of the pliers, determine the clamping force exerted on the smooth pipe $B$ and the magnitude of the resultant force that one of the members exerts on pin $A$.



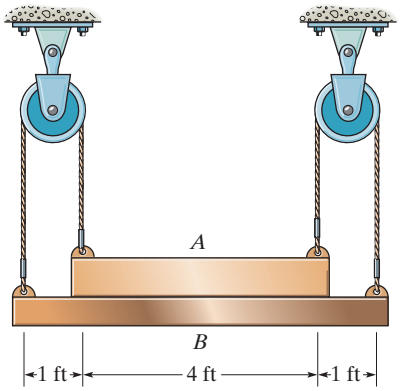

**Prob. F6–15**

**F6–16.** Determine the horizontal and vertical components of reaction at pin $C$.



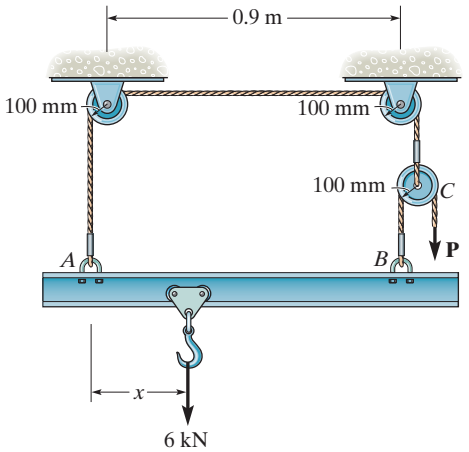

**Prob. F6–16**

**F6–17.** Determine the normal force that the 100-lb plate $A$ exerts on the 30-lb plate $B$.



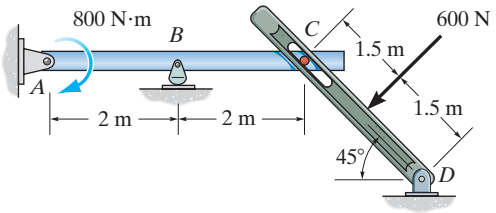

**Prob. F6–17**

**F6–18.** Determine the force $P$ needed to lift the load. Also, determine the proper placement $x$ of the hook for equilibrium. Neglect the weight of the beam.



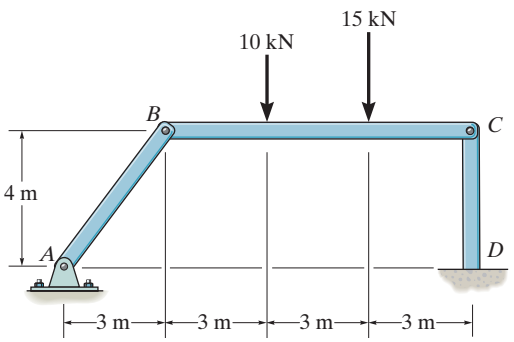

**Prob. F6–18**

**F6–19.** Determine the components of reaction at $A$ and $B$.



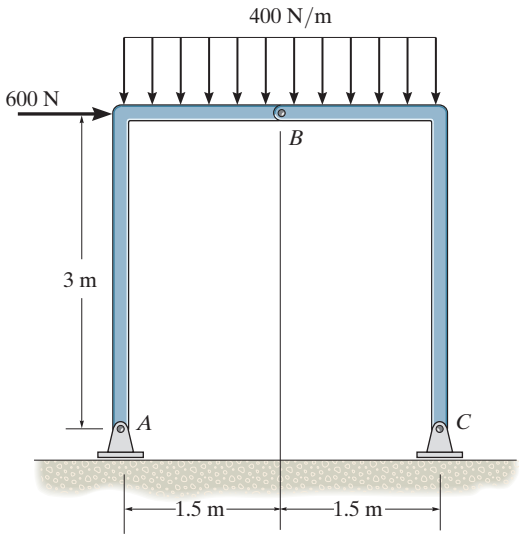

**Prob. F6–19**

**F6–20.** Determine the reactions at $D$.



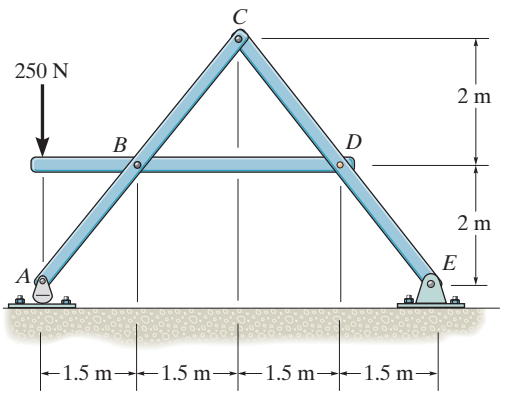

**Prob. F6–20**

**F6–21.** Determine the components of reaction at *A* and *C*.



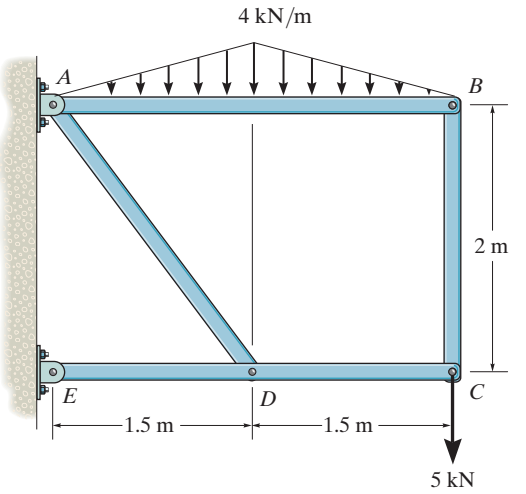

**Prob. F6–21**

**F6–22.** Determine the components of reaction at *C*.



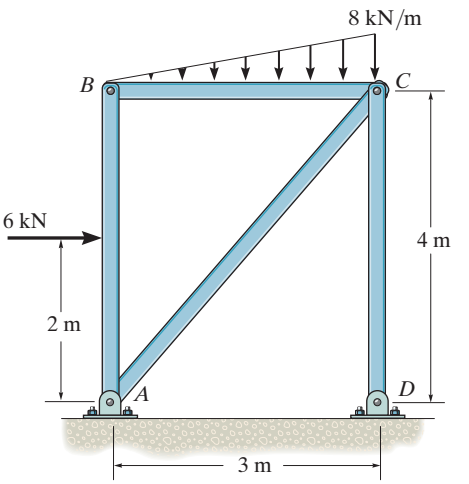

**Prob. F6–22**

**F6–23.** Determine the components of reaction at *E*.




**Prob. F6–23**

**F6–24.** Determine the components of reaction at *D* and the components of reaction the pin at *A* exerts on member *BA*.

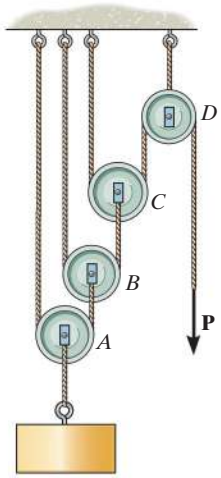



**Prob. F6–24**

## PROBLEMS

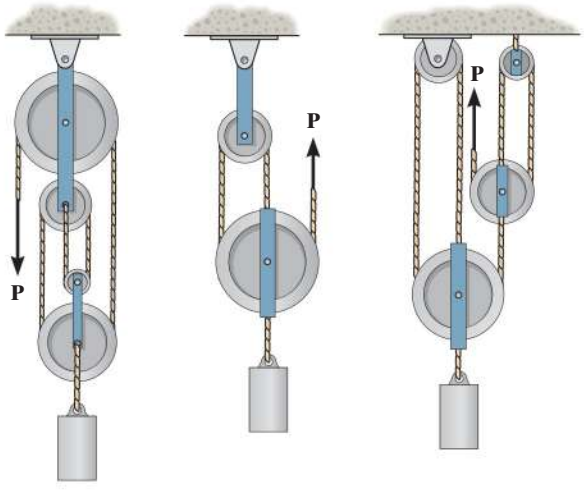
***All problem solutions must include FBDs.***

**6–61.** Determine the force **P** required to hold the 100-lb weight in equilibrium.



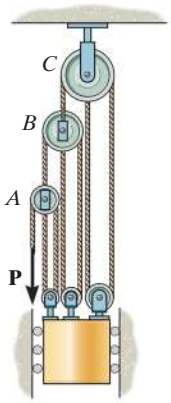

**Prob. 6–61**

**6–62.** In each case, determine the force **P** required to maintain equilibrium. The block weighs 100 lb.



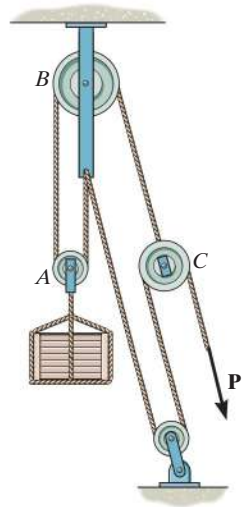

**Prob. 6–62**

**6–63.** Determine the force **P** required to hold the 50-kg mass in equilibrium.



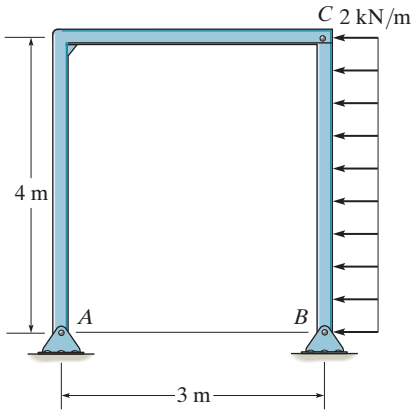

**Prob. 6–63**

***6–64.** Determine the force **P** required to hold the 150-kg crate in equilibrium.



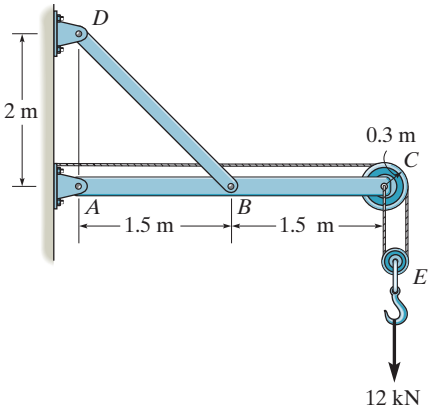

**Prob. 6–64**

**6–65.** Determine the horizontal and vertical components of force that pins *A* and *B* exert on the frame.



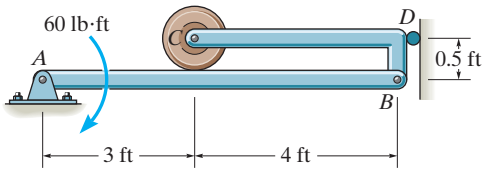

**Prob. 6–65**

**6–66.** Determine the horizontal and vertical components of force at pins *A* and *D*.



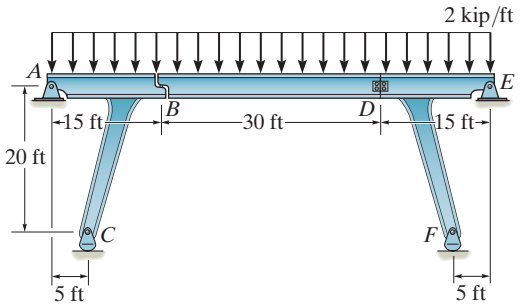

**Prob. 6–66**

**6–67.** Determine the force that the smooth roller *C* exerts on member *AB*. Also, what are the horizontal and vertical components of reaction at pin *A*? Neglect the weight of the frame and roller.



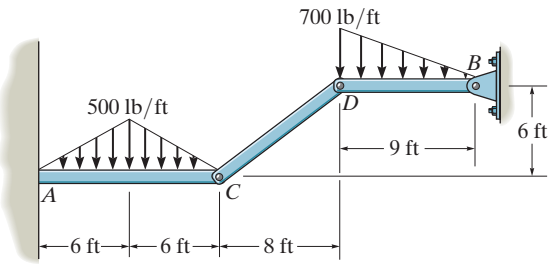

**Prob. 6–67**

***6–68.** The bridge frame consists of three segments which can be considered pinned at *A*, *D*, and *E*, rocker supported at *C* and *F*, and roller supported at *B*. Determine the horizontal and vertical components of reaction at all these supports due to the loading shown.



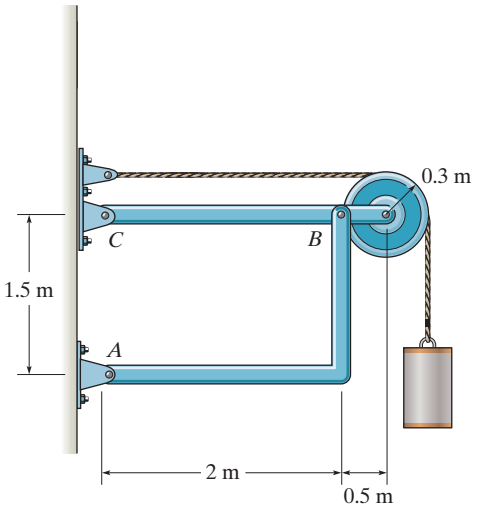

**Prob. 6–68**

**6–69.** Determine the reactions at supports *A* and *B*.



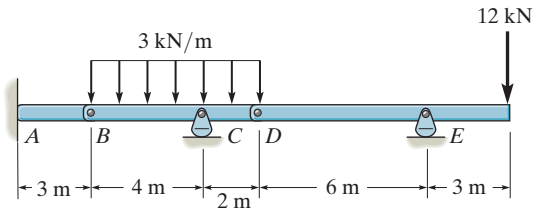

**Prob. 6–69**

**6–70.** Determine the horizontal and vertical components of force at pins *B* and *C*. The suspended cylinder has a mass of 75 kg.



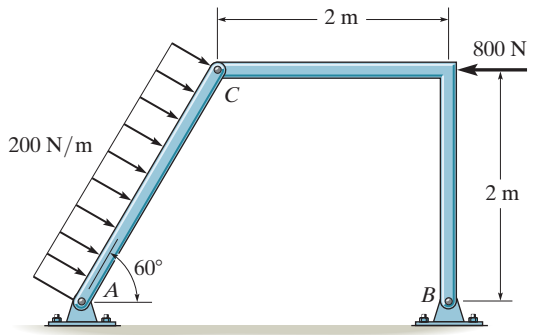

**Prob. 6–70**

**6–71.** Determine the reactions at the supports $A$, $C$, and $E$ of the compound beam.



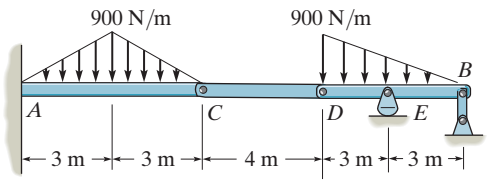

**Prob. 6–71**

***6–72.** Determine the resultant force at pins $A$, $B$, and $C$ on the three-member frame.




**Prob. 6–72**

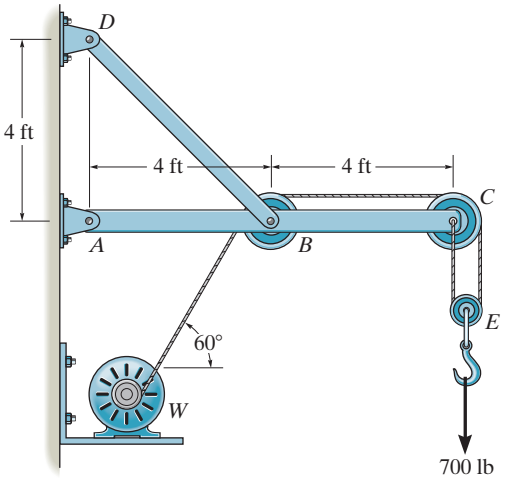
**6–73.** Determine the reactions at the supports at $A$, $E$, and $B$ of the compound beam.




**Prob. 6–73**

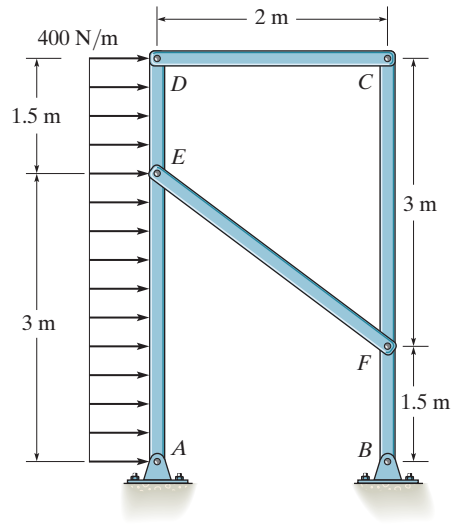
**6–74.** The wall crane supports a load of 700 lb. Determine the horizontal and vertical components of reaction at the pins $A$ and $D$. Also, what is the force in the cable at the winch $W$?

**6–75.** The wall crane supports a load of 700 lb. Determine the horizontal and vertical components of reaction at the pins $A$ and $D$. Also, what is the force in the cable at the winch $W$? The jib $ABC$ has a weight of 100 lb and member $BD$ has a weight of 40 lb. Each member is uniform and has a center of gravity at its center.



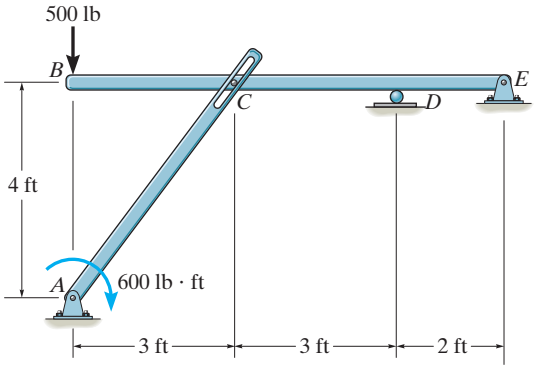

**Probs. 6–74/75**

***6–76.** Determine the horizontal and vertical components of force which the pins at $A$ and $B$ exert on the frame.



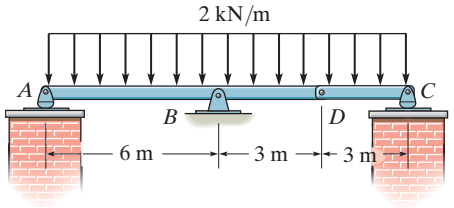

**Prob. 6–76**

**6–77.** The two-member structure is connected at $C$ by a pin, which is fixed to $BDE$ and passes through the smooth slot in member $AC$. Determine the horizontal and vertical components of reaction at the supports.



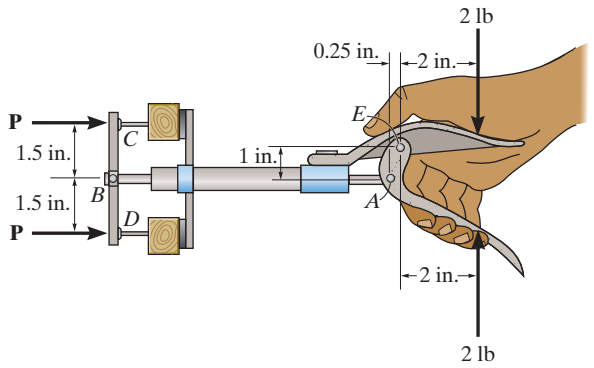

**Prob. 6–77**

**6–78.** The compound beam is pin supported at $B$ and supported by rockers at $A$ and $C$. There is a hinge (pin) at $D$. Determine the reactions at the supports.



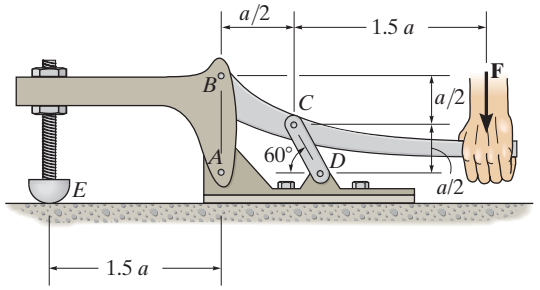

**Prob. 6–78**

**6–79.** When a force of 2 lb is applied to the handles of the brad squeezer, it pulls in the smooth rod $AB$. Determine the force **P** exerted on each of the smooth brads at $C$ and $D$.



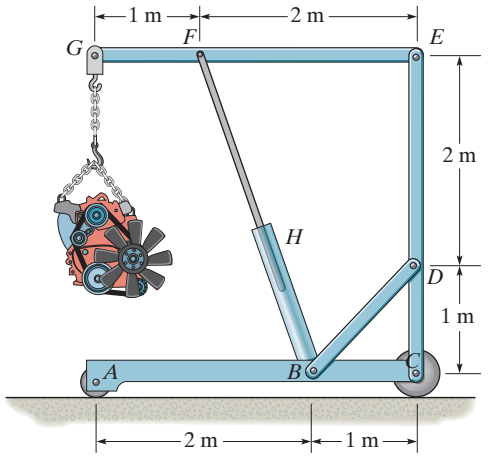

**Prob. 6–79**

***6–80.** The toggle clamp is subjected to a force **F** at the handle. Determine the vertical clamping force acting at $E$.



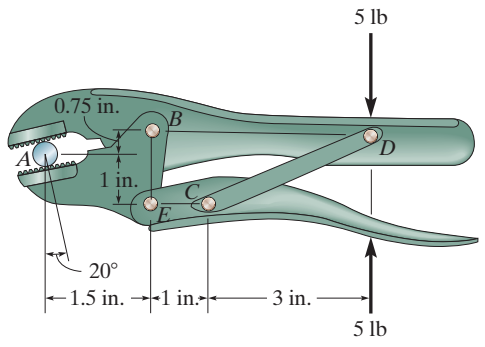

**Prob. 6–80**

**6–81.** The hoist supports the 125-kg engine. Determine the force the load creates in member $DB$ and in member $FB$, which contains the hydraulic cylinder $H$.



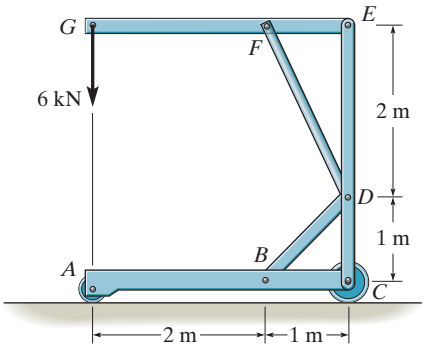

**Prob. 6–81**

**6–82.** A 5-lb force is applied to the handles of the vise grip. Determine the compressive force developed on the smooth bolt shank $A$ at the jaws.



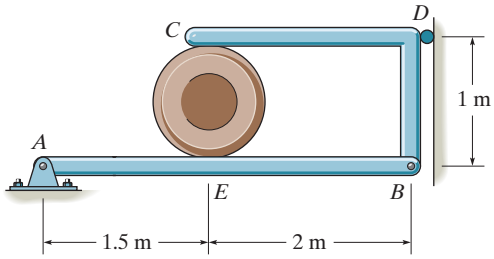

**Prob. 6–82**

**6–83.** Determine the force in members *FD* and *DB* of the frame. Also, find the horizontal and vertical components of reaction the pin at *C* exerts on member *ABC* and member *EDC*.



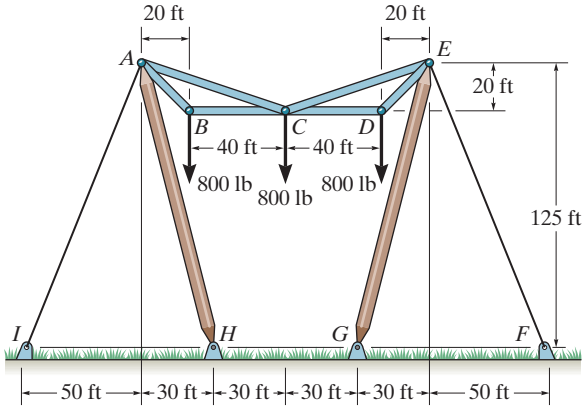

**Prob. 6–83**

***6–84.** Determine the force that the smooth 20-kg cylinder exerts on members *AB* and *CDB*. Also, what are the horizontal and vertical components of reaction at pin *A*?



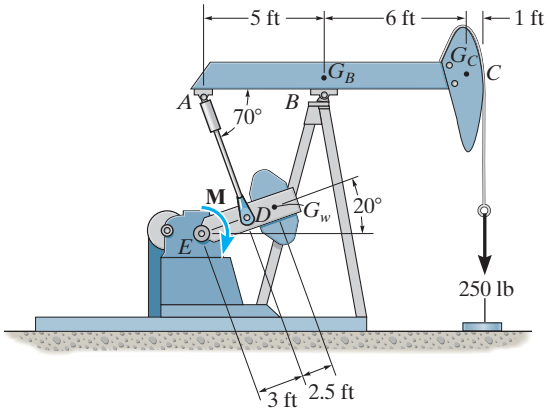

**Prob. 6–84**

**6–85.** The three power lines exert the forces shown on the pin-connected members at joints *B*, *C*, and *D*, which in turn are pin connected to the poles *AH* and *EG*. Determine the force in the guy cable *AI* and the pin reaction at the support *H*.



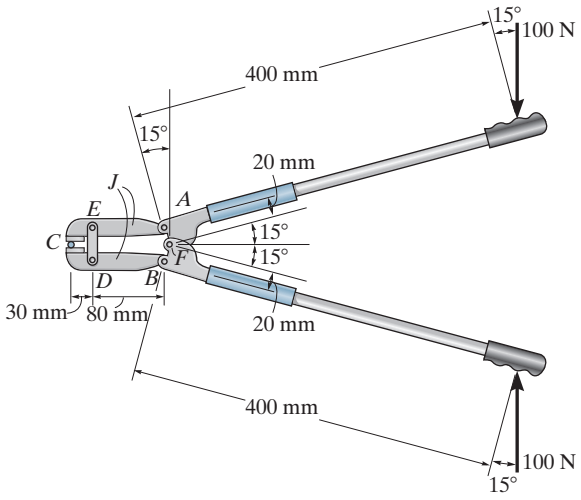

**Prob. 6–85**

**6–86.** The pumping unit is used to recover oil. When the walking beam *ABC* is horizontal, the force acting in the wireline at the well head is 250 lb. Determine the torque **M** which must be exerted by the motor in order to overcome this load. The horse-head *C* weighs 60 lb and has a center of gravity at $G_C$. The walking beam *ABC* has a weight of 130 lb and a center of gravity at $G_B$, and the counterweight has a weight of 200 lb and a center of gravity at $G_W$. The pitman, *AD*, is pin connected at its ends and has negligible weight.



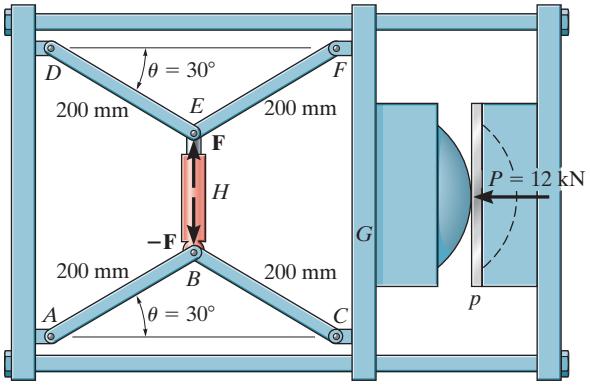

**Prob. 6–86**

**6–87.** Determine the force that the jaws *J* of the metal cutters exert on the smooth cable *C* if 100-N forces are applied to the handles. The jaws are pinned at *E* and *A*, and *D* and *B*. There is also a pin at *F*.



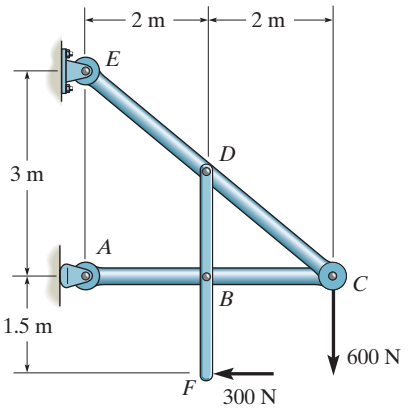

**Prob. 6–87**

***6–88.** The machine shown is used for forming metal plates. It consists of two toggles *ABC* and *DEF*, which are operated by the hydraulic cylinder *H*. The toggles push the movable bar *G* forward, pressing the plate *p* into the cavity. If the force which the plate exerts on the head is $P = 12$ kN, determine the force *F* in the hydraulic cylinder when $\theta = 30°$.



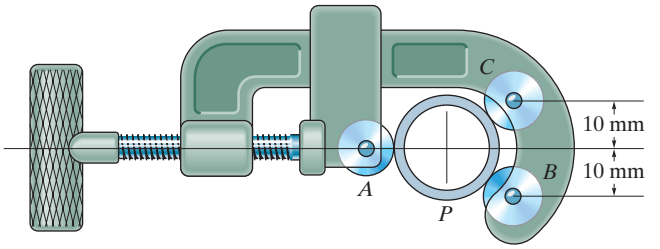

**Prob. 6–88**

**6–89.** Determine the horizontal and vertical components of force which pin *C* exerts on member *ABC*. The 600-N load is applied to the pin.



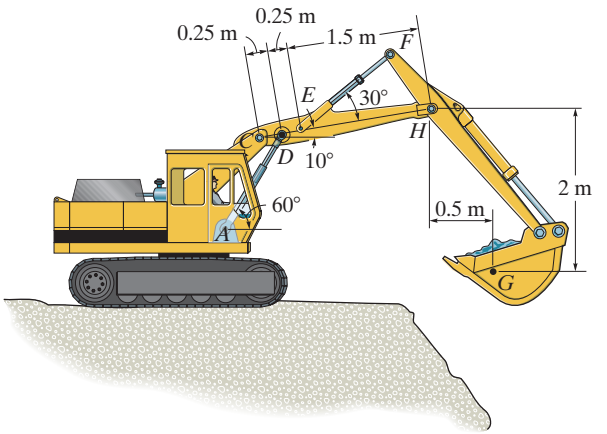

**Prob. 6–89**

**6–90.** The pipe cutter is clamped around the pipe *P*. If the wheel at *A* exerts a normal force of $F_A = 80$ N on the pipe, determine the normal forces of wheels *B* and *C* on the pipe. Also compute the pin reaction on the wheel at *C*. The three wheels each have a radius of 7 mm and the pipe has an outer radius of 10 mm.



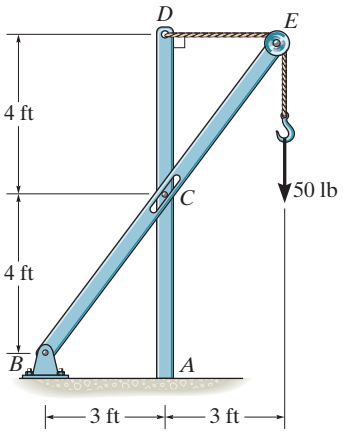

**Prob. 6–90**

**6–91.** Determine the force created in the hydraulic cylinders *EF* and *AD* in order to hold the shovel in equilibrium. The shovel load has a mass of 1.25 Mg and a center of gravity at *G*. All joints are pin connected.



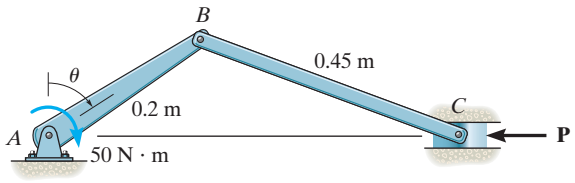

**Prob. 6–91**

*6–92. Determine the horizontal and vertical components of force at pin $B$ and the normal force the pin at $C$ exerts on the smooth slot. Also, determine the moment and horizontal and vertical reactions of force at $A$. There is a pulley at $E$.



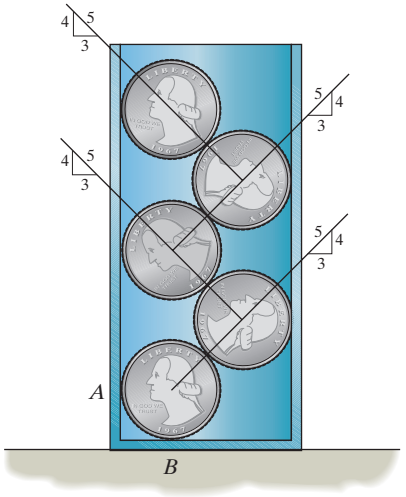

Prob. 6–92

6–93. The constant moment of 50 N · m is applied to the crank shaft. Determine the compressive force $P$ that is exerted on the piston for equilibrium as a function of $\theta$. Plot the results of $P$ (vertical axis) versus $\theta$ (horizontal axis) for $0° \leq \theta \leq 90°$.



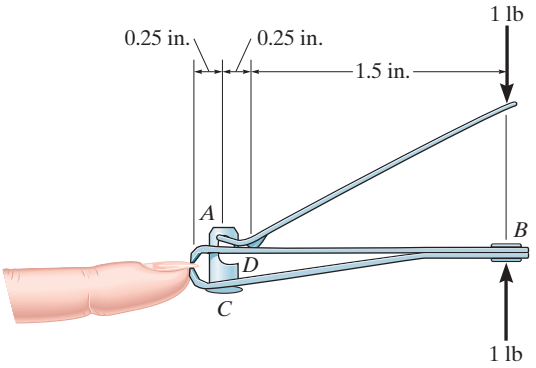

Prob. 6–93

6–94. Five coins are stacked in the smooth plastic container shown. If each coin weighs 0.0235 lb, determine the normal reactions of the bottom coin on the container at points $A$ and $B$.



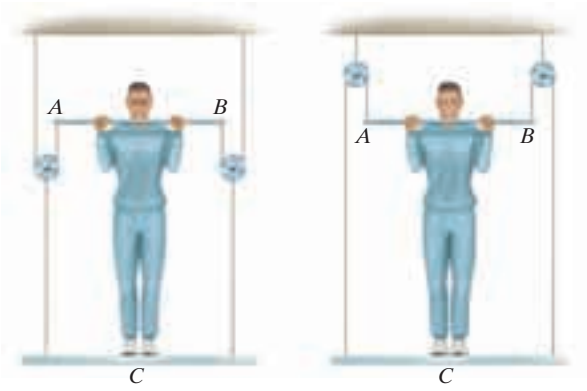

Prob. 6–94

6–95. The nail cutter consists of the handle and the two cutting blades. Assuming the blades are pin connected at $B$ and the surface at $D$ is smooth, determine the normal force on the fingernail when a force of 1 lb is applied to the handles as shown. The pin $AC$ slides through a smooth hole at $A$ and is attached to the bottom member at $C$.




Prob. 6–95

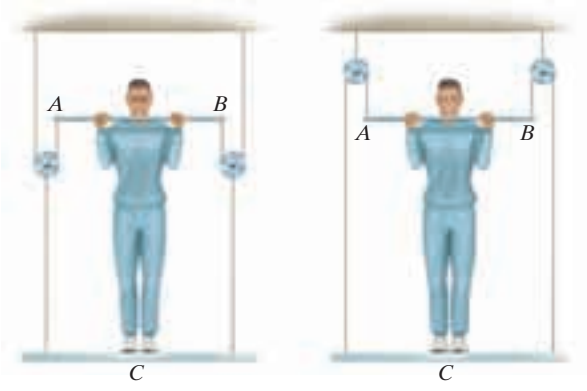
***6–96.** A man having a weight of 175 lb attempts to hold himself using one of the two methods shown. Determine the total force he must exert on bar *AB* in each case and the normal reaction he exerts on the platform at *C*. Neglect the weight of the platform.




(a) (b)

**Prob. 6–96**

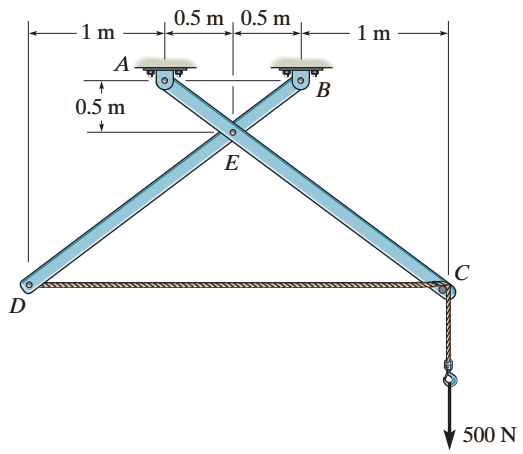
**6–97.** A man having a weight of 175 lb attempts to hold himself using one of the two methods shown. Determine the total force he must exert on bar *AB* in each case and the normal reaction he exerts on the platform at *C*. The platform has a weight of 30 lb.




(a) (b)

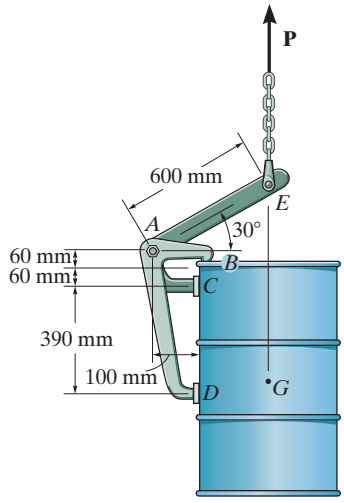
**Prob. 6–97**

**6–98.** The two-member frame is pin connected at *E*. The cable is attached to *D*, passes over the smooth peg at *C*, and supports the 500-N load. Determine the horizontal and vertical reactions at each pin.



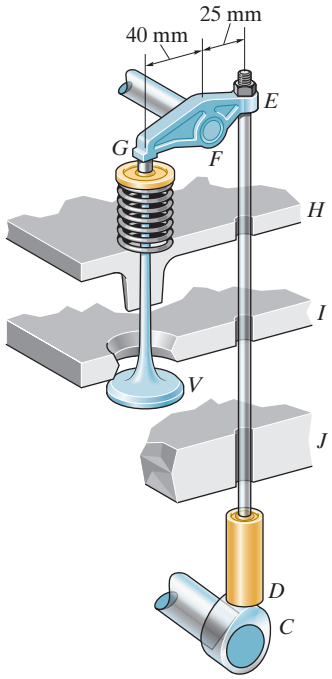

**Prob. 6–98**

**6–99.** If the 300-kg drum has a center of mass at point *G*, determine the horizontal and vertical components of force acting at pin *A* and the reactions on the smooth pads *C* and *D*. The grip at *B* on member *DAB* resists both horizontal and vertical components of force at the rim of the drum.



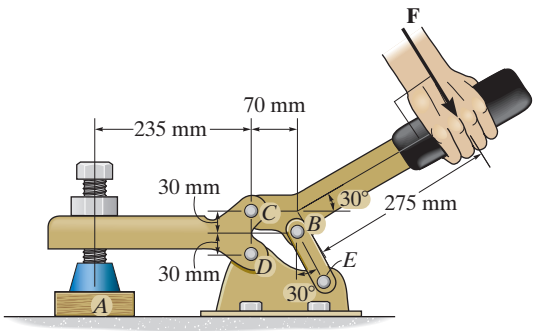

**Prob. 6–99**

*6–100. Operation of exhaust and intake valves in an automobile engine consists of the cam $C$, push rod $DE$, rocker arm $EFG$ which is pinned at $F$, and a spring and valve, $V$. If the compression in the spring is 20 mm when the valve is open as shown, determine the normal force acting on the cam lobe at $C$. Assume the cam and bearings at $H$, $I$, and $J$ are smooth. The spring has a stiffness of 300 N/m.



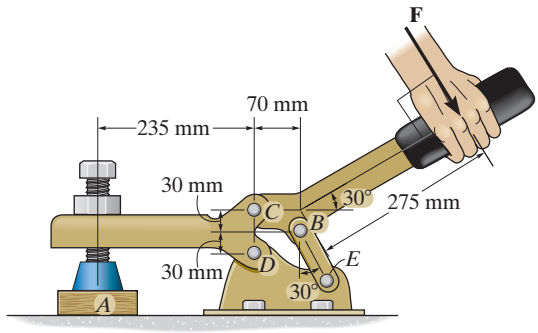

Prob. 6–100

6–101. If a clamping force of 300 N is required at $A$, determine the amount of force **F** that must be applied to the handle of the toggle clamp.



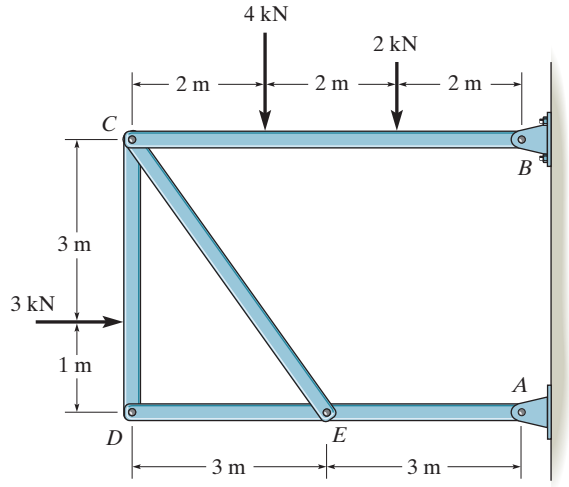

Prob. 6–101

6–102. If a force of $F = 350$ N is applied to the handle of the toggle clamp, determine the resulting clamping force at $A$.



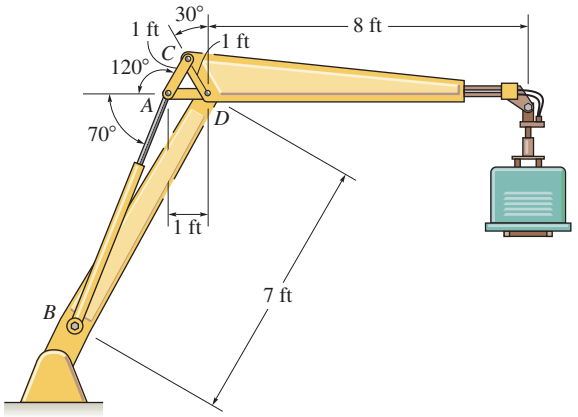

Prob. 6–102

6–103. Determine the horizontal and vertical components of force that the pins at $A$ and $B$ exert on the frame.



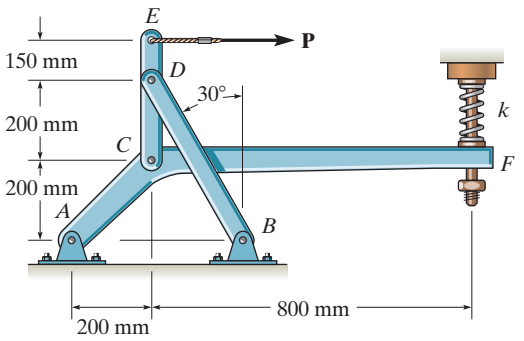

Prob. 6–103

***6–104.** The hydraulic crane is used to lift the 1400-lb load. Determine the force in the hydraulic cylinder $AB$ and the force in links $AC$ and $AD$ when the load is held in the position shown.



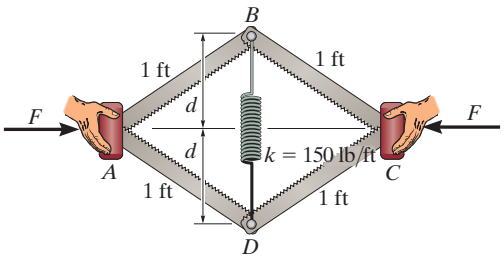

**Prob. 6–104**

**6–105.** Determine force **P** on the cable if the spring is compressed 0.025 m when the mechanism is in the position shown. The spring has a stiffness of $k = 6$ kN/m.



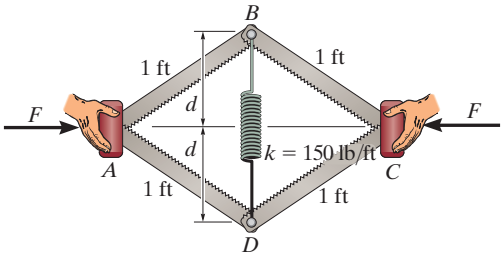

**Prob. 6–105**

**6–106.** If $d = 0.75$ ft and the spring has an unstretched length of 1 ft, determine the force $F$ required for equilibrium.



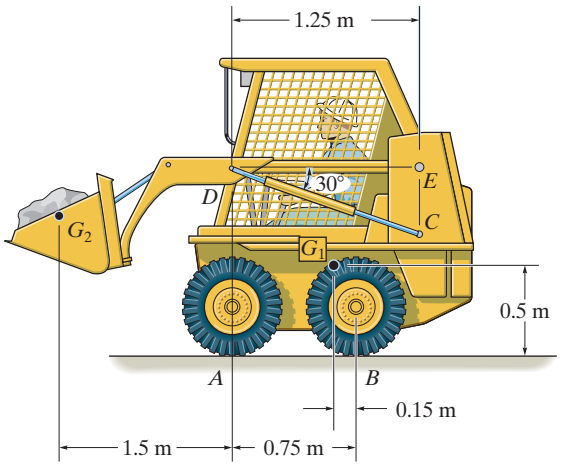

**Prob. 6–106**

**6–107.** If a force of $F = 50$ lb is applied to the pads at $A$ and $C$, determine the smallest dimension $d$ required for equilibrium if the spring has an unstretched length of 1 ft.



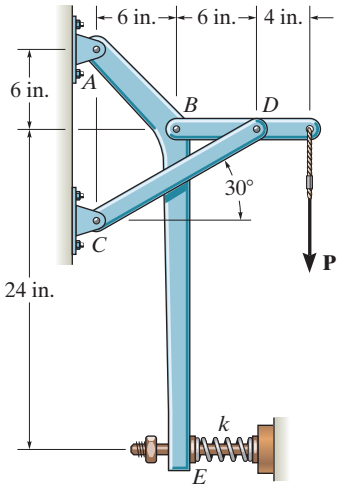

**Prob. 6–107**

***6–108.** The skid-steer loader has a mass of 1.18 Mg, and in the position shown the center of mass is at $G_1$. If there is a 300-kg stone in the bucket, with center of mass at $G_2$, determine the reactions of each pair of wheels $A$ and $B$ on the ground and the force in the hydraulic cylinder $CD$ and at the pin $E$. There is a similar linkage on each side of the loader.




**Prob. 6–108**

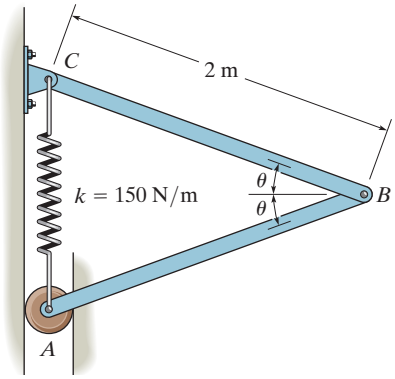
**6–109.** Determine the force **P** on the cable if the spring is compressed 0.5 in. when the mechanism is in the position shown. The spring has a stiffness of $k = 800$ lb/ft.




Prob. 6–109

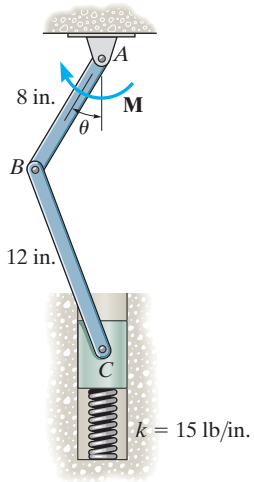
**6–110.** The spring has an unstretched length of 0.3 m. Determine the angle $\theta$ for equilibrium if the uniform bars each have a mass of 20 kg.

**6–111.** The spring has an unstretched length of 0.3 m. Determine the mass $m$ of each uniform bar if each angle $\theta = 30°$ for equilibrium.




Probs. 6–110/111

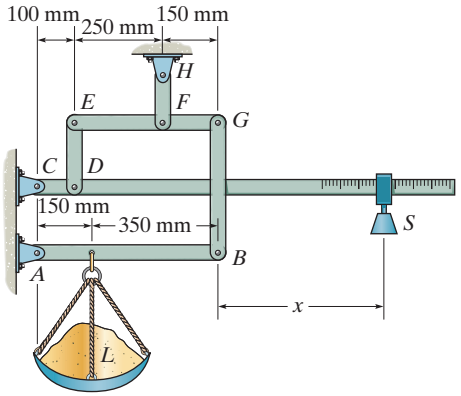
***6–112.** The piston $C$ moves vertically between the two smooth walls. If the spring has a stiffness of $k = 15$ lb/in., and is unstretched when $\theta = 0°$, determine the couple **M** that must be applied to $AB$ to hold the mechanism in equilibrium when $\theta = 30°$.




Prob. 6–112

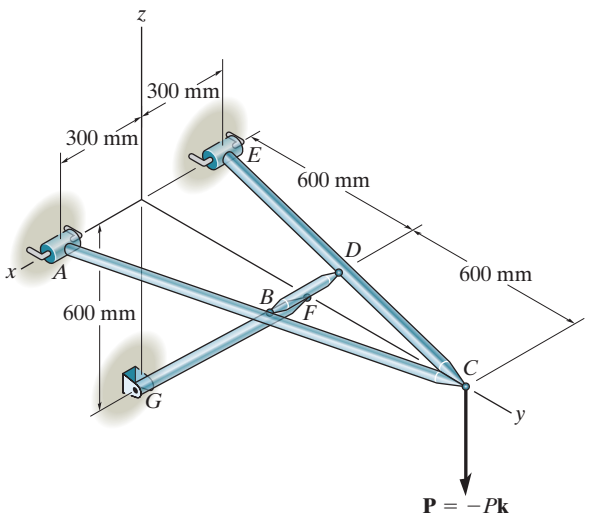
**6–113.** The platform scale consists of a combination of third and first class levers so that the load on one lever becomes the effort that moves the next lever. Through this arrangement, a small weight can balance a massive object. If $x = 450$ mm, determine the required mass of the counterweight $S$ required to balance a 90-kg load, $L$.

**6–114.** The platform scale consists of a combination of third and first class levers so that the load on one lever becomes the effort that moves the next lever. Through this arrangement, a small weight can balance a massive object. If $x = 450$ mm, and the mass of the counterweight $S$ is 2 kg, determine the mass of the load $L$ required to maintain the balance.



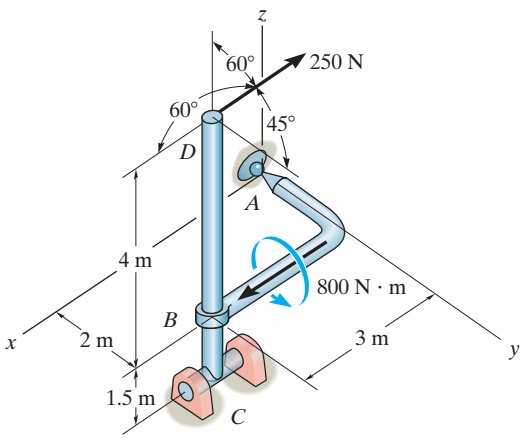

Probs. 6–113/114

**6–115.** The four-member "A" frame is supported at $A$ and $E$ by smooth collars and at $G$ by a pin. All the other joints are ball-and-sockets. If the pin at $G$ will fail when the resultant force there is 800 N, determine the largest vertical force $P$ that can be supported by the frame. Also, what are the $x, y, z$ force components which member $BD$ exerts on members $EDC$ and $ABC$? The collars at $A$ and $E$ and the pin at $G$ only exert force components on the frame.



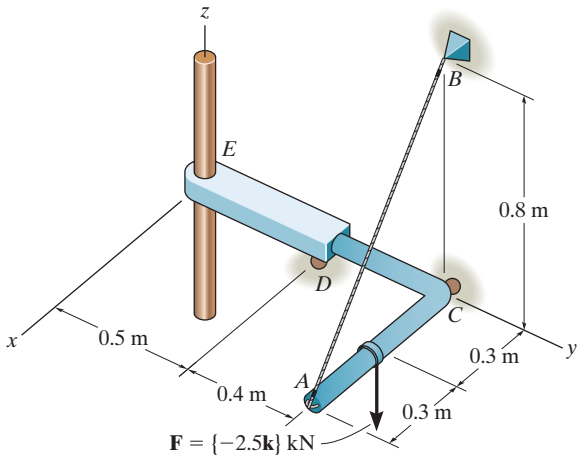

Prob. 6–115

***6–116.** The structure is subjected to the loadings shown. Member $AB$ is supported by a ball-and-socket at $A$ and smooth collar at $B$. Member $CD$ is supported by a pin at $C$. Determine the $x, y, z$ components of reaction at $A$ and $C$.



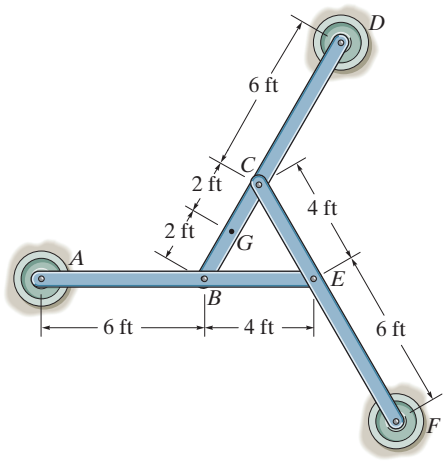

Prob. 6–116

**6–117.** The structure is subjected to the loading shown. Member $AD$ is supported by a cable $AB$ and roller at $C$ and fits through a smooth circular hole at $D$. Member $ED$ is supported by a roller at $D$ and a pole that fits in a smooth snug circular hole at $E$. Determine the $x, y, z$ components of reaction at $E$ and the tension in cable $AB$.




Prob. 6–117

**6–118.** The three pin-connected members shown in the *top view* support a downward force of 60 lb at $G$. If only vertical forces are supported at the connections $B, C, E$ and pad supports $A, D, F$, determine the reactions at each pad.

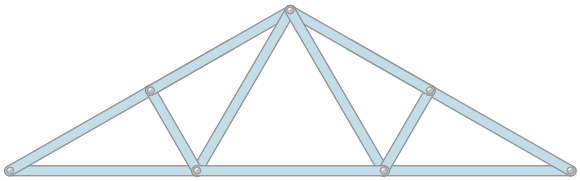



Prob. 6–118

# CHAPTER REVIEW

### Simple Truss

A simple truss consists of triangular elements connected together by pinned joints. The forces within its members can be determined by assuming the members are all two-force members, connected concurrently at each joint. The members are either in tension or compression, or carry no force.



Roof truss

### Method of Joints

The method of joints states that if a truss is in equilibrium, then each of its joints is also in equilibrium. For a plane truss, the concurrent force system at each joint must satisfy force equilibrium.

To obtain a numerical solution for the forces in the members, select a joint that has a free-body diagram with at most two unknown forces and one known force. (This may require first finding the reactions at the supports.)

Once a member force is determined, use its value and apply it to an adjacent joint.

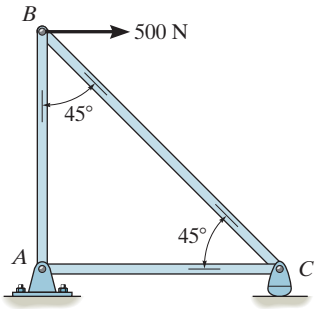
Remember that forces that are found to *pull* on the joint are *tensile forces*, and those that *push* on the joint are *compressive forces*.

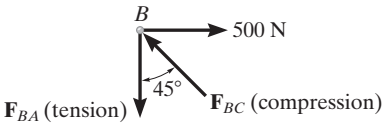
To avoid a simultaneous solution of two equations, set one of the coordinate axes along the line of action of one of the unknown forces and sum forces perpendicular to this axis. This will allow a direct solution for the other unknown.

The analysis can also be simplified by first identifying all the zero-force members.

$$\Sigma F_x = 0$$

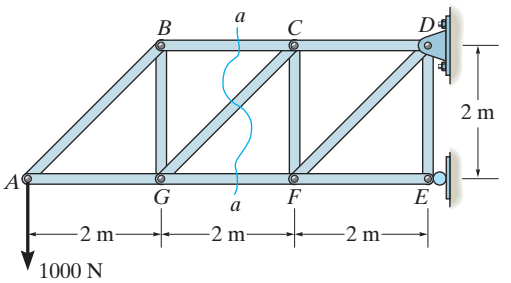
$$\Sigma F_y = 0$$



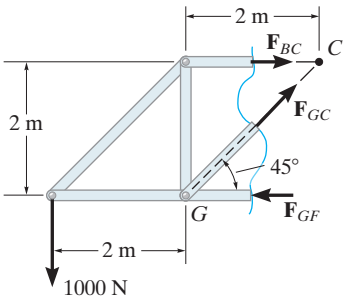



**Method of Sections**

The method of sections states that if a truss is in equilibrium, then each segment of the truss is also in equilibrium. Pass a section through the truss and the member whose force is to be determined. Then draw the free-body diagram of the sectioned part having the least number of forces on it.




Sectioned members subjected to *pulling* are in *tension*, and those that are subjected to *pushing* are in *compression*.




Three equations of equilibrium are available to determine the unknowns.

$$\Sigma F_x = 0$$
$$\Sigma F_y = 0$$
$$\Sigma M_O = 0$$

If possible, sum forces in a direction that is perpendicular to two of the three unknown forces. This will yield a direct solution for the third force.

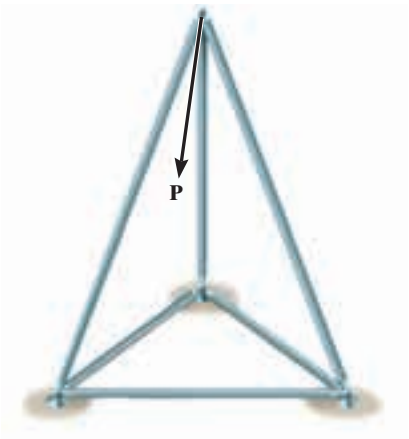
$$+\uparrow \Sigma F_y = 0$$
$$-1000\ \text{N} + F_{GC} \sin 45° = 0$$
$$F_{GC} = 1.41\ \text{kN (T)}$$

Sum moments about the point where the lines of action of two of the three unknown forces intersect, so that the third unknown force can be determined directly.

$$\circlearrowleft + \Sigma M_C = 0$$
$$1000\ \text{N}(4\ \text{m}) - F_{GF}(2\ \text{m}) = 0$$
$$F_{GF} = 2\ \text{kN (C)}$$

**Space Truss**

A space truss is a three-dimensional truss built from tetrahedral elements, and is analyzed using the same methods as for plane trusses. The joints are assumed to be ball-and-socket connections.

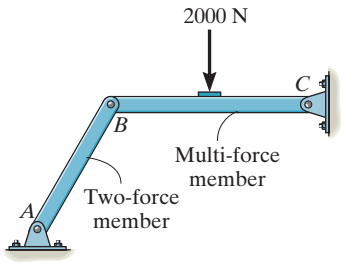


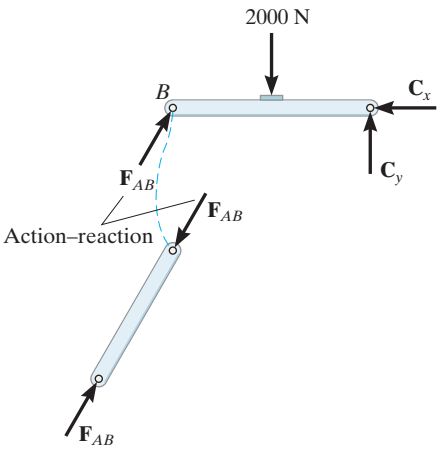

**Frames and Machines**

Frames and machines are structures that contain one or more multiforce members, that is, members with three or more forces or couples acting on them. Frames are designed to support loads, and machines transmit and alter the effect of forces.

The forces acting at the joints of a frame or machine can be determined by drawing the free-body diagrams of each of its members or parts. The principle of action–reaction should be carefully observed when indicating these forces on the free-body diagram of each adjacent member or pin. For a coplanar force system, there are three equilibrium equations available for each member.

To simplify the analysis, be sure to recognize all two-force members. They have equal but opposite collinear forces at their ends.

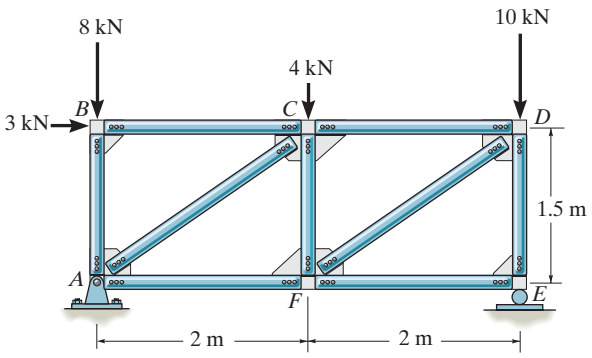


# REVIEW PROBLEMS

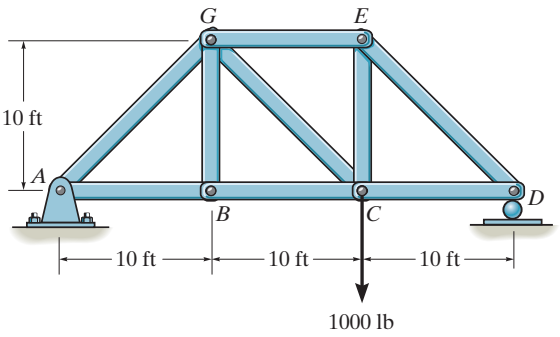
***All problem solutions must include FBDs.***

**R6–1.** Determine the force in each member of the truss and state if the members are in tension or compression.



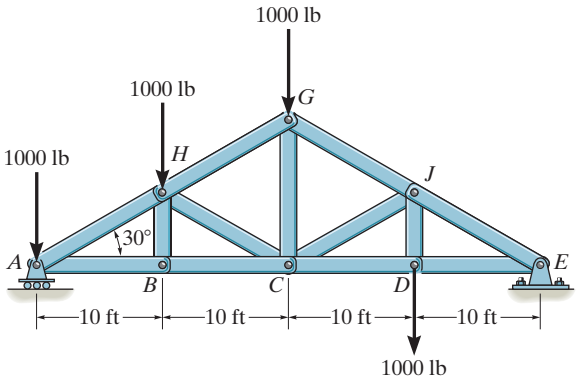

**Prob. R6–1**

**R6–2.** Determine the force in each member of the truss and state if the members are in tension or compression.



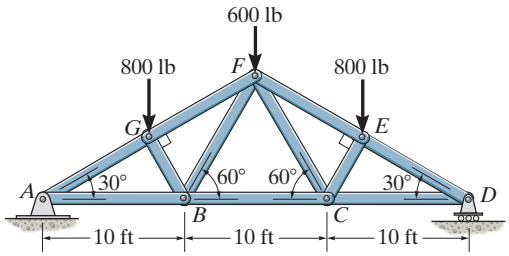

**Prob. R6–2**

**R6–3.** Determine the force in member *GJ* and *GC* of the truss and state if the members are in tension or compression.



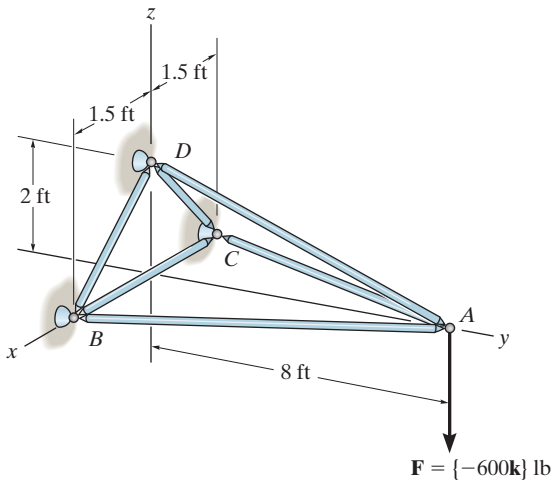

**Prob. R6–3**

**R6–4.** Determine the force in members *GF*, *FB*, and *BC* of the *Fink truss* and state if the members are in tension or compression.



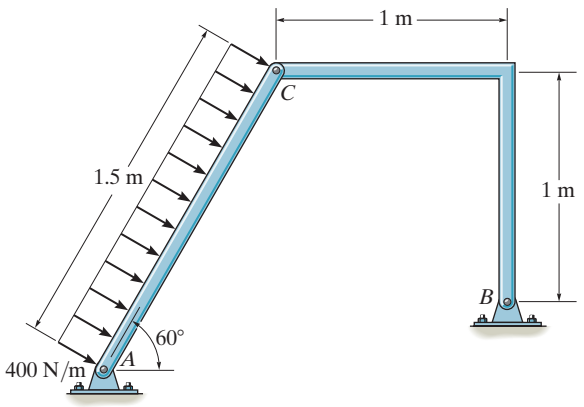

**Prob. R6–4**

**R6–5.** Determine the force in members $AB$, $AD$, and $AC$ of the space truss and state if the members are in tension or compression.



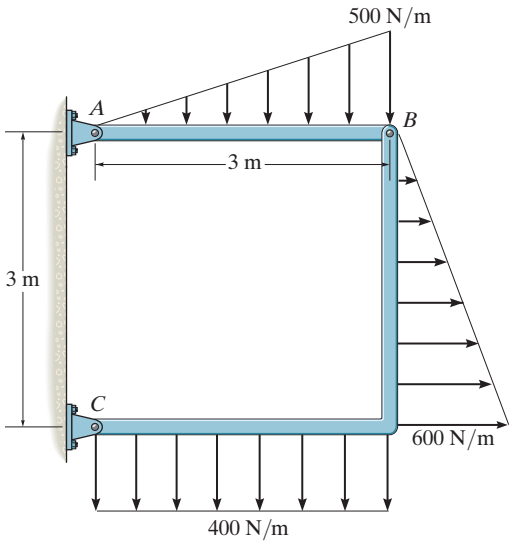

**Prob. R6–5**

**R6–6.** Determine the horizontal and vertical components of force that the pins $A$ and $B$ exert on the two-member frame.



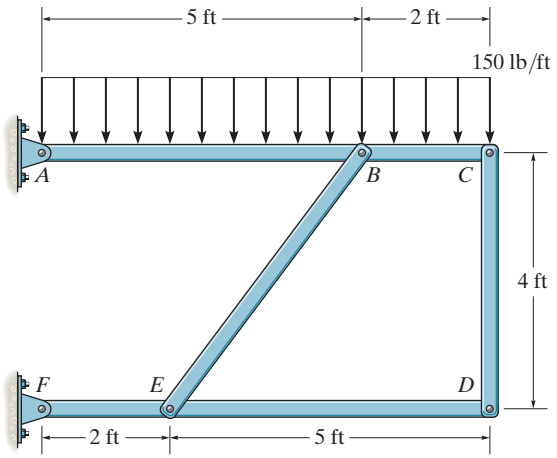

**Prob. R6–6**

**R6–7.** Determine the horizontal and vertical components of force at pins $A$ and $C$ of the two-member frame.




**Prob. R6–7**

**R6–8.** Determine the resultant forces at pins $B$ and $C$ on member $ABC$ of the four-member frame.




**Prob. R6–8**